

Mathieu-Harmonic Mass Spectrum of Elementary Particles in Induced Scalar-Pressure Gravity

George Lotishvili

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Introduction

Physical picture.—Consider a guitar string. When plucked, it vibrates at a definite note (frequency) with large amplitude. The vibration transfers energy to the guitar body, which radiates it as sound. Over time the string grows quieter—but the **note never changes**. Only the amplitude (energy) decreases; the frequency is invariant.

The ISPG universe behaves the same way. Space is a medium vibrating at a single fundamental carrier frequency $\nu_0 \approx 1.91 \times 10^{30}$ Hz—the MASS/Mathieu-calibrated “note” of the cosmos. This vibration is energy; its macroscopic manifestation is *pressure*. Through nonlinear self-interaction, the vibration spontaneously condenses into localized resonances (oscillons)—these are particles. The particle spectrum is treated as a Mathieu-organized resonance ladder built on this carrier; this quantitative ladder is not derived by the frequency-invariance theorem alone. The oscillon, in turn, generates a static gravitational field—the “sound” radiated into the surrounding medium. The original ν_0 -mode vibration depletes as it condenses into matter, but the matter itself radiates faint resonant echoes back into the medium. The accumulated echoes *are* dark energy—a slowly growing stochastic background (w slightly below -1) whose growth decelerates via negative feedback. The note ν_0 remains the same; the partition of energy between the fundamental mode, matter, and the echo background evolves (§4.4).

In the ISPG (Induced Scalar-Pressure Gravity) framework [20, 49, 51], space is described as a scalar-metric medium possessing pressure, flow, and the capacity for wave propagation. Particles (electrons, quarks, bosons) are **oscillons** of this medium—localized resonant excitations [96, 97].

The central question of this work is: **if all particles are excitations of a single medium, are their masses determined by the fundamental properties of that medium?**

The answer is affirmative. We show that:

- the medium possesses a single universal proper-time carrier frequency ν_0 ;
- the massive-particle spectrum is organized as a Mathieu resonance ladder built on this carrier; neutrinos are identified as *polarisation-rotation waves*—helical excitations whose oscillation plane rotates as they propagate—whose tiny mass reflects the suppressed torsional rigidity of the nonlinear $G_2(X)$ coupling (§5);
- the mass formula $m_N = m_1 b_N(q)$ (Mathieu characteristic value [86, 87]) compresses the charged leptons, quarks, and selected heavy bosons onto a common integer ladder; in the current PDG-style scan, 10/12 entries are below 1% and 11/12 below 5%, with the u quark (5.62%) as the outlier and b, H, c, t all below 0.07% [88]; e and μ are calibration anchors, and the physical rule selecting each particle’s N remains an open derivation target;
- the electron is the first fully stable occupied charged Mathieu mode under the adopted gap-threshold criterion: $N = 1\text{--}3$ modes lie in the wide-gap instability zone, while $N = 4$ is a barely-stable resonance candidate predicted at ~ 329 keV (Sec. 3.15);

- the parameter q follows from the shift-symmetric $G_2(X)$ extension of the ISPG action [98, 99]—the same extension independently established for ghost-free UV completion in Appendix 11 (Stability)—without introducing new fields or breaking any symmetry;
- the integer-harmonic structure of the scalar sector (j_0 zeros $= n\pi$) is the spectral signature of a one-dimensional substrate; the requirement for tensor ($\ell \geq 2$) modes to propagate selects $d = 3$ as the minimal spatial dimension, while exact $d = 3$ rather than $d > 3$ requires the minimality/empirical input stated in §7.

Epistemic status of the above.—The mass formula reproduces the observed spectrum numerically, but several ingredients are working hypotheses rather than first-principles derivations: (i) the reduction from the full three-dimensional cavity problem of the Quantum paper to the one-dimensional Mathieu equation used here is not formally established and is left as an open calculation (§3.13); (ii) the numerical value of ν_0 is a calibrated MASS/Mathieu value, while its first-principles origin remains open (§3); (iii) the identification of neutrinos as polarisation-rotation waves (§5) and of the $W^\pm/Z^0$ bosons as bridge modes (§5.6) is heuristic, with the Standard-Model quantum numbers not derived here; (iv) the three-dimensional selection argument relies on the assumption that tensor ($\ell \geq 2$) modes are physically required, which the main theory invokes but does not prove from the action alone (§7). Each of these items is flagged locally in the corresponding section; their resolution is deferred to future work.

1 Frequency Invariance and Derivation of Time Dilation

1.1 Starting Point in the Theory

In the ISPG framework the bi-conformal metric is built from two operational postulates (Appendix 3):

$$d\tau = e^{\phi/2} dt \quad (\text{clock postulate}), \quad (1)$$

$$d\ell = e^{-\phi/2} d\sigma \quad (\text{rod postulate}), \quad (2)$$

while the effective mass obeys:

$$m_{\text{eff}} = m_0 e^{\phi/2}. \quad (3)$$

All three—time, length, mass—are manifestations of a single conformal factor ($e^{\phi/2}$). Currently all three are treated as *parallel* consequences of a single metric.

1.2 Logical Necessity: Why an Invariant Must Exist

Before calling the invariant frequency an *assumption*, let us consider a more fundamental question: is “time slowing down” even possible if **nothing in the universe remains unchanged**?

The answer is no.

“Slowing down” is a *comparative* notion. To say “this clock runs slowly” one needs a reference—something that *did not* slow down, against which to compare. If everything changes equally, no one notices anything. This is the deep content of the local undetectability theorem.

Therefore time dilation logically requires **two elements**:

1. **A variable element**—something that changes with the pressure state (ϕ). In ISPG this is the oscillon’s *size* and *mass*.
2. **An invariant element**—something that *never* changes, relative to which the variable element’s change becomes perceptible.

What is this invariant element? Consider the candidates:

- Size? No, it varies: $d\ell = e^{-\phi/2} d\sigma$.
- Mass? No, it varies: $m_{\text{eff}} = m_0 e^{\phi/2}$.
- Clock rate? No, that is precisely what we seek to explain.
- Local speed of light? It is invariant ($c_{\text{local}} = c$), but this is a *consequence*, not a cause.

One candidate remains: the **fundamental vibration frequency** of the medium, ν_0 . If it does not change, then variations in size and mass become *perceptible* relative to it—and this perception is what we call “time dilation.”

1.3 Circle Analogy

Imagine a circular track of fixed circumference L . Two runners stand on it: one tall (stride length λ_1), the other exactly half as tall (stride $\lambda_2 = \lambda_1/2$). Both run at the same cadence—the same number of strides per second.

The tall runner completes the lap in $N_1 = L/\lambda_1$ strides. The short one in $N_2 = L/\lambda_2 = 2N_1$ strides—twice as many strides, hence twice as much time.

Now imagine each runner drags the hand of a clock. The tall runner turns the hand fast—“time passes quickly.” The short runner turns it slowly—“time passes slowly.”

Three elements are present:

- **Variable:** the runner’s size (stride length)—this is m_{eff} .
- **Invariant:** the track length L —this is the fundamental frequency ν_0 (the “cycle” that is the same for everyone).
- **Result:** the clock rate—this is $d\tau/dt$.

If the track length also changed (proportionally to the runner’s size), both runners would finish the lap simultaneously—no “slowing down” would occur. **It is precisely the invariance of the track (frequency invariance) that makes slowing down possible.**

1.4 Theorem: Invariance of the Fundamental Frequency

The logical necessity argument (§1) identified a fundamental carrier frequency ν_0 as the invariant element required by the very concept of time dilation. The theorem below proves the narrower statement needed here: once a localized matter-sector resonator with proper-time frequency ν_0 is given, a static background ϕ_{bg} does not change that proper-time frequency. The theorem does not derive the existence, uniqueness, or numerical value of ν_0 ; the value used in this paper is the separate MASS/Mathieu calibration discussed in §3.

Theorem (frequency invariance). *The proper-time oscillation frequency of any localized resonant excitation in Φ (oscillon, atomic clock, or other matter-sector resonator) is position-independent: if such a resonator oscillates at frequency ν_0 in proper time at $\phi = 0$, it oscillates at the same frequency ν_0 (measured in local proper time of an observer co-located with the resonator) in any region with static background potential $\phi_{\text{bg}} \neq 0$. The substrate itself is not assigned its own clock; the statement is about matter-sector resonators living in the substrate.*

Proof.—Consider a region where the static gravitational background is approximately constant: $\phi_{\text{bg}} \approx \text{const}$. The bi-conformal metric is

$$ds^2 = -e^{\phi_{\text{bg}}} c^2 dt^2 + e^{-\phi_{\text{bg}}} d\sigma^2. \quad (4)$$

Introduce locally Minkowski coordinates of an observer co-located with the resonator at fixed ϕ_{bg} :

$$\tau \equiv e^{\phi_{\text{bg}}/2} t, \quad \xi^i \equiv e^{-\phi_{\text{bg}}/2} \sigma^i. \quad (5)$$

In these coordinates the metric becomes Minkowski: $ds^2 = -c^2 d\tau^2 + d\xi^2$. The transformation is a mathematical convenience for analysing the field equation of the oscillon solution; it does not assign a hidden time variable to the substrate.

The kinetic scalar $X = -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$ is invariant under coordinate transformations (it is a scalar). Therefore the $G_2(X)$ field equation

$$\nabla_\mu \left[\frac{\partial G_2}{\partial X} \nabla^\mu \phi \right] = 0 \quad (6)$$

takes, in the (τ, ξ) chart, exactly its flat-space form:

$$\partial_\mu \left[\frac{\partial G_2}{\partial X} \eta^{\mu\nu} \partial_\nu \phi \right] = 0. \quad (7)$$

Equation (7) contains no reference to ϕ_{bg} : its coefficients are the coupling constants M_{Pl} , α , M appearing in $G_2(X)$. A localized oscillon therefore keeps the same flat-space proper-time eigenfrequency $\omega_0 = 2\pi\nu_0$ when placed in a static background. This proves background independence of the local proper-time frequency, not a first-principles computation of its numerical value. $\square$

Coordinate frequency and gravitational redshift.—In coordinate time t , the oscillon appears to oscillate at frequency

$$\nu_{\text{coord}} = \nu_0 e^{\phi_{\text{bg}}/2}, \quad (8)$$

since $\tau = e^{\phi_{\text{bg}}/2} t$. An asymptotic observer ($\phi_{\text{bg}} = 0$) receiving signals from a region with $\phi_{\text{bg}} < 0$ detects a lower frequency—this is the gravitational redshift, here *derived* from the resonator’s intrinsic frequency invariance rather than postulated.

Epoch independence.—For cosmological evolution, $\phi_{\text{bg}}(t)$ varies on timescales $\sim H_0^{-1} \sim 10^{17}$ s, while a typical oscillon resonates at $\nu_0 \sim 10^{30}$ Hz. The adiabatic ratio $\nu_0/H_0 \sim 10^{47}$ ensures that the local proper-time frequency tracks its flat-space value to fractional accuracy $\sim (H_0/\nu_0)^2 \sim 10^{-94}$. Over the entire Hubble time, the cumulative phase drift is $\Delta\omega/\omega_0 \sim H_0/(2\pi\nu_0) \sim 10^{-48}$ —undetectable by any conceivable experiment. This is an adiabatic extension of the static theorem, not an independent proof for arbitrary dynamical backgrounds.

Physical content.—What changes is not the frequency but the “stride size” of each vibration—the oscillon’s effective mass and spatial scale. The frequency is the invariant “track” against which size changes are perceived as “time dilation.”

Central thesis. Time is the balance between oscillon size and the medium’s intrinsic frequency. This frequency, measured in local proper time, is universal—it is the same at every point in space and is adiabatically stable over cosmological evolution. The particle mass spectrum is then modeled by the Mathieu resonance program built on this carrier.

1.5 Quantum Origin: The ISPG Vacuum as a Relativistic Condensate

The local proof of §1.4 established static-background invariance of a given proper-time carrier frequency. A deeper question remains: *why* does the medium have a single, universal carrier frequency at all? The answer is that the ISPG vacuum is a macroscopic quantum coherent state—a relativistic Bose–Einstein condensate [94, 95] of the archion field.

Quantization of the archion field. Promote the classical scalar ϕ to a quantum field operator and decompose into Fourier modes in a large comoving volume V :

$$\hat{\phi}(\mathbf{x}, t) = \sum_{\mathbf{k}} \frac{1}{\sqrt{2\omega_k V}} \left[a_{\mathbf{k}} e^{i(\mathbf{k} \cdot \mathbf{x} - \omega_k t)} + a_{\mathbf{k}}^\dagger e^{-i(\mathbf{k} \cdot \mathbf{x} - \omega_k t)} \right], \quad (9)$$

with canonical commutation relation $[a_{\mathbf{k}}, a_{\mathbf{k}'}^\dagger] = \delta_{\mathbf{k}\mathbf{k}'}$.

Condensate ground state. The ISPG vacuum is defined as a coherent state of the zero-momentum ($\mathbf{k} = 0$) mode:

$$a_0 |\alpha_0\rangle = \alpha_0 |\alpha_0\rangle, \quad N_0 \equiv |\alpha_0|^2 \gg 1, \quad (10)$$

where N_0 is the macroscopic occupation number. All N_0 quanta share the single-particle ground-state energy $\varepsilon_0 = \hbar\omega_0$. The expectation value of the field operator is

$$\langle \hat{\phi} \rangle = \frac{2|\alpha_0|}{\sqrt{2\omega_0 V}} \cos(\omega_0 t + \theta_0) \equiv \Phi_{\text{bg}} \cos(\omega_0 t), \quad (11)$$

which is precisely the classical oscillating background assumed in the oscillon ansatz (82). The classical field used throughout this work is the condensate mean field.

Frequency as chemical potential. In a non-relativistic BEC with contact interaction $g|\Psi|^2$, the chemical potential $\mu = g n_0$ determines the condensate frequency: $\omega = \mu/\hbar$. In ISPG the analogous role is played by the nonlinear kinetic self-interaction $G_2(X) \supset \alpha X^2/M^2$: the X^2 term acts as the self-interaction that sets the oscillation frequency. From the self-consistency condition (87),

$$\Phi_{\text{bg}}^2 \omega_0^2 = \frac{2 M^2 M_{\text{Pl}}^2}{3 \alpha}, \quad (12)$$

the product $\Phi_{\text{bg}}^2 \omega_0^2$ is fixed entirely by the coupling constants α , M , M_{Pl} . Given the total number of quanta $N_0 = \Phi_{\text{bg}}^2 \omega_0 V / (2\hbar)$ (from equating the classical energy density $\rho = \frac{1}{2} \Phi_{\text{bg}}^2 \omega_0^2$ with $N_0 \hbar \omega_0 / V$), the frequency is

$$\boxed{\omega_0 = \frac{M^2 M_{\text{Pl}}^2 V}{3 \alpha \hbar N_0} \propto \frac{1}{n_0}}, \quad (13)$$

where $n_0 = N_0/V$ is the archion number density. This is a *structural* analog of $\mu = g n_0$ in a standard BEC: in both systems the oscillation frequency is set by the condensate density and coupling constants—not by position. The functional dependence is, however, *inverted*: in a repulsive BEC the chemical potential *increases* with density ($\mu \propto n_0$), whereas in ISPG the proper-time frequency *decreases* ($\omega_0 \propto 1/n_0$) because the X^2 self-interaction raises the effective inertia of the oscillating field at higher amplitude. Since α , M , M_{Pl} are spacetime-constant parameters and n_0 is homogeneous on scales where $\phi_{\text{bg}} \approx \text{const}$, the frequency ω_0 is the same at every point.

Why ν_0 invariance is automatic. In a coherent state (10), every quantum carries the same energy $\hbar\omega_0$. Spatial variation of the condensate (Bernoulli depletion near a massive body) changes the *local amplitude* $\Phi_{\text{bg}}(x)$ and the local number density $n_0(x)$, but not the single-particle energy. This is exact: in a superfluid, a density dip (vortex, depletion layer) does not shift the chemical potential—it is the chemical potential that is globally uniform, while the density adjusts. In ISPG, the chemical potential is ω_0 :

$$\nu_0 = \frac{\omega_0}{2\pi} = \text{const (in proper time)} \quad \text{everywhere.} \quad (14)$$

The coordinate-frame manifestation $\nu_{\text{coord}} = \nu_0 e^{\phi_{\text{bg}}/2}$ (Eq. 8) is the standard relation between proper-time and coordinate frequencies.

Measurable-existence guardrail. The measurable-existence identification gives an operational reading of condensate depletion and measurable-content redistribution: a local change in active carrier content is seen by matter standards as a change of measured scale, density, or mass. It does not replace the separate MASS/Mathieu calibration of ν_0 . In particular, this identification does not derive ν_0 , does not derive 1.91×10^{30} Hz, does not prove the mass ladder, and does not close the Mathieu q derivation. The notation $\mathcal{N}_{\text{act}}$ used below denotes the matter-standard side of this measurable-existence identification: the effective amount of interaction-active resonant participation carried by a localized oscillon or measuring standard. It is distinct from the condensate occupation N_0 and from the density $n_0 = N_0/V$. The latter describe the background condensate; $\mathcal{N}_{\text{act}}$ describes the operational content of a resonant standard and is normalized through ratios such as $\mathcal{N}_{\text{act}}(\phi)/\mathcal{N}_{\text{act}}(0)$. This is an identification, not a synonym: in the local/static sector the effective mass m_{eff} , the matter-standard length L , and the clock rate are read as projections of the same active measurable content, with m_{eff} playing the role of the asymptotic mass/energy projection rather than an independent renaming of $\mathcal{N}_{\text{act}}$.

Oscillons as condensate excitations. Particles (oscillons) are localized, nonlinear excitations on top of the condensate background, analogous to vortices in He-II, solitons in nonlinear optics, or rotons in superfluid helium. In the second-quantized language:

$$\hat{\phi}(x, t) = \underbrace{\Phi_{\text{bg}} \cos(\omega_0 t)}_{\text{condensate (vacuum)}} + \underbrace{\delta\hat{\phi}(x, t)}_{\text{excitations (oscillons, waves)}}. \quad (15)$$

The condensate determines the arena—background pressure, c_{coord} , effective mass scaling. The excitations are what we call matter. This split is the quantum analog of the classical decomposition already used in §3.13.

Summary of the condensate identification.

Standard BEC	ISPG condensate	Physical meaning
$\Psi(\mathbf{x}, t)$	$\phi(\mathbf{x}, t)$	Macroscopic order parameter
$\mu = g n_0$	$\hbar\omega_0 \propto 1/n_0$	Chemical potential = frequency
$ \Psi ^2 = n_0$	$\Phi_{\text{bg}}^2 \omega_0 / (2\hbar V)$	Number density
$g \Psi ^2$	$\alpha \bar{X}^2 / M^2$	Nonlinear self-interaction
Phonons, vortices	Oscillons, gravitational waves	Elementary excitations
Local n varies	Local $\Phi_{\text{bg}}(x)$ varies	Density adjusts; ω_0 does not

The “one archion everywhere” of the intuitive picture is, in formal language, the statement that N_0 quanta occupy a single macroscopic quantum state with one frequency. This is neither a postulate nor a metaphor—it is the standard quantum-field-theoretic description of a condensate, realized here for the ISPG scalar sector.

1.6 Practical Note: Searching for ν_0

This subsection gives a search heuristic, not an independent proof. Every particle has a Compton frequency $\nu_C = mc^2/h$ (de Broglie, 1924; Müller, 2013). If a single carrier frequency underlies the MASS/Mathieu ladder, the observed Compton frequencies should display a commensurate pattern when expressed through the ladder’s dimensionless indices. The exact integer/harmonic status of that pattern belongs to the Mathieu analysis, not to the frequency-invariance theorem itself.

Concretely: rational or near-rational mass ratios are evidence for a resonant organization of the spectrum and guide the calibration of ν_0 in the MASS/Mathieu model. They do not by themselves derive ν_0 from first principles.

1.7 Guitar String Analogy

Recall a guitar string. If we loosen it (reduce tension), the same fret produces a different note. To maintain the same note (frequency) one must:

- shorten the vibrating length (size changes),
- reduce the amplitude (mass/energy changes),
- but keep the frequency the same.

Analogously: when the medium's background pressure decreases, the oscillon "shrinks in size and mass" in order to maintain the fundamental frequency.

1.8 Tall and Short Person Analogy

Two people run at the same cadence (one frequency ν_0). The tall person has long legs (large m_{eff} , high pressure)—covers a large distance per stride. The short person has short legs (small m_{eff} , low pressure)—covers less distance at the same cadence.

"Time rate" = cadence $\times$ stride size. The cadence (ν_0) is the same. The stride (m_{eff}) differs. Therefore the rate differs.

1.9 Derivation of the Clock Postulate

Given:

1. Rod postulate: $d\ell = e^{-\phi/2} d\sigma$.
2. Effective mass scaling: $m_{\text{eff}} = m_0 e^{\phi/2}$ (follows from the bi-conformal metric's matter coupling).
3. Frequency invariance (Theorem 1.4): proper-time frequency $\nu_0 = \text{const}$; coordinate frequency $\nu_{\text{coord}} = \nu_0 e^{\phi/2}$.

To derive: Clock postulate $d\tau = e^{\phi/2} dt$.

Derivation.

An oscillon-based clock counts oscillations. In coordinate time dt , the number of oscillations is $N = \nu_{\text{coord}} dt = \nu_0 e^{\phi/2} dt$ (Eq. 8). Each oscillation takes proper time $1/\nu_0$ (since ν_0 is the proper-time frequency). Therefore:

$$d\tau = N \times \frac{1}{\nu_0} = \nu_0 e^{\phi/2} dt \times \frac{1}{\nu_0} = e^{\phi/2} dt. \quad (16)$$

Hence:

$$\boxed{d\tau = e^{\phi/2} dt} \quad (17)$$

which is exactly the clock postulate (1). $\square$

Note on circularity. The coordinate frequency $\nu_{\text{coord}} = \nu_0 e^{\phi/2}$ (Eq. 8) was derived using the bi-conformal metric, in which the clock postulate is already embedded. This derivation is therefore a self-consistency check rather than an independent proof.

This problem is fully resolved in Section 2.6, where mass scaling is **independently** derived from two-plate collision physics.

1.10 Derivation of the Local Speed of Light

The locally measured speed of light:

$$c_{\text{local}} = \frac{d\ell}{d\tau} = \frac{e^{-\phi/2} d\sigma}{e^{\phi/2} dt} = e^{-\phi} \frac{d\sigma}{dt}. \quad (18)$$

The coordinate speed of light in the bi-conformal metric ($ds^2 = 0$):

$$\frac{d\sigma}{dt} = c e^{\phi}. \quad (19)$$

Substituting (19) into (18):

$$c_{\text{local}} = e^{-\phi} \cdot c e^{\phi} = c. \quad (20)$$

$$\boxed{c_{\text{local}} = c} \quad (21)$$

The locally measured speed of light is invariant once the rod and clock scalings are used. The clock scaling's status is the self-consistency check discussed above and qualified below. $\square$

1.11 Explanation of Redshift

Light is emitted at pressure state ϕ_{emit} and received at ϕ_{recv} .

The fundamental frequency ν_0 does not change. But a local observer measures frequency with their own clock, whose “stride size” is proportional to $e^{\phi/2}$. Therefore:

$$\frac{\nu_{\text{obs}}}{\nu_{\text{emit}}} = \frac{e^{\phi_{\text{emit}}/2}}{e^{\phi_{\text{recv}}/2}} = e^{(\phi_{\text{emit}} - \phi_{\text{recv}})/2}. \quad (22)$$

Gravitational redshift: Light is emitted at low pressure (in a gravitational well, $\phi_{\text{emit}} < \phi_{\text{recv}}$). Then $\nu_{\text{obs}} < \nu_{\text{emit}}$ —redshift.

Light “loses” nothing. The fundamental oscillation is the same. What changed is the *measuring ruler's size*: at the emission point the clock “stride” was small (small m_{eff}), at the reception point it is large. We perceive this difference as “redshift.”

Cosmological redshift: In the past, background pressure was high—oscillons were “large,” the clock stride was large. Today background pressure has decreased—oscillons are “small,” the clock stride is small. Distant light was emitted in the “large stride” epoch and arrives in the “small stride” epoch—we measure a large-ruler result with a small ruler, and we see redshift.

1.12 Summary: One Postulate Instead of Two

If the scoped proper-time frequency invariance above is used, then:

Given	Rod postulate + mass scaling + $\nu_0 = \text{const}$
Self-consistency check	Clock postulate: $d\tau = e^{\phi/2} dt$
Consequence	$c_{\text{local}} = c$ once the rod and clock scalings are used
Consequence	Gravitational and cosmological redshift

Instead of treating the clock law as an unrelated operational postulate, this section records its oscillon self-consistency route from the rod scaling and frequency invariance. The caveat below spells out the status: this is a self-consistency check, not an independent kinematic derivation from frequency data alone.

Time does not “slow down.” The stride shrinks. The rhythm is the same.

1.13 Status

The derivation (16) above relies on the interpretation that proper-time accumulation is proportional to the oscillation’s energetic “weight.”

Caveat (self-consistency, not independent derivation). Step (3) of the derivation uses $\nu_{\text{coord}} = \nu_0 e^{\phi/2}$, which itself follows from the metric/clock law of the bi-conformal background. Equation (16) should therefore be read as a *self-consistency check* of the clock postulate within the oscillon picture, not as an independent derivation of $d\tau = e^{\phi/2} dt$ from kinematics alone.

Update. The two-plate model (§2.6) **confirmed** this interpretation: $m_{\text{eff}} = m_0 e^{\phi/2}$ follows from collision kinematics (not analogical reasoning), and the clock postulate $d\tau = e^{\phi/2} dt$ follows logically from it. The numerical Mathieu solution (§3.12) provided quantitative confirmation: muon mass to $< 0.001\%$ accuracy.

2 Harmonic Resonance: The Universe as a Chord

2.1 From One Frequency to Many

In the previous section a single proper-time carrier frequency ν_0 was identified as invariant under static backgrounds. This does not by itself determine the particle mass ladder. Nature contains different particles with different masses—electrons, muons, quarks, bosons. If mass and frequency are related ($E = mc^2 = h\nu$), different masses imply different Compton frequencies. How is this compatible with a single carrier frequency?

The answer comes from the same guitar string considered above, but now examined more deeply.

2.2 Guitar String Harmonics

When a guitar string vibrates, it does not produce only one note. Alongside the fundamental, an entire spectrum of *harmonics*—accompanying frequencies that are integer multiples of the fundamental—sounds:

$$\nu_n = n \nu_{\text{fund}}, \quad n = 1, 2, 3, \dots \quad (23)$$

But string physics is richer. Pressing the string at different points yields *musical intervals*:

- At the midpoint—**octave**: ratio 2:1.
- At $2/3$ —**fifth**: ratio 3:2.
- At $3/4$ —**fourth**: ratio 4:3.
- At $4/5$ —**major third**: ratio 5:4.

All these intervals are *compatible* with the fundamental: they are rational fractions, i.e. every n -th vibration coincides exactly with the fundamental cycle.

This is precisely why these intervals sound “harmonious”—there is no dissonance between them because they are *commensurate sub-modes* of the same fundamental vibration.

2.3 The Medium as a String—Oscillons as Harmonics

The ISPG medium is far richer than a one-dimensional string—it is three-dimensional, has pressure, compressibility, and topological structure (vortices). But the fundamental principle is the same: the medium can support only certain resonant modes—those oscillations that “fit” local conditions and do not vanish through destructive interference.

This means that alongside the fundamental ν_0 , the medium may host *harmonically compatible* resonances:

$$\nu_{n,m} = \frac{n}{m} \nu_0, \quad n, m \in \mathbb{Z}^+, \quad (24)$$

where n/m is a rational fraction, analogous to a musical interval.

Different particles would be different “notes” in this spectrum:

- One oscillon “lives” on every fundamental click—the fundamental mode.
- Another responds every *second* click—half frequency, twice the mass.
- A third responds every *third* click.
- Some respond every 2 out of 3 clicks (the analogue of a fifth), or 3 out of 4 (a fourth).

All modes “live” in the same fundamental rhythm, but each “listens” to it differently—one on every beat, another on every second, a third on every third.

2.4 Two Plates and Bouncing Balls

The guitar-string analogy has one weakness: a string possesses *all* harmonics simultaneously. In nature, however, only specific particles with specific masses exist. What *selection rule* determines which harmonics are “allowed” and which are not?

A different, more precise analogy answers this question.

Picture: two planes of a wave. The medium’s fundamental vibration ν_0 creates a standing wave. This wave has two extrema: the **crest** (upper plate) and the **trough** (lower plate). The two plates stand at a fixed “distance” L —the space defined by the wave amplitude.

A particle (oscillon) bounces *between* these two plates: the lower plate throws it upward, it crosses the distance, the upper plate “throws” it back down, it returns, the lower plate throws it up again—and so on indefinitely.

The particle **never rests**. It is in perpetual motion between the plates—corresponding to the fact that real elementary particles never “stop”: their internal oscillation (Compton frequency, Zitterbewegung) is constant.

Resonance condition. A ball bouncing between two plates is stable only when its round-trip time (bottom to top + top to bottom = one full cycle) *exactly* fits an integer multiple of the plate vibration period:

$$t_{\text{cycle}} = t_{\uparrow} + t_{\downarrow} = \frac{n}{\nu_0}, \quad n = 1, 2, 3, \dots \quad (25)$$

If the ball reaches a plate at the “wrong” moment—the plate is already moving the other way—the ball loses energy, and after a few cycles the bouncing ceases. This is an **unstable mode** that “dies.”

Mass = collision intensity. Different balls cross the distance between the two plates at different speeds:

- A **fast** ball collides frequently with the plates $\Rightarrow$ large impulse per collision $\Rightarrow$ strong interaction with the external world $\Rightarrow$ **large mass**.
- A **slow** ball collides rarely $\Rightarrow$ small impulse $\Rightarrow$ weak interaction $\Rightarrow$ **small mass**.

In this picture **mass is not a property of a “thing”**—mass is the intensity of the **bouncing**. It is the rate of impulse transfer to the plates. A light particle (electron) crosses slowly, collides rarely. A heavy particle (top quark) crosses fast, collides often.

Meanwhile *all* plates vibrate at the same frequency ν_0 . One ball keeps up with every vibration; another needs several vibrations. But all share the same rhythm.

Energy conservation: the necessity of balance. The upper plate “throws” the ball downward with exactly the same force the lower plate threw it upward. The ball cannot “stick” to the upper plate—it necessarily returns. This symmetry is the energy conservation law in this picture.

If the lower impulse $\neq$ the upper impulse, the ball gradually drifts to one side and bouncing ceases. Only **perfect balance** is stable.

Different balls—different sub-harmonics.

- Fast ball: completes a full cycle on every vibration ($n = 1$, frequency ν_0 , maximum mass in this class).
- Medium ball: completes a cycle every *second* vibration ($n = 2$, frequency $\nu_0/2$).
- Slow ball: every *third* vibration ($n = 3$, frequency $\nu_0/3$).
- Very slow: every n -th vibration.

Complex mass ratios. The ball’s transit time depends not only on its size but also on elasticity, the character of plate impulses, and wave amplitude. This complex dependence naturally produces complicated mass ratios (e.g. $m_\mu/m_e \approx 206.8$ —not a simple 2 or 3), yet *all* are still determined by a single frequency between two plates.

Physical precedents. The two-plate bouncing picture has direct parallels in known physics:

1. **Quantum particle in a box.** A particle between two walls—the most fundamental model in quantum mechanics. Energy levels $E_n = n^2\pi^2\hbar^2/(2mL^2)$ are discrete and determined by the box size L .
2. **Zitterbewegung.** Schrödinger (1930) showed that the Dirac electron oscillates between positive and negative energy states—effectively “bouncing” between matter and antimatter at frequency $\omega = 2mc^2/\hbar$. The factor 2 arises because one cycle involves *two* crossings: up and down.
3. **Casimir effect.** Between two metal plates, vacuum fluctuations produce a measurable force. Only certain wavelengths “fit” between the plates—the rest are forbidden. This is a natural selection rule between two boundaries.

Experimental confirmation: Couder’s bouncing droplets. In 2005 Yves Couder and Emmanuel Fort (Paris) demonstrated an experiment in which an oil droplet bounces on a vibrating oil bath. The droplet bounces on every second vibration (sub-harmonic $\nu_0/2$) and each bounce creates a surface wave—a *pilot wave*.

This classical system **reproduced quantum-mechanical phenomena**: double-slit interference, quantized orbits, tunneling, and spin analogues, as reported in the experimental literature in 2005–2006.

ISPG analogue.

Two-plate model	↔	ISPG
Upper/lower plate		Wave crest/trough in medium
Plate frequency ν_0		Fundamental frequency ν_0
Bouncing ball		Oscillon (particle)
Transit speed		Effective mass m_{eff}
Collision impulse		Inertia / mass
Stable bouncing		Stable particle
Unstable bouncing		Non-existent particle
Plate balance		Energy conservation

Mathematical formalization. The dynamics of a particle oscillating between two boundaries is described by the **Mathieu equation** [86, 87]:

$$\frac{d^2x}{dt^2} + (a - 2q \cos(2\nu_0 t)) x = 0, \quad (26)$$

where x is the particle’s position between the two plates, and a, q depend on the particle’s properties and plate vibration amplitude.

The Mathieu equation possesses **stability zones**—specific (a, q) combinations where the solution is periodic (the ball bounces stably), and **instability zones** where the solution grows exponentially or decays (the ball “dies”).

Stability zones = allowed harmonics = existing particles. Instability zones = forbidden modes = non-existent particles.

If the oscillon wave equation in the ISPG medium yields a Mathieu-type structure, the particle mass spectrum follows automatically from the stability zones.

2.5 What Happens When the Plates Move Closer

Now let us pose the decisive question: if the plates move closer ($L \rightarrow L' < L$) and the ball *proportionally* shrinks, what changes?

From the ball’s perspective: nothing. The ball’s size and its “habitat” (distance between plates) shrank equally. The ball cannot tell it is smaller—its ruler is smaller too. This is **local undetectability**: local physics is unchanged.

From an external observer’s perspective: everything. The external observer’s plates are unchanged. They see:

1. **Rhythm is the same**— ν_0 did not change. The ball still completes a cycle every n -th vibration.
2. **Distance is less**—plates are closer; each jump covers *less* distance.
3. **Speed is less**—less distance in the same time = less speed.
4. **Impulse is less**—slow collision = less impulse transfer = **less mass**.
5. **Time rate is less**—same rhythm $\times$ smaller stride = **slower clock**.

This is time dilation. Denote the plate-separation reduction factor by α : $L' = \alpha L$, where $\alpha < 1$. Then:

- Stride length: $\lambda' = \alpha \lambda$ (rod postulate),

- Speed: $v' = L'/t = \alpha v$,
- Collision impulse: $p' = \alpha p \Rightarrow m' = \alpha m$ (mass scaling),
- Time rate: $\nu_0 \times \lambda' = \alpha (\nu_0 \times \lambda) \Rightarrow d\tau' = \alpha d\tau$ (clock postulate).

In ISPG, $\alpha = e^{\phi/2}$. Everything follows from a single factor—the plate separation.

Resolving the circularity. This picture resolves the circularity problem described earlier. When deriving the clock postulate we used $m_{\text{eff}} = m_0 e^{\phi/2}$, which was itself derived from the clock postulate—a circle.

In the two-plate model, mass scaling **does not require the clock postulate**. It follows directly from collision physics:

Plates closer $\Rightarrow$ less distance $\Rightarrow$ less speed $\Rightarrow$ less impulse $\Rightarrow$ less mass.

This is an *independent* causal chain that does not require the clock postulate. Mass scaling becomes a *consequence*, not an *assumption*.

Then: mass + invariant frequency $\Rightarrow$ clock postulate—and this chain is no longer circular.

Complete causal chain.

$\nu_0 = \text{const}$ (fundamental frequency) $+$ Plate separation $L(\phi)$ (rod postulate) $\Downarrow$ Less $L \Rightarrow$ less impulse $\Rightarrow m_{\text{eff}} = m_0 e^{\phi/2}$ (mass scaling) $\Downarrow$ $\nu_0 \times m_{\text{eff}} \Rightarrow d\tau = e^{\phi/2} dt$ (clock postulate) $\Downarrow$ $d\ell/d\tau = c$ (light-speed invariance)

Everything flows from two elements: invariant frequency + plate separation. Everything else—mass, clock, speed of light—is a *consequence*.

2.6 Formal Derivation from the Two-Plate Model

Given:

1. Fundamental frequency: $\nu_0 = \text{const}$.
2. Rod postulate: the oscillon's local size ℓ_0 is unchanged; its coordinate size (plate separation) is

$$L(\phi) = \ell_0 e^{\phi/2}. \quad (27)$$

(From $d\ell = e^{-\phi/2} d\sigma$ inversion: $d\sigma = e^{\phi/2} d\ell$.) In the measurable-existence identification we may also write $L_{\text{std}}(\phi) \equiv L(\phi)$ and

$$\frac{\mathcal{N}_{\text{act}}(\phi)}{\mathcal{N}_{\text{act}}(0)} = \frac{L_{\text{std}}(\phi)}{L_{\text{std}}(0)} = e^{\phi/2}. \quad (28)$$

This is an operational identification of $\mathcal{N}_{\text{act}}$ through the two-plate matter-standard ratio, not an independent number-density count of medium nodes, not a three-dimensional volume count, and not a new observable.

3. Resonance condition: the ball completes a full cycle (up + down) every n/ν_0 coordinate seconds, where $n \in \mathbb{Z}^+$ is a fixed harmonic number.

To derive: $m_{\text{eff}}(\phi) = m_0 e^{\phi/2}$ and $d\tau = e^{\phi/2} dt$.

Step 1. Coordinate velocity.

In one cycle the ball covers $2L(\phi)$ coordinate distance (up L + down L) in coordinate time $T = n/\nu_0$:

$$v(\phi) = \frac{2L(\phi)}{n/\nu_0} = \frac{2\ell_0 \nu_0}{n} e^{\phi/2} \equiv v_0 e^{\phi/2}, \quad (29)$$

where $v_0 \equiv 2\ell_0 \nu_0/n$.

Step 2. Impulse rate on the plates (effective mass derivation).

Each collision transfers impulse $\Delta p = 2\mu v(\phi)$ to a plate, where μ is the ball's internal inertia—an oscillon “core” parameter determined by *local* physics and therefore ϕ -independent (local undetectability).

One cycle involves exactly 2 collisions (upper + lower plate). Collision frequency: $f_{\text{coll}} = 2\nu_0/n$.

Average impulse transfer rate (force):

$$F(\phi) = \Delta p \times f_{\text{coll}} = 2\mu v(\phi) \times \frac{2\nu_0}{n} = \frac{4\mu \nu_0}{n} v(\phi). \quad (30)$$

Substituting $v(\phi)$ from (29):

$$F(\phi) = \frac{8\mu \ell_0 \nu_0^2}{n^2} e^{\phi/2} \equiv F_0 e^{\phi/2}. \quad (31)$$

The effective mass is what the external world “feels”—the impulse rate. Equivalently, it is the asymptotic mass/energy projection of the same operational active measurable content that controls the effective matter-standard extension in (28). By normalization:

$$\boxed{m_{\text{eff}}(\phi) = m_0 e^{\phi/2}} \quad (32)$$

where $m_0 \equiv F_0 \times (\text{normalization factor})$.

The clock postulate was not used anywhere in this derivation. Mass scaling follows solely from: (a) the coordinate plate separation (27) and (b) collision kinematics.

Step 3. Derivation of the clock postulate.

An oscillon-based clock counts oscillations at frequency ν_0/n . Each oscillation's “weight” in proper-time accumulation is proportional to the oscillation energy: $E(\phi) = m_{\text{eff}}(\phi) c^2$.

Proper-time accumulation rate:

$$\frac{d\tau}{dt} = \frac{E(\phi)}{E(0)} = \frac{m_{\text{eff}}(\phi)}{m_{\text{eff}}(0)} = e^{\phi/2}. \quad (33)$$

$$\boxed{d\tau = e^{\phi/2} dt} \quad (34)$$

Step 4. Local speed of light.

Locally measured speed:

$$c_{\text{local}} = \frac{d\ell}{d\tau} = \frac{e^{-\phi/2} d\sigma}{e^{\phi/2} dt} = e^{-\phi} \frac{d\sigma}{dt}. \quad (35)$$

Coordinate speed of light in the bi-conformal metric: $d\sigma/dt = c e^{\phi}$. Therefore:

$$\boxed{c_{\text{local}} = e^{-\phi} \cdot c e^{\phi} = c} \quad (36)$$

Step 5. Local undetectability check.

Local plate separation:

$$\ell_0 = e^{-\phi/2} \cdot L(\phi) = e^{-\phi/2} \cdot \ell_0 e^{\phi/2} = \ell_0. \quad (37)$$

(a) Mass-ratio invariance. Consider two oscillons—harmonics n_1 and n_2 . From Step 2:

$$\frac{m_1(\phi)}{m_2(\phi)} = \frac{F_1(\phi)}{F_2(\phi)} = \frac{v_1(\phi) f_1}{v_2(\phi) f_2} = \frac{(2L\nu_0/n_1) \cdot (2\nu_0/n_1)}{(2L\nu_0/n_2) \cdot (2\nu_0/n_2)} = \frac{n_2^2}{n_1^2}. \quad (38)$$

$L(\phi)$ and ν_0 cancel! The ratio is a function of integers only and is ϕ -invariant.

(b) Compton-ratio invariance. In the bi-conformal frame, dimensionful constants scale as:

$$m_{\text{eff}} = m_0 e^{\phi/2} \quad [\text{mass}], \quad (39)$$

$$\hbar_{\text{eff}} = \hbar_0 e^{-\phi} \quad [\text{action} = ML^2T^{-1}], \quad (40)$$

$$\nu_{\text{proper}} = (\nu_0/n) e^{-\phi/2} \quad [\text{proper frequency}]. \quad (41)$$

Checking the Compton ratio:

$$\hbar_{\text{eff}} (2\pi\nu_{\text{proper}}) = \hbar_0 e^{-\phi} \cdot \frac{2\pi\nu_0}{n} e^{-\phi/2} = \frac{2\pi\hbar_0\nu_0}{n} e^{-3\phi/2}. \quad (42)$$

On the other hand:

$$m_{\text{eff}} c^2 = m_0 c^2 e^{\phi/2}. \quad (43)$$

The local observer measures a *dimensionless* quantity:

$$\frac{\hbar_{\text{eff}} \omega_{\text{proper}}}{m_{\text{eff}} c^2} = \frac{2\pi\hbar_0\nu_0}{n m_0 c^2} e^{-2\phi}. \quad (44)$$

At first glance this depends on ϕ .

Resolution: ν_0 is a *coordinate* frequency. A local observer cannot measure this number—they measure only *ratios* among local quantities. All local frequencies (of all oscillons and photons) scale identically as $e^{-\phi/2}$ in proper time, so their *ratios* are ϕ -invariant.

(c) Summary. A local observer cannot distinguish the value of ϕ by any experiment, because:

- Plate separation: ℓ_0 (unchanged),
- Mass ratios: n_2^2/n_1^2 (integers),
- Speed of light: c (unchanged),
- All frequencies co-scale—mutual ratios are fixed.

This is the local undetectability theorem in the two-plate model. $\square$

Summary.

Quantity	Formula	Source
Plate separation	$L(\phi) = \ell_0 e^{\phi/2}$	Rod postulate
Ball velocity	$v(\phi) = v_0 e^{\phi/2}$	L + resonance
Impulse rate	$F(\phi) = F_0 e^{\phi/2}$	Collision physics
Effective mass	$m_{\text{eff}} = m_0 e^{\phi/2}$	$\equiv F/(\text{norm.})$
Active measurable existence	$\mathcal{N}_{\text{act}}(\phi)/\mathcal{N}_{\text{act}}(0) = L_{\text{std}}(\phi)/L_{\text{std}}(0) = e^{\phi/2}$	Operational reading of $L(\phi)$
Clock rate	$d\tau/dt = e^{\phi/2}$	$\nu_0 \times m_{\text{eff}}$
Speed of light	$c_{\text{local}} = c$	$d\ell/d\tau$

One assumption (rod) + **one principle** ($\nu_0 = \text{const}$) + **collision kinematics** $\Rightarrow$ the entire bi-conformal structure.

2.7 Why This Is Physically Non-Trivial

If particle masses are harmonics of a fundamental frequency, then the mass spectrum is determined not by accident but by the medium’s resonant structure. This potentially answers one of the deepest unanswered questions in physics: **why do particles have the masses they have?**

In the Standard Model, particle masses (electron, muon, quarks, W and Z bosons) are 20+ free parameters inserted “by hand” from experiment—the theory does not explain their origin.

If these masses are *different harmonics of a single fundamental frequency*, then 20+ parameters may follow from a single principle—just as all harmonics of a guitar string follow from one fundamental note.

2.8 Precedents in Physics

The harmonic resonance principle is not new in physics—it has been discovered repeatedly at different levels:

1. **Bohr atom (1913).** The electron in hydrogen occupies only “allowed” orbits—those fitting an integer number of wavelengths. Energy levels $E_n = E_0/n^2$ form a harmonic spectrum yielding hydrogen’s entire spectral series (Balmer, Lyman, Paschen).
2. **Nuclear shell model.** Nucleons organize into energetic “shells,” analogous to electronic orbitals. “Magic numbers” (2, 8, 20, 28, 50, 82, 126) mark especially stable configurations—resonant modes of the nuclear medium.
3. **Koide formula [85].** Among the electron (m_e), muon (m_μ), and tau lepton (m_τ) masses a remarkably precise empirical relationship exists:

$$\frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3}, \quad (45)$$

holding to 0.01% accuracy. The ratio 2/3 is the inverse of the musical fifth (3:2). This formula remains unexplained within the Standard Model but naturally fits the harmonic resonance picture.

4. **String theory.** Particles are interpreted as different vibrational modes of a single “string.” Particle mass = frequency ($E = mc^2 = h\nu$). The entire mass spectrum is a spectrum of harmonics—exactly the same philosophy, except in ISPG the “string” is a scalar-metric *medium*.

5. **De Broglie (1924).** Every massive particle has a wave of frequency $\nu = mc^2/h$. Matter vibrates. Matter = wave. This foundational quantum-mechanical discovery directly states: **mass is frequency**.
6. **Müller’s experiment (2013).** Holger Müller (UC Berkeley) experimentally demonstrated that a cesium atom’s matter wave can serve as a clock—a “Compton clock” counting oscillations at frequency mc^2/h . His famous phrase: “a rock is a clock.” This experimentally confirms that every massive particle “ticks” at a fundamental frequency.

2.9 ISPG Context: Topology and Harmonics

In the ISPG framework oscillons are already classified by topological winding number n : fermions (vortex winding $n = \pm 1$), opposite-winding antiparticle labels, and bosons ($n = 0$). The topological classification labels the particle’s *topological class*. The harmonic resonance principle complements this: within a single topological class, different particles (e.g. electron, muon, and tau—all fermions with $n = 1$) may be *different harmonics* of the same topology:

$$\nu_{\text{electron}} : \nu_{\text{muon}} : \nu_{\text{tau}} = n_e : n_\mu : n_\tau, \quad (46)$$

where n_e, n_μ, n_τ are integers or rationals determined by the medium’s resonant structure.

Topology + harmonic together give a particle its full identity: topology says *what* it is (fermion or boson); the harmonic says *which* it is (electron, muon, or tau).

2.10 Metaphor: The Universe as a Chord

The universe is not a single note—it is a **chord**.

All notes come from one “string” (the medium). Different particles are different notes in this chord. All notes are harmonically compatible because all are multiples or rational fractions of one fundamental rhythm.

Particle interactions are the chord’s *consonance*. Consonance (stable binding) occurs when frequency ratios form simple rational fractions. Dissonance (instability, decay) occurs when the ratio is non-integer, highly incommensurate, or very complex.

For a musician this is intuitive: a fifth (3:2) is harmonious; a tritone ($\sqrt{2}$:1) is dissonant. Perhaps nature also “chooses” which particle combinations are stable by the same principle.

2.11 Status

Program completed:

- ✓ The oscillon wave equation has been solved— particle masses: $m_N = m_1 b_N(q)$, where $b_N(q)$ is a Mathieu characteristic value (§3.10, §3.12).
- ✓ The mass spectrum has been compared with experiment as a nearest- N scan: in the current PDG-style table, 10/12 entries lie below 1% and 11/12 below 5%; the muon anchor is exact at the fitted q , while the physical N -selection rule remains an open derivation target.
- ✓ The Koide formula (45) **automatically** holds from the integer harmonics (5, 72, 295) (0.028% accuracy).

The ISPG framework not only describes particles’ gravitational behavior (which it already does) but **explains** their *existence*, their *mass spectrum*, and the *electron’s special status as the first fully stable charged Mathieu mode in the adopted gap-threshold scheme*— all from the resonant structure of a single medium.

3 Calibration and Historical Estimates of the Universal Frequency ν_0

Goal: separate the historical classical estimate of ν_0 from the calibrated MASS/Mathieu value used in the quantitative mass spectrum. The frequency-invariance theorem fixes the background independence of a given proper-time carrier frequency; it does not by itself compute this numerical value.

3.1 Bohr–Sommerfeld Condition in the Two-Plate Model

In Section 2.6 mass scaling was derived from classical collision kinematics. We now add a **quantum condition**: the ball’s action over one full cycle is quantized.

A ball (at $\phi = 0$ background) moves between the plates (local separation ℓ_0) at speed $v = 2\ell_0\nu_0/r$ (where r is the harmonic parameter). A full cycle (up + down) covers distance $2\ell_0$.

Bohr–Sommerfeld quantization:

$$\oint p dx = N h, \quad N = 1, 2, 3, \dots \quad (47)$$

The ball’s momentum is $p = \mu v$, where μ is its internal inertia (a local parameter). Over one cycle: $\oint p dx = \mu v \times 2\ell_0$. Hence:

$$\mu \cdot \frac{2\ell_0\nu_0}{r} \cdot 2\ell_0 = N h \quad \implies \quad \nu_0 = \frac{N r h}{4\mu \ell_0^2}. \quad (48)$$

3.2 Quantum Mass Formula

The classical mass formula (from Section 2.6): $m = 8\mu \ell_0^2 \nu_0^2 / (r^2 c^2)$. Substituting ν_0 from (48):

$$\begin{aligned} m &= \frac{8\mu \ell_0^2}{r^2 c^2} \cdot \left(\frac{N r h}{4\mu \ell_0^2} \right)^2 = \frac{8\mu \ell_0^2}{r^2 c^2} \cdot \frac{N^2 r^2 h^2}{16\mu^2 \ell_0^4} \\ &= \frac{N^2 h^2}{2\mu \ell_0^2 c^2}. \end{aligned} \quad (49)$$

This is the standard quantum “particle in a box” result. **Mass is determined by the Bohr–Sommerfeld quantum number N** , not the harmonic number r .

Important clarification: In the classical model, r determined mass ($m \propto 1/r^2$). In the quantum model, r and N are linked: for each r , only specific N values are stable (from the Mathieu equation’s stability zones). Therefore the *allowed* (r, N) pairs produce a **discrete mass spectrum**.

3.3 Relation Between Mass and Frequency

From the Compton relation every particle has a fundamental frequency:

$$\nu_C = \frac{mc^2}{h}. \quad (50)$$

In the two-plate model the ball’s bounce frequency is ν_0/r . Via the Zitterbewegung interpretation, this bouncing **is** the particle’s internal oscillation:

$$\frac{\nu_0}{r} = \frac{2mc^2}{h} = 2\nu_C. \quad (51)$$

(The factor of 2 comes from the Zitterbewegung $\omega_Z = 2mc^2/\hbar$: in one cycle the ball bounces *twice*—up and down.)

Therefore:

$$\nu_0 = 2r \frac{mc^2}{h} = 2r \nu_C \quad (52)$$

This general relation links ν_0 to any particle's mass m and its classical harmonic number r .

3.4 Historical Planck-Scale Estimate

The following calculation records the older classical estimate. In that classical picture, the fundamental mode ($r = 1$) corresponds to the **maximum mass**. Physically, this is the mass above which an oscillon collapses into a black hole—the **Planck mass**:

$$m_{r=1} = m_P = \sqrt{\frac{\hbar c}{G}} = 2.176 \times 10^{-8} \text{ kg} = 1.221 \times 10^{19} \text{ GeV}/c^2. \quad (53)$$

From (52) with $r = 1$:

$$\nu_0^{(\text{cl})} = \frac{2m_P c^2}{h} = \frac{\nu_P}{\pi} \quad (54)$$

where $\nu_P = 1/t_P = 1.855 \times 10^{43} \text{ Hz}$ is the Planck frequency.

Numerical value:

$$\nu_0^{(\text{cl})} = 5.904 \times 10^{42} \text{ Hz}. \quad (55)$$

Historical status.—This classical estimate $\nu_0^{(\text{cl})}$ is based on the linear dispersion $m \propto 1/r$. The quantum Mathieu model (§3.12) replaces this with $m \propto N^2$, yielding a corrected value $\nu_0 = 1.91 \times 10^{30} \text{ Hz}$ —thirteen orders of magnitude lower (Eq. 74). **Only the quantum value ν_0 is used in all subsequent calculations;** $\nu_0^{(\text{cl})}$ serves solely as a historical/pedagogical stepping stone.

Corresponding period of the classical estimate:

$$T_0^{(\text{cl})} = 1/\nu_0^{(\text{cl})} = \pi t_P = 1.694 \times 10^{-43} \text{ s}. \quad (56)$$

Corresponding wavelength:

$$\lambda_0^{(\text{cl})} = c/\nu_0^{(\text{cl})} = \pi \ell_P = 5.08 \times 10^{-35} \text{ m}. \quad (57)$$

3.5 Classical Particle Harmonic Map

In the historical classical map, $r = 1$ fundamental mode = Planck mass. Each particle's harmonic number:

$$r = \frac{m_P}{m} \quad (58)$$

(follows from (52) with $\nu_0^{(\text{cl})} = 2m_P c^2/h$.)

Particle	Mass (MeV/c ²)	ν_C (Hz)	$r = m_P/m$	$\nu_0^{(\text{cl})}/r$ (Hz)
Top quark	172 760	4.18×10^{25}	7.07×10^{16}	8.35×10^{25}
Higgs boson	125 100	3.02×10^{25}	9.76×10^{16}	6.05×10^{25}
Z boson	91 188	2.20×10^{25}	1.34×10^{17}	4.41×10^{25}
W boson	80 379	1.94×10^{25}	1.52×10^{17}	3.89×10^{25}
b quark	4 180	1.01×10^{24}	2.92×10^{18}	2.02×10^{24}
τ lepton	1 776.9	4.30×10^{23}	6.87×10^{18}	8.59×10^{23}
c quark	1 270	3.07×10^{23}	9.61×10^{18}	6.14×10^{23}
μ lepton	105.66	2.55×10^{22}	1.16×10^{20}	5.11×10^{22}
s quark	93.4	2.26×10^{22}	1.31×10^{20}	4.51×10^{22}
d quark	4.67	1.13×10^{21}	2.62×10^{21}	2.26×10^{21}
u quark	2.16	5.23×10^{20}	5.65×10^{21}	1.05×10^{21}
Electron	0.511	1.24×10^{20}	2.39×10^{22}	2.47×10^{20}
Planck mass	1.221×10^{19}	2.95×10^{42}	1	5.90×10^{42}

The hierarchy problem in ISPG language. The table shows that all known particles have very large r (10^{16} – 10^{22}). The fundamental mode ($r = 1$, Planck mass) versus the lightest known particle (electron)— $r \sim 10^{22}$.

This is particle physics' well-known **hierarchy problem**: why are elementary particles 10^{17} – 10^{22} times lighter than the fundamental scale?

In the historical ISPG map this means: particles are **extremely high classical harmonics** of the medium—very slow bouncing between very rapidly vibrating plates. Modes close to the fundamental ($r \sim 1$ – 10^{10}) are unstable—they cannot maintain resonance outside the Mathieu equation's stability zones.

Note: the harmonic numbers r in this table are computed from the classical estimate $\nu_0^{(\text{cl})}$. In the quantum Mathieu model (§3.12), the relevant quantum numbers are $N = 1, 2, 3, \dots$ with a corrected $\nu_0 = 1.91 \times 10^{30}$ Hz. The physical content (discrete spectrum, hierarchy) is retained as a qualitative motivation, but the final labeling and numerical carrier frequency come from the calibrated MASS/Mathieu model.

3.6 Harmonic Structure of Lepton Masses

Lepton mass ratios:

$$\frac{m_\mu}{m_e} = 206.768 \quad \sqrt{\frac{m_\mu}{m_e}} = 14.38, \quad (59)$$

$$\frac{m_\tau}{m_e} = 3\,477.5 \quad \sqrt{\frac{m_\tau}{m_e}} = 58.97, \quad (60)$$

$$\frac{m_\tau}{m_\mu} = 16.82 \quad \sqrt{\frac{m_\tau}{m_\mu}} = 4.101. \quad (61)$$

These numbers are close to integers but **not exactly integers**. We seek the best approximations:

Approximation 1: $\sqrt{m_\tau/m_e} \approx 59 \Rightarrow m_\tau/m_e \approx 59^2 = 3481$ vs. actual 3477.5 (0.1% error).

Approximation 2: $m_\mu/m_e \approx (72/5)^2 = 207.36$ vs. actual 206.77 (0.3% error).

Approximation 3: $m_\tau/m_\mu \approx (41/10)^2 = 16.81$ vs. actual 16.82 (0.06% error).

These approximations show that $\sqrt{m_1/m_2}$ tends toward *rational* numbers, consistent with the harmonic-rational resonance model. Exact values require solving the Mathieu equation.

3.7 Mathieu Equation: Stability Zones and Mass Selection

The two-plate model is analogous to the Fermi–Ulam problem (a ball between oscillating walls). In the linear approximation, the dynamics is described by the Mathieu equation:

$$\frac{d^2\psi}{d\tau^2} + [a - 2q \cos(2\tau)] \psi = 0, \quad (62)$$

where

$$a = \left(\frac{E}{h\nu_0} \right)^2 = \left(\frac{E}{\hbar\omega_0} \right)^2 = \left(\frac{mc^2}{h\nu_0} \right)^2, \quad \omega_0 \equiv 2\pi\nu_0, \quad q \approx 1 + \delta q, \quad (63)$$

where q arises from the nonlinear kinetic self-interaction $G_2(X) \supset \alpha X^2/M^2$ of the ISPG scalar field. q **does not depend on** N —this is a consequence of ISPG’s conformal coupling (detailed derivation: §3.13).

The Mathieu equation’s **stability map** in the (a, q) plane creates zones where solutions are bounded (stable oscillon) and zones where solutions grow (unstable—particle decays).

Allowed particle masses = stability zone a -values:

$$m_n = \frac{h\nu_0}{c^2} \sqrt{a_n(q)}. \quad (64)$$

The Planck-mass form belongs only to the historical Planck-anchored estimate of $\nu_0^{(\text{cl})}$ below. The calibrated MASS/Mathieu construction uses the cyclic frequency ν_0 with $h\nu_0 = \hbar\omega_0$ and does not assume the historical m_P/π normalization.

After fixing q by calibration, or eventually deriving it from the full oscillon dynamics, the Mathieu characteristic values $a_n(q)$ yield the **particle mass spectrum** in this one-dimensional EFT limit, analogous to Bohr’s formula $E_n = E_0/n^2$ for hydrogen.

Estimate of ε . If $\varepsilon \ll 1$ (weak fluctuation), the principal stability zones are centered at $a = n^2$ ($n = 0, 1, 2, \dots$) and mass ratios are approximately n_2/n_1 . If $\varepsilon \sim 1$ (strong fluctuation), stability zones shift dramatically and produce a *complex* mass spectrum—possibly explaining why elementary particle masses appear in “unexpected” ratios.

3.8 Historical Interpretation of $\nu_0^{(\text{cl})}$

Historical classical estimate

$$\nu_0^{(\text{cl})} = \frac{\nu_P}{\pi} \approx 5.9 \times 10^{42} \text{ Hz}$$

This is not the current calibrated carrier frequency. It is the older Planck-anchored estimate obtained from the classical $m \propto 1/r$ map. The present paper uses the calibrated MASS/Mathieu value $\nu_0 = 1.91 \times 10^{30}$ Hz for oscillonic physics.

- **Historical particle = sub-harmonic picture.** The electron “bounces” every $r_e \approx 2.4 \times 10^{22}$ medium vibrations. The top quark—every $r_t \approx 7 \times 10^{16}$ vibrations. This is the classical intuition, not the final Mathieu calibration.
- **Classical Planck limit.** $r = 1$ —bouncing on every vibration—yields $m_P = 1.22 \times 10^{19}$ GeV. A heavier classical oscillon would collapse in this picture.

- **Classical wavelength** $= \pi \ell_P \approx 5 \times 10^{-35} \text{ m}$. This belongs to the historical estimate $\lambda_0^{(\text{cl})} = c/\nu_0^{(\text{cl})}$. It should not be read as the wavelength of the calibrated ν_0 .
- **Historical mass-frequency map**. In the classical picture, $m = m_P/r$. The quantitative particle spectrum used in this paper is instead the calibrated MASS/Mathieu spectrum.

3.9 Status of the Program

The historical value $\nu_0^{(\text{cl})}$ (55) relies on the assumption that $r = 1$ corresponds to the Planck mass. The quantum MASS/Mathieu model performs the recalibration used below. The following status summary separates completed results from open closure tasks:

- ✓ **Derivation of q from ISPG $G_2(X)$ extension at leading order**— $q_0 = 1$ follows from flat-space self-consistency.
- **Nonlinear correction to the fitted Mathieu parameter**— the fitted reference value is $q_{\text{fit}} = 1.853$; the present semi-analytic correction bundle gives $q_{\text{semi}} \simeq 1.907$, within 2.9% of that value. Exact closure of δq requires the full lattice/PDE oscillon calculation (§3.13).
- ✓ **Solution of Mathieu stability zones**— $b_N(q)$ for all particles (§3.12).
- ✓ **Mass ratio comparison**— muon: $< 0.001\%$; leptons: $< 0.2\%$ (§3.12).
- ✓ **Reproduction of the Koide formula**— 0.028% accuracy (§3.12).

3.10 Numerical Results of the Quantum Model

By Bohr–Sommerfeld quantization (Eq. 49), mass scales as N^2 :

$$m = m_1 \times N^2, \quad m_1 \equiv \frac{h^2}{2\mu \ell_0^2 c^2}, \quad N \in \mathbb{Z}^+. \quad (65)$$

The electron is one of the harmonics: $m_e = m_1 N_e^2$. **Goal:** find N_e and then all other particles' N .

Determining N_e from lepton masses.

Mass ratio: $m_\mu/m_e = N_\mu^2/N_e^2$. Hence $N_\mu/N_e = \sqrt{m_\mu/m_e} = \sqrt{206.768} = 14.380$.

We seek a rational approximation:

$$14.380 \approx \frac{72}{5} = 14.400 \quad (\text{error } 0.14\%). \quad (66)$$

This gives: $N_e = 5$, $N_\mu = 72$.

For the tau lepton: $N_\tau/N_e = \sqrt{m_\tau/m_e} = \sqrt{3477.5} = 58.97 \approx 59$. Hence $N_\tau = 5 \times 59 = 295$.

$$\boxed{N_e = 5, \quad N_\mu = 72, \quad N_\tau = 295} \quad (67)$$

Prediction vs. experiment (leptons):

	N	$m_{\text{predicted}}$ (MeV)	m_{actual} (MeV)	Error
e	5	0.511 (reference)	0.5110	—
μ	72	105.96	105.658	0.29%
τ	295	1778.8	1776.86	0.11%

Koide consistency on the selected lepton ladder.

The Koide formula (45) with $m \propto N^2$ becomes:

$$\frac{N_e^2 + N_\mu^2 + N_\tau^2}{(N_e + N_\mu + N_\tau)^2} = \frac{25 + 5184 + 87025}{(5 + 72 + 295)^2} = \frac{92\,234}{138\,384} = 0.66654. \quad (68)$$

Exact value: $2/3 = 0.66667$. **Error:** 0.02%!

This means: once the lepton integers (5, 72, 295) are selected from the mass pattern, the Koide formula—an unexplained empirical relation since 1981—is reproduced by the N^2 ladder to 0.024%. A rank-1/E1D consistency argument supporting this assignment is given in ISPG_Quantum, Sec. 7.2; the derivation of the lepton integers (5, 72, 295) from oscillon dynamics, and the formal 3D-to-1D bridge to first-principles charged-lepton masses, remain open.

Harmonic map of all elementary particles.

Fundamental mass: $m_1 = m_e/N_e^2 = m_e/25 = 20.44$ keV. The harmonic quantum number of each particle is $N = \text{round}(5\sqrt{m/m_e})$.

Methodological note.—The integer N is extracted from measured masses; the table below is therefore a *one-parameter compression* of the mass spectrum, not yet a blind prediction. The non-trivial content is threefold: (i) twelve quark, lepton and boson masses, which in the Standard Model require twelve independent Yukawa couplings, here collapse onto a single fundamental scale m_1 and a single nonlinearity parameter q ; (ii) the integers N are not arbitrary—they coincide with the Mathieu stability chart (unstable/wide-gap for $N = 1\text{--}3$, barely stable for $N = 4$, fully stable for $N \geq 5$ under the adopted threshold convention), which explains why the electron is the first fully stable occupied charged Mathieu mode in this scheme; (iii) the residual errors are case-by-case after the nearest- N selection, with the current PDG-style scan giving 10/12 entries below 1% and 11/12 below 5%. Light-quark masses depend on the renormalization scheme (PDG running $\overline{\text{MS}}$ masses at 2 GeV versus the paper-target values), while the high- N heavy entries are empirical placements until a physical N -selection principle is derived. The programme becomes fully predictive once the fitted reference value of q is replaced by an independently derived value from the lattice simulation of $G_2(X)$ (future work, §??).

Particle	m (MeV)	N	m_{pred} (MeV)	Error
e	0.511	5	0.511	—
u quark	2.16	10	2.04	5.4%*
d quark	4.67	15	4.60	1.5%*
s quark	93.4	68	94.5	1.2%*
μ	105.66	72	105.96	0.29%
c quark	1 270	249	1 267	0.24%
τ	1 776.9	295	1 778.8	0.11%
b quark	4 180	452	4 176	0.10%
W	80 379	1 983	80 363	0.02%
Z	91 188	2 112	91 215	0.03%
H	125 100	2 474	125 132	0.03%
t quark	172 760	2 907	172 686	0.04%

*

Light quark masses are poorly known experimentally ($\sim 10\text{--}30\%$ uncertainty).

Assessment: The table is a compact empirical ladder placement: one scale (m_1), one fitted nonlinearity (q), and one integer per particle compress the mass data, but the integer choice is not yet predicted. The light-quark entries require an explicit mass convention, and the heavy high- N entries are close nearest- N scan hits rather than first-principles electroweak/Higgs mass derivations.

Physical interpretation of m_1 . The fundamental mass $m_1 = 20.44$ keV corresponds to the “slowest bouncing” ($N = 1$) energy in the box. This is ~ 40 times lighter than the electron. In the adopted gap-threshold scheme, $N = 1$ and $N = 2$ are unstable (gaps ~ 3.5 and ~ 1.3), $N = 3$ is marginal/short-lived (gap ~ 0.18), $N = 4$ is the first barely-stable candidate (gap $\sim 10^{-2}$), and $N = 5$ is the first fully stable mode (gap $< 10^{-3}$):

$$m_{N=2} = 4 m_1 = 81.8 \text{ keV} \quad (\text{unstable Mathieu mode}), \quad (69)$$

$$m_{N=3} = 9 m_1 = 184 \text{ keV} \quad (\text{marginal/short-lived resonance}). \quad (70)$$

3.11 Recalibration of ν_0 via the Quantum Model

From the Bohr–Sommerfeld condition (48): $\nu_0 = N r h / (4 \mu \ell_0^2)$, where $N r = K$ (constant for all particles).

The Planck limit ($m_P = \text{maximum mass}$) now corresponds to $N = N_{\text{max}}$:

$$N_{\text{max}} = \frac{N_e m_P^{1/2}}{m_e^{1/2}} = 5 \sqrt{\frac{m_P}{m_e}} = 5 \times 1.546 \times 10^{11} = 7.73 \times 10^{11}. \quad (71)$$

ν_0 in the quantum model:

$$\nu_0 = \frac{\sqrt{m_P m_e} c^2}{10 \hbar} = \frac{\sqrt{m_P m_e}}{10} \nu_C \left(\frac{1}{\sqrt{m_P m_e}} \right). \quad (72)$$

Numerically:

$$\begin{aligned} \sqrt{m_P m_e} &= \sqrt{2.176 \times 10^{-8} \times 9.109 \times 10^{-31}} = 1.408 \times 10^{-19} \text{ kg} \\ &= 7.90 \times 10^4 \text{ MeV}/c^2 \approx 79 \text{ GeV}/c^2. \end{aligned} \quad (73)$$

$$\boxed{\nu_0 \equiv \nu_0^{(\text{quantum})} = 1.91 \times 10^{30} \text{ Hz}} \quad (74)$$

This is the calibrated MASS/Mathieu value of ν_0 used throughout this paper. It supersedes the historical classical estimate $\nu_0^{(\text{cl})} = 5.9 \times 10^{42} \text{ Hz}$ (Eq. 55), which is 10^{13} times larger. The difference arises because the classical model has $m \propto 1/r$, while the quantum model has $m \propto N^2 \propto 1/r^2$. The quantum model yields **significantly better** mass ratios. The frequency-invariance theorem does not compute this number; it establishes the background independence of the proper-time carrier once the carrier is given. A first-principles derivation of the calibrated value remains open.

Interesting coincidence: The geometric mean $\sqrt{m_P m_e} \approx 79 \text{ GeV}$ is very close to the electroweak scale ($v_{\text{EW}} = 246 \text{ GeV}$, $m_W = 80 \text{ GeV}$). This is not accidental: the electroweak scale is expected, in the ISPG completion, to be fixed by the same oscillon-amplitude normalization that determines Φ_0 ; the explicit coarse-graining map from the dimensionless amplitude Φ_0 to the imported energy scale $v_{\text{EW}} = 246 \text{ GeV}$ remains open (§7.6).

3.12 Numerical Solution of the Mathieu Equation

The Mathieu equation’s (62) quantum characteristic values $b_N(q)$ are computed using the `scipy` library.

Fitting the reference value of q from the muon mass.

We define the fitted reference value q_{fit} by requiring $b_{72}(q)/b_5(q) = m_\mu/m_e = 206.768$:

$$\boxed{q_{\text{fit}} = 1.853} \quad (75)$$

At this fitted value the muon/electron ratio is reproduced by construction ($< 0.001\%$ numerical error). The independent theory task is then to derive a value of q from the oscillon dynamics that closes on this reference value.

Stability analysis: why $N_e = 5$?

Instability zone widths at $q_{\text{fit}} = 1.853$:

N	$b_N(q)$	$a_N(q)$	Gap width	Status	
1	−1.19	2.33	3.52	Unstable	$b_1 < 0$: does not exist
2	3.72	5.04	1.33	Unstable	
3	9.13	9.31	0.185	Marginal	
4	16.11	16.12	0.010	Stable	
5	25.07	25.07	0.0003	Stable	← Electron
6	36.05	36.05	5×10^{-6}	Stable	
7	49.04	49.04	$< 10^{-7}$	Stable	

Result:

- $N = 1$: **does not exist**— $b_1(q) < 0$; no bound state corresponds to this mode.
- $N = 2$: **unstable**— instability zone is wide (1.33); the particle decays instantly.
- $N = 3$: **marginal**— zone width 0.185; very short-lived resonance.
- $N = 4$: stable, but zone width 0.01—possibly a “barely stable” particle.
- $N \geq 5$: **fully stable**. Zone width $< 10^{-3}$.

Status of the gap-width criterion.—The designations “marginal”, “barely stable”, and “fully stable” above are based on the Mathieu gap width (a measure of instability-zone size in (a, q) space). The thresholds separating “marginal” ($\sim 10^{-1}$), “barely stable” ($\sim 10^{-2}$), and “fully stable” ($< 10^{-3}$) are adopted here for concreteness and are not derived from first principles. A rigorous mapping from the gap width to a physical decay rate requires a full Floquet analysis of the driven oscillon including its dissipation channels; that mapping is not derived in the present section, so the lifetime estimate $\tau \lesssim 10^{-20}$ s for the $N = 4$ resonance (below) should be read as an order-of-magnitude guideline rather than a derived prediction.

The electron ($N = 5$) is the first fully stable charged Mathieu mode under the adopted gap-threshold scheme; $N = 1$ –3 are unstable/marginal, and $N = 4$ is a barely-stable resonance candidate predicted at ~ 329 keV.

This explains one of nature’s fundamental questions: **why does a lightest charged particle exist, and why does it have this particular mass?**

The neutrino question. For neutrino masses ($m_\nu \sim 0.05$ eV): $N_\nu = 5\sqrt{m_\nu/m_e} \approx 0.002 \ll 1$. This means neutrinos **are not** ordinary harmonics of the two-plate model. In the ISPG framework, neutrinos are identified as **polarisation-rotation waves**—helical excitations whose oscillation plane progressively rotates—a fundamentally different vibration mode whose suppressed torsional rigidity supplies a partial mass suppression; reaching the sub-eV scale additionally requires the open trapping factor η_T derived in §5.

3.13 Derivation of q from the ISPG Action: Shift-Symmetric $G_2(X)$ Extension

In the previous section $q_{\text{fit}} = 1.853$ was fixed from the muon/electron ratio as a reference calibration. We now show that the **leading-order value** $q_0 = 1$ **follows from first principles**, and identify three calculable correction channels (metric backreaction, spatial gradients, higher harmonics) that drive the theory toward the fitted remainder $\delta q_{\text{fit}} = 0.853$ —all without introducing new fields or breaking the shift symmetry $\phi \rightarrow \phi + \text{const}$ that protects gravity’s long-range nature. The present semi-analytic evaluation gives $q_{\text{semi}} \simeq 1.907$, within 2.9% of q_{fit} ; exact closure of δq awaits lattice computation of the oscillon profile (§3.13, Step 6).

Relation to the Quantum paper’s mass program.—The Quantum paper (Sec. 7) derives the mass hierarchy from the eigenvalue spectrum of three-dimensional resonant cavities embedded in the scalar-metric medium. We regard the one-dimensional Mathieu reduction used here as a candidate EFT limit of that 3D problem, motivated by the following heuristic step: when the oscillon core is approximated as spatially homogeneous (Step 3 below), the radial eigenvalue equation has the same structural form as a one-dimensional parametric oscillator, i.e. a Mathieu equation. We note, however, that a *formal* reduction from the 3D cavity problem to the one-dimensional Mathieu equation is not established here: the QUANTUM (n, ℓ) lepton assignments $\tau = (0, 0)$, $\mu = (1, 0)$, $e = (0, 2)$ are not in one-to-one correspondence with the N assignments used below, and the quark/gauge-boson sectors addressed by the present one-dimensional model do not appear as cavity modes in QUANTUM Sec. 7. The two routes are therefore better described as *parallel frameworks* with overlapping but non-identical scope, whose eventual unification via a formal 3D-to-1D reduction (or via lattice simulation of the full 3D system) is left as an open calculation.

Step 1: The ISPG action and its natural extension.

The baseline ISPG action (Appendix 1 of the main theory) is

$$S = \frac{1}{16\pi G} \int d^4x \sqrt{-g} \left[R - \frac{M_{\text{Pl}}^2}{2} X \right] + S_m, \quad X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi, \quad (76)$$

where ϕ is the *single* scalar field whose bi-conformal coupling $ds^2 = -e^\phi c^2 dt^2 + e^{-\phi} d\sigma^2$ determines all of gravity. The vacuum field equation is $\square\phi = 0$.

The action (76) is the leading term of a derivative expansion. The next shift-symmetric operator compatible with the Horndeski structure (Appendix 11, Stability) is X^2 . This is the same $G_2(X)$ extension that was independently established for ghost-free UV completion of the phantom kinetic sector; here it acquires a second physical role—generating the oscillon nonlinearity that produces particle masses:

$$\boxed{G_2(X) = -\frac{M_{\text{Pl}}^2}{2} X + \frac{\alpha}{M^2} X^2, \quad \alpha > 0,} \quad (77)$$

where M is the UV scale at which the nonlinearity becomes important, and α is a dimensionless coupling constant.

Three properties are essential:

1. **Same field.** No new field is introduced. The action contains only ϕ , the same field that generates the bi-conformal metric, mediates gravity, and forms oscillons.
2. **Shift symmetry preserved.** $X = -\frac{1}{2}(\partial\phi)^2$ depends only on *derivatives* of ϕ , so $\phi \rightarrow \phi + c$ leaves the action invariant. No mass term $\mu^2\phi^2$ is generated, and gravity remains long-range.
3. **Weak-field transparency.** For $X \ll M^2$ (weak gravity, PPN regime), $X^2/M^2 \ll X$ and the action reduces to the baseline ISPG. All existing observational tests are unaffected.

Step 2: Field equation from $G_2(X)$.

The Euler–Lagrange equation for the scalar sector with $\mathcal{L}_\phi = G_2(X)$ is

$$\nabla_\mu \left(\frac{\partial G_2}{\partial X} \nabla^\mu \phi \right) = 0. \quad (78)$$

Computing the derivative:

$$\frac{\partial G_2}{\partial X} = -\frac{M_{\text{Pl}}^2}{2} + \frac{2\alpha}{M^2} X. \quad (79)$$

Define the *effective sound-speed function*

$$\mathcal{K}(\phi) \equiv \frac{\partial G_2}{\partial X} = -\frac{M_{\text{Pl}}^2}{2} + \frac{2\alpha}{M^2} X. \quad (80)$$

The field equation becomes

$$\mathcal{K} \square \phi + \frac{2\alpha}{M^2} (\nabla^\mu \phi) (\nabla^\nu \phi) \nabla_\mu \nabla_\nu \phi = 0. \quad (81)$$

In the weak-field limit ($X \rightarrow 0$): $\mathcal{K} \rightarrow -M_{\text{Pl}}^2/2$ and the second term vanishes, recovering $\square \phi = 0$.

Step 3: Oscillon ansatz.

We seek a localized oscillating solution of the same form used in the ISPG quantum paper:

$$\phi(t, \mathbf{x}) = \Phi_0(r) \cos(\omega_0 t), \quad (82)$$

where $\Phi_0(r)$ is a localized profile and $\omega_0 = 2\pi\nu_0$.

For this ansatz:

$$X = -\frac{1}{2}(\partial\phi)^2 = \frac{1}{2}[\dot{\phi}^2 - (\nabla\phi)^2] = \frac{1}{2}[\Phi_0^2\omega_0^2\sin^2(\omega_0 t) - (\nabla\Phi_0)^2\cos^2(\omega_0 t)]. \quad (83)$$

At the oscillon core ($r \approx 0$) where $|\nabla\Phi_0| \ll \Phi_0\omega_0$, the kinetic term dominates:

$$X \approx \frac{1}{2}\Phi_0^2\omega_0^2\sin^2(\omega_0 t) \equiv X_0\sin^2(\omega_0 t), \quad X_0 \equiv \frac{1}{2}\Phi_0^2\omega_0^2. \quad (84)$$

Step 4: Self-consistency condition.

Substituting into (81) and keeping the leading terms at the oscillon core, the self-consistency condition for the fundamental harmonic $\cos(\omega_0 t)$ requires balancing the linear and cubic terms. The X^2 contribution generates an effective cubic nonlinearity:

$$\dot{\phi}^2 \cdot \square \phi \supset \Phi_0^2\omega_0^2\sin^2(\omega_0 t) \cdot [-\omega_0^2\Phi_0\cos(\omega_0 t)] = -\Phi_0^3\omega_0^4\sin^2 \cdot \cos, \quad (85)$$

and after Fourier projection onto $\cos(\omega_0 t)$ (using $\langle \sin^2 \cdot \cos^2 \rangle = \frac{1}{8}$ for the relevant cross-term), the self-consistency condition becomes

$$-\omega_0^2 + \frac{3\alpha\Phi_0^2\omega_0^4}{4M^2 \cdot M_{\text{Pl}}^2/2} \cdot \omega_0^2 = 0. \quad (86)$$

This determines the oscillon amplitude:

$$\boxed{\Phi_0^2 = \frac{2 M^2 M_{\text{Pl}}^2}{3 \alpha \omega_0^2}} \quad (87)$$

Step 5: Perturbation equation $\rightarrow$ Mathieu.

Consider a small perturbation on the oscillon background: $\phi = \phi_{\text{bg}} + \delta\phi$. Linearizing (81), the perturbation sees a time-dependent effective “mass” from the oscillating $\mathcal{K}$:

$$\mathcal{K}(t) = -\frac{M_{\text{Pl}}^2}{2} + \frac{2\alpha}{M^2} \cdot X_0 \sin^2(\omega_0 t) = -\frac{M_{\text{Pl}}^2}{2} \left[1 - \frac{2\alpha \Phi_0^2 \omega_0^2}{M^2 M_{\text{Pl}}^2} \sin^2(\omega_0 t) \right]. \quad (88)$$

Define the *nonlinearity parameter*

$$\varepsilon \equiv \frac{2\alpha \Phi_0^2 \omega_0^2}{M^2 M_{\text{Pl}}^2} = \frac{2\alpha X_0}{M^2 \cdot M_{\text{Pl}}^2/2} = \frac{4}{3}, \quad (89)$$

where the last equality uses the self-consistency condition (87).

Using $\sin^2(\omega_0 t) = \frac{1}{2}[1 - \cos(2\omega_0 t)]$, the perturbation equation becomes, after rescaling $\tau = \omega_0 t$:

$$\frac{d^2 \delta\phi_k}{d\tau^2} + [a_k - 2q \cos(2\tau)] \delta\phi_k = 0, \quad (90)$$

which is the **Mathieu equation**, with the parameter

$$\boxed{q = \frac{\varepsilon}{2(2-\varepsilon)} = \frac{2/3}{2-4/3} = \frac{2/3}{2/3} = 1} \quad (91)$$

at leading order.

Step 6: Bi-conformal corrections to q .

The leading-order result $q = 1$ was derived in flat spacetime. On the ISPG bi-conformal background, $ds^2 = -e^\phi c^2 dt^2 + e^{-\phi} d\sigma^2$, three effects shift q upward. We now compute each explicitly.

(a) Metric backreaction.

The d’Alembertian on the bi-conformal background (established in the Quantum paper, Eq. (3.14)) is

$$\square\phi = e^\phi [-e^{-2\phi} \ddot{\phi} + \nabla^2 \phi], \quad (92)$$

where the spatial Laplacian carries *no* ϕ -dependent prefactor: the identity $\sqrt{-g} g^{ij} = \delta^{ij}$ is an exact property of the bi-conformal structure.

For the oscillon background $\phi_{\text{bg}} = \Phi_0 \cos(\omega_0 t)$, the temporal coefficient becomes $e^{-2\Phi_0 \cos(\omega_0 t)}$. This factor modulates the kinetic energy of perturbations at all harmonics of ω_0 . The Jacobi–Anger expansion gives:

$$e^{-2\Phi_0 \cos(\omega_0 t)} = I_0(2\Phi_0) + 2 \sum_{n=1}^{\infty} (-1)^n I_n(2\Phi_0) \cos(n\omega_0 t), \quad (93)$$

where I_n are modified Bessel functions of the first kind. The component that directly contributes to the Mathieu driving ($\cos 2\omega_0 t$) is

$$\delta q_{\text{metric}} \sim \frac{I_2(2\Phi_0)}{I_0(2\Phi_0)}. \quad (94)$$

For $\Phi_0 \ll 1$: $I_2(z) \approx z^2/8$, $I_0(z) \approx 1$, so $\delta q_{\text{metric}} \approx \Phi_0^2/2$. For $\Phi_0 \sim 1$ (the oscillon regime), the full Bessel ratio saturates toward unity:

Φ_0	$I_0(2\Phi_0)$	$I_2(2\Phi_0)$	I_2/I_0
0.5	1.266	0.136	0.107
1.0	2.280	0.689	0.302
1.5	4.880	2.244	0.460
2.0	11.30	6.422	0.568

The metric backreaction alone accounts for a fraction of the total correction; the remainder comes from gradients and higher harmonics.

(b) *Gradient corrections.*

The oscillon spatial profile established in the Quantum paper has the form $\Phi_0(r) = A \operatorname{sech}^2(r/R_c)$, where $R_c \sim \hbar/(mc)$ is the Compton-scale core radius and $A \sim \mathcal{O}(1)$ is the peak amplitude. The gradient terms $(\nabla\Phi_0)^2 \sim \Phi_0^2/R_c^2$, neglected in the homogeneous approximation (84), modify the kinetic variable:

$$X = \frac{1}{2}\Phi_0^2\omega_0^2 \sin^2(\omega_0 t) - \frac{1}{2}(\nabla\Phi_0)^2 \cos^2(\omega_0 t). \quad (95)$$

The $\cos^2(\omega_0 t) = \frac{1}{2}[1 + \cos(2\omega_0 t)]$ part of the gradient term directly shifts the Mathieu parameter:

$$\delta q_{\text{grad}} \sim \frac{(\nabla\Phi_0)^2}{\Phi_0^2\omega_0^2} \sim \frac{1}{(R_c\omega_0)^2}. \quad (96)$$

For a Compton-scale oscillon with $R_c\omega_0 \sim c/c = \mathcal{O}(1)$, this correction is also $\mathcal{O}(\Phi_0^2)$.

(c) *Higher harmonics.*

The cubic nonlinearity from X^2 generates $\cos(3\omega_0 t)$ components via the identity $\cos^3\theta = \frac{3}{4}\cos\theta + \frac{1}{4}\cos 3\theta$. These third-harmonic terms couple back to the fundamental through the parametric driving, shifting the effective frequency and contributing

$$\delta q_{\text{harm}} \sim \mathcal{O}(\Phi_0^2). \quad (97)$$

Combined result.

All three corrections are positive and of order $\Phi_0^2 \sim \mathcal{O}(1)$ in the strongly nonlinear oscillon regime ($\varepsilon = 4/3$). The total:

$$q = 1 + \delta q_{\text{metric}} + \delta q_{\text{grad}} + \delta q_{\text{harm}} = 1 + \beta \Phi_0^2, \quad (98)$$

where $\beta > 0$ is an $\mathcal{O}(1)$ coefficient that absorbs all three contributions. Using the self-consistency condition (87):

$$\Phi_0^2 = \frac{2 M^2 M_{\text{Pl}}^2}{3 \alpha \omega_0^2}, \quad \beta_{\text{fit}} \Phi_0^2 = q_{\text{fit}} - 1 = 0.853. \quad (99)$$

Fitted target: if the correction coefficient is read as an effective parameter, then the peak dimensionless oscillon amplitude must satisfy $\Phi_0 = \sqrt{0.853/\beta_{\text{fit}}}$ in order to close exactly on q_{fit} . For $\beta_{\text{fit}} = 1$ this gives $\Phi_0 \approx 0.924$, already in the nonlinear regime where both the $G_2(X)$ and metric effects are $\mathcal{O}(1)$. The actual Φ_0 and β must be fixed by the full oscillon calculation, not by this calibration identity alone.

Key structural result: The $G_2(X)$ extension of ISPG predicts $q = 1$ from *first principles* at leading order (no free parameters). The bi-conformal metric backreaction, spatial gradients, and harmonic coupling each contribute corrections of order Φ_0^2 , shifting q toward the fitted reference value $q_{\text{fit}} = 1.853$. The present semi-analytic bundle lands nearby, while the coefficient β and the oscillon amplitude Φ_0 are *independently determinable* by lattice simulation of the coupled $G_2(X) + \text{bi-conformal}$ system, providing a **falsifiable** test of the entire framework.

Numerical verification.—Using the 3D spherical oscillon profile at $\Phi_0 = 2.35$, $\alpha = 1/2$ (the background established in the Quantum paper, Sec. 7), a shooting computation yields $\Omega = 0.86594$ ($< 0.008\%$ from the reference value 0.866). The three correction channels evaluate to:

$$\delta q_{\text{metric}} = \frac{I_2(2\Phi_0)}{I_0(2\Phi_0)} = 0.623, \quad \delta q_{\text{grad}} = 0.177, \quad \delta q_{\text{harm}} = 0.107, \quad (100)$$

giving a total $q_{\text{semi}} = 1 + 0.907 = 1.907$, within 2.9% of the fitted reference value $q_{\text{fit}} = 1.853$. The dominant correction (metric backreaction, 73% of δq) is computed exactly; the remainder comes from volume-averaged gradient and order-of-magnitude harmonic estimates. This is therefore near-support for the mechanism, not an exact derivation of q_{fit} . At $q = 1.907$ the muon/electron mass ratio is $b_{72}(q)/b_5(q) = 206.73$, matching the observed 206.77 to 0.02%. The residual 2.9% gap in q maps to a $< 0.02\%$ shift in the mass ratio and is within the expected precision of the semi-analytical gradient and harmonic formulas; it remains an open target for full lattice PDE evolution of the oscillon profile (future work). The simulation code is provided as `MASS/python/ispg_oscillon_simulation.py`.

Step 7: Why a common q is plausible in the Mathieu ladder.

The parameter q depends on the background oscillon properties (ε), not on the perturbation mode number N . This mode-independence is suggested directly by the structure of $G_2(X)$: the X^2 term couples to *all* perturbation wavelengths equally because X is a scalar constructed from ϕ alone. In the bi-conformal metric, the conformal factor $e^{\phi/2}$ acts identically on all length scales, so the same oscillon background naturally leads to a mode-independent q within this one-dimensional Mathieu EFT limit.

This is numerically consistent with the fitted ladder (§3.12): a single constant $q_{\text{fit}} = 1.853$ organizes the charged lepton masses and, with re-optimized N assignments, the heavy particle table. This should not yet be read as a proof that sector-specific effective q values are excluded in the still-unified 3D theory; that stronger universality claim belongs to the later unification work.

Step 8: Physical interpretation.

The $G_2(X)$ extension says: the ISPG medium has **nonlinear elasticity**. When deformations are small (weak gravitational fields, PPN regime), the medium responds linearly: $\square\phi = 0$. When deformations are large (inside an oscillon), the nonlinear X^2 term creates a self-confining potential that traps resonant modes—these are particles.

The two EFT parameters α and M characterize the medium’s nonlinear response:

- α : the dimensionless strength of the nonlinearity.
- M : the energy scale at which the nonlinear regime activates.

From the oscillon amplitude (87) and the quantum model frequency $\nu_0 = 1.91 \times 10^{30}$ Hz:

$$\frac{\alpha}{M^2} = \frac{2M_{\text{Pl}}^2}{3\Phi_0^2\omega_0^2}. \quad (101)$$

With $\alpha \sim \mathcal{O}(1)$, the nonlinearity scale is

$$M \sim \Phi_0\omega_0/M_{\text{Pl}} \sim \sqrt{X_0}/M_{\text{Pl}}. \quad (102)$$

This is far below the Planck scale, consistent with the X^2 term being an EFT correction that is irrelevant in the gravitational sector but dominant inside oscillons.

Summary: the main chain.

$$G_2(X) \xrightarrow[\text{self-consistency}]{\text{oscillon}} \varepsilon = \frac{4}{3} \xrightarrow[\text{flat space}]{\text{Mathieu}} q = 1 \xrightarrow{+\delta q_{\text{bi-conf}}} q_{\text{semi}} \simeq 1.907 \approx q_{\text{fit}} = 1.853 \xrightarrow{b_N(q)} m_N = m_1 b$$

One field ϕ , one EFT extension $G_2(X)$ (dual role: ghost-free UV completion + oscillon nonlinearity) $\rightarrow$ all particle masses.

Shift symmetry: **preserved**. New fields: **none**. Weak-field gravity: **unchanged**.
 $\delta q_{\text{bi-conf}} = 0.853$: fitted target for metric backreaction + gradients + harmonics. The present semi-analytic bundle gives $\delta q_{\text{semi}} \simeq 0.907$; exact closure is lattice-determinable future work.

3.14 Summary of Results

Result	Accuracy	Ref.
Fitted Mathieu ladder scan: $m_N = m_1 b_N(q)$	10/12 < 1%, 11/12 < 5%; light-quark scheme caveat	§3.10
Muon mass ($q_{\text{fit}} = 1.853$)	< 0.001%	§3.12
Selected heavy entries (b, t, W, Z, H)	nearest- N scan; case-by-case	§3.12
Koide formula (inherited from individually-fit lepton masses; <i>not</i> an independent derivation — see QUANTUM Sec. 7 for the rank-1/E1D consistency argument on the MASS/Mathieu ladder; the formal 3D-to-1D bridge to first-principles charged-lepton masses remains open)	0.028%	§3.12
Electron—first fully stable charged Mathieu mode	adopted gap threshold	§3.12
$q = 1$ from $G_2(X)$ self-consistency (leading order)	analytic	§3.13
$q_{\text{semi}} \simeq 1.907$ from bi-conformal corrections	near-support; open closure	§3.13
Common q in the fitted Mathieu ladder	numeric; sector universality open	§3.12
$G_2(X)$ dual role: UV completion + mass generation	structural	§3.13

Epistemic status of the table below.—At fixed $q_{\text{fit}} = 1.853$, the integer N associated with each particle is determined by selecting the nearest integer whose Mathieu characteristic value reproduces the observed mass ($\Delta N \approx 1.5N/1000$). The < 0.3% per-particle residuals quoted below therefore represent a fit in which one integer degree of freedom per particle is consumed, on top of the single global parameter q : given the observed masses, the nearest- N procedure yields these values by construction. The framework does not predict which N each species must take; it predicts that charged-sector masses fall on the $b_N(q)$ ladder for some q close to 1.853, and the empirical monotonicity $N(\text{species}) \uparrow$ with mass is the non-trivial content.

Reoptimized N assignments (at $q_{\text{fit}} = 1.853$, $\Delta N \approx 1.5N/1000$):

Particle	N	m_{pred} (MeV)	m_{exp} (MeV)	Error
e	5	0.511	0.511	exact
μ	72	105.659	105.658	0.001%
τ	295	1773.7	1776.9	0.177%
c	250	1273.9	1273.0	0.067%
b	453	4182.5	4183.0	0.012%
W	1986	80389.1	80369.2	0.025%
Z	2115	91171.6	91188.0	0.018%
H	2479	125254.0	125250.0	0.003%
t	2910	172593.6	172570.0	0.014%

3.15 Testable Predictions

The model makes the following **experimentally verifiable** predictions:

1. $N = 4$ **resonance:** $m_4 = m_1 \times b_4(1.853) \approx 329$ **keV**. The Mathieu instability zone width at this index is $\sim 10^{-2}$, giving a gap-threshold lifetime estimate $\tau \lesssim 10^{-20}$ s (order-of-magnitude only; the gap-to-decay-rate map is not first-principles). Possible search channels:

- Resonance peak in e^+e^- scattering at $\sqrt{s} \approx 329$ keV (low-energy collider regime).
- Hard-X / soft-gamma photon line at $m_4/2 \approx 165$ keV from a radiative decay channel, if such a channel exists; instrument matching requires the ~ 100 – 300 keV band (e.g. INTEGRAL/SPI), *not* the 3–20 keV NuSTAR sterile-neutrino-DM band.

A KATRIN-type beta-spectrum search is *not* accessible here: the tritium beta endpoint is $E_0 \approx 18.57$ keV, well below 329 keV, so a 329 keV kink is kinematically impossible in tritium decay. A beta source with endpoint $\gg 329$ keV would be required. A $\tau \lesssim 10^{-20}$ s state cannot simultaneously serve as a cosmological dark-matter candidate; the DM-candidate language is dropped here.

2. $N = 3$ **ultra-short resonance:** $m_3 \approx 186$ **keV**. Instability zone 0.185—practically instantaneous decay. May appear as a virtual state in electron–positron annihilation channels.
3. **Nonlinearity scale M of $G_2(X)$.** The EFT scale M at which the X^2 term activates determines the oscillon core size. If M can be constrained independently (e.g. from primordial gravitational wave spectrum or scalar radiation in neutron star mergers), this provides a direct test of the $G_2(X)$ extension.
4. **Light quark mass verification.** The assignments $N = 10$ and $N = 15$ give the paper-target values $m_u \approx 2.04$ MeV and $m_d \approx 4.59$ MeV. The live PDG-style central-mass scan with $\overline{\text{MS}}$ running masses at 2 GeV gives a u residual 5.62% and a d residual 1.80%. A consistent comparison requires an explicit renormalization scheme and scale, since light-quark masses are running parameters rather than pole masses; this remains a concrete lattice/PDG-scheme test, not a closed percent-level prediction.
5. **Cosmological frequency $\nu_0 = 1.91 \times 10^{30}$ Hz.** This is the calibrated value of the medium’s fundamental proper-time carrier frequency. Indirect test:
 - Primordial gravitational wave spectrum— a cutoff or characteristic structure is expected at wavelengths associated with ν_0 .

- Cosmological redshift anisotropy—the invariant frequency model predicts specific correlations in the CMB.
6. **k-Oscillon in $G_2(X)$: closing q_{semi} onto $q_{\text{fit}} = 1.853$.** Lattice simulation of k-oscillons with $G_2(X) = -M_{\text{Pl}}^2 X/2 + \alpha X^2/M^2$ on the bi-conformal background determines β and Φ_0 independently (Eq. 99). Three independent checks:
- Leading order: $q = 1$ from self-consistency alone.
 - Corrections: δq_{metric} from $I_2(2\Phi_0)/I_0(2\Phi_0)$, δq_{grad} from the sech^2 profile, δq_{harm} from cubic coupling—the present semi-analytic bundle gives $\delta q \simeq 0.907$ and must be tested against the fitted target $\delta q_{\text{fit}} = 0.853$.
 - Dual role of $G_2(X)$: the same extension that provides ghost-free UV completion (Appendix 11) must also produce the correct q .

4 Ontological Synthesis: Vibration, Pressure, and Gravity

The results of this paper, combined with the Bernoulli pressure-deficit mechanism (Quantum paper, §3) and the $G_2(X)$ extension (§3.13), close a conceptual loop that identifies the fundamental ontology of the ISPG framework. The entire construction rests on a single empirical input: the fundamental frequency $\nu_0 \approx 1.91 \times 10^{30}$ Hz, whose value is calibrated in the MASS/Mathieu spectrum (§3.10). *What determines ν_0 from first principles* remains an open question; it may ultimately follow from the discrete micro-structure of the medium (see Future Directions).

4.1 Pressure Is Vibrational Energy

In kinetic theory, gas pressure is the macroscopic manifestation of molecular vibration: $P = \frac{1}{3}\rho\langle v^2 \rangle$. Temperature measures the vibration intensity.

In ISPG the identification is exact. The kinetic variable

$$X \equiv -\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \quad (103)$$

is the **vibrational energy density** of the scalar field ϕ oscillating at the fundamental frequency ν_0 . The scalar-sector Lagrangian

$$\mathcal{L}_\phi = G_2(X) = -\frac{M_{\text{Pl}}^2}{2} X + \frac{\alpha}{M^2} X^2 \quad (104)$$

is simultaneously the **pressure equation of state** of the medium, expressed as a function of its vibrational energy:

- The linear term $-\frac{M_{\text{Pl}}^2}{2} X$: pressure from weak vibrations (linear elasticity $\rightarrow$ ordinary gravity).
- The quadratic term $\frac{\alpha}{M^2} X^2$: pressure from strong vibrations (nonlinear elasticity $\rightarrow$ oscillon self-trapping $\rightarrow$ particle masses).

Ontological identification:

Pressure $\equiv$ vibrational energy density $\equiv G_2(X)$.

The medium’s “pressure” is not an abstract thermodynamic quantity—it *is* the energy of the ϕ -field vibrating at ν_0 .

4.2 The Frequency Cascade

All gravitational phenomena in ISPG arise through a **nonlinear frequency cascade**—a downward transfer of vibrational energy from the fundamental frequency to zero frequency:

$$\boxed{
 \underbrace{\nu_0}_{\substack{\text{medium} \\ \text{fundamental} \\ \sim 10^{30} \text{ Hz}}} \xrightarrow[\substack{\text{oscillon formation} \\ G_2(X) \text{ nonlinearity}}]{\substack{\nu_N = \frac{N}{m} \nu_0 \\ \text{particle resonance} \\ \sim 10^{20} \text{ Hz}}} \xrightarrow[\text{time-averaging}]{\langle \phi^2 \rangle \neq 0} \underbrace{\omega = 0}_{\substack{\text{static } \phi(r) \\ = \text{gravity}}}
 } \quad (105)$$

1. **Stage 1:** $\nu_0 \rightarrow \nu_N$. The medium vibrates at ν_0 . The X^2 nonlinearity in $G_2(X)$ amplifies small inhomogeneities, causing the uniform vibration to condense into localized resonances—oscillons. Each oscillon vibrates at a harmonic ν_N determined by the Mathieu stability zones (§3.12). The oscillon’s mass is $m_N c^2 = \hbar \nu_N \times b_N(q)$.
2. **Stage 2:** $\nu_N \rightarrow \omega = 0$. The oscillon is a nonlinear object: its $\cos(\omega_N t)$ oscillation, passed through the X^2 and $e^{-2\phi}$ nonlinearities, generates a *time-averaged* (DC) component $\langle \phi \rangle \neq 0$ that extends as $\phi(r) \sim -GM/(rc^2)$ far from the core. This static profile *is* the Newtonian gravitational potential.
3. **Stage 3:** $\nabla \phi \neq 0 \rightarrow \text{force}$. The gradient of the static profile creates a spatial variation in the medium’s pressure: $\nabla P = (dG_2/dX) \nabla X \neq 0$. This pressure gradient is the gravitational force—the Bernoulli mechanism of the Quantum paper.

The cascade explains **why uniform vibration does not gravitate**: a spatially constant X has $\nabla X = 0$, hence $\nabla P = 0$, hence no force—regardless of how large the vibrational amplitude is. Gravity requires **inhomogeneity**. Two sources of inhomogeneity operate in ISPG:

- **Microscopic (local Bernoulli channel):** an oscillon, once formed, is a spatially localized object whose field amplitude falls off as a standing-wave tail. In the tail region the gradient kinetic energy density $|\nabla \varphi|^2 \neq 0$; by the Bernoulli identity (Quantum paper, §3) this kinetic energy comes at the expense of the medium’s static pressure. The result is a pressure deficit that extends outward as $\sim 1/r$ and whose gradient is the Newtonian gravitational field g_N . (The *formation* of the oscillon—the cascade $\nu_0 \rightarrow \nu_N$ —is a separate, prior step; what produces gravity is the *spatial structure* of the already-formed resonance.)
- **Macroscopic (transported channel):** cosmological flows of the ϕ -medium form vortices (via Kelvin–Helmholtz and related instabilities; see Appendix 9, MOND). Centrifugal acceleration physically transports the vibrating substance outward, depleting the centre—a rarefaction by mass transport rather than energy conversion. This produces the additional gravitational gradient g_h responsible for MOND phenomenology.

Both mechanisms reduce the medium’s background vibrational energy at a given location—one by *mode conversion*, the other by *physical transport*—but the outcome is the same: a pressure deficit that acts as gravity.

4.3 Spontaneous Symmetry Breaking: From Uniform Vibration to Matter

The uniform ν_0 -vibration is a **metastable** state. The X^2 term in $G_2(X)$ introduces a nonlinear instability analogous to the Jeans instability in self-gravitating fluids:

1. **Linear regime** ($X^2/M^2 \ll X$): small perturbations $\delta\phi$ propagate as waves satisfying $\square\phi = 0$. The background is stable.

2. **Nonlinear regime** ($X^2/M^2 \sim X$): perturbations with amplitude exceeding a critical threshold Φ_c experience self-focusing—the X^2 term increases the local energy density, which further increases X , which further amplifies the perturbation. This positive feedback concentrates energy into a localized region.
3. **Saturation:** the self-consistency condition (87) halts the collapse at the oscillon amplitude $\Phi_0 = \sqrt{2M^2 M_{\text{Pl}}^2 / (3\alpha\omega_0^2)}$. The result is a stable, localized, oscillating bound state—a **particle**.

This is the ISPG mechanism for **matter creation**: the uniform vibration of the medium spontaneously breaks translational symmetry by condensing into localized oscillons. No external “seed” is required—quantum fluctuations in the ϕ -field provide the initial perturbation (as detailed in the Quantum paper’s nucleation mechanism).

4.4 Dark Energy as Accumulated Echo Background

The Quantum paper (Sec. 8) identifies the stochastic background vibration $\langle\delta\varphi^2\rangle$ as the source of dark energy. The frequency cascade provides a complementary perspective:

- **Before condensation:** the medium’s full vibrational energy E_{ν_0} is distributed nearly uniformly at frequency ν_0 . No particles exist. No *localized* (Newtonian) gravity exists—only large-scale pressure gradients from primordial inhomogeneities.
- **During condensation:** the $G_2(X)$ instability converts a fraction of E_{ν_0} into oscillons (matter) and their gravitational tails ($\phi(r) \sim 1/r$). Two competing energy flows coexist:
 1. **Depletion:** the ν_0 -mode background loses energy to oscillon formation (the guitar string quietens).
 2. **Feedback:** every oscillon continuously broadcasts resonant-tail echoes back into the medium, feeding a stochastic background vibration $\langle\delta\varphi^2\rangle$ that accumulates globally (Quantum paper, §8).
- **Late universe:** the Quantum paper gives a conditional perturbative-branch estimate in which the net balance is dominated by the feedback channel: the echo-background energy density $\varepsilon(t)$ (= the modeled dark-energy component) grows *logarithmically* ($\varepsilon \sim \varepsilon_0 + A \ln(t/t_0)$), making the effective equation of state **slightly phantom**:

$$w(z) = -1 - \delta w_0 \frac{(1+z)^3}{E(z)}, \quad \delta w_0 \sim 10^{-2} - 10^{-1}. \quad (106)$$

The deviation $|\delta w|$ was larger at early times (more particles, stronger emission) and shrinks toward zero at late times: the dark energy asymptotically **approaches Λ from the phantom side** ($w < -1$).

Guitar-string analogy, refined. The guitar string loses energy (amplitude decreases) and the note fades. But in the ISPG universe the “sound” radiated by matter (resonant-tail echoes) is itself a form of background vibration that accumulates in the medium. The string quietens in one mode (ν_0), while the acoustic energy in the room (stochastic $\langle\delta\varphi^2\rangle$) slowly builds. The net dark-energy density (echo background) grows, but the growth decelerates via negative feedback (Quantum paper, §8.2): lighter particles emit less, so the process is self-regulating.

Connection to DESI (2024–2025): The DESI baryon acoustic oscillation data (DR1 at 2.6σ ; DR2 at 3.1σ , rising to 4.2σ with DES-SN5YR) favor time-varying dark energy over Λ CDM. The source-driven ISPG branch gives $w_0 < -1$ and a small negative CPL slope $w_a < 0$.

DESI DR2 therefore provides a mixed comparison rather than a confirmation: the sign of w_a is aligned, while the central CPL sign of $w_0 + 1$ is opposite. The illustrative values $w_0 \approx -1.06$ and $w_a \approx -0.15$ should be read as a conditional branch estimate whose magnitude inherits the tail-power normalization discussed in the companion quantum analysis.

4.5 The Closed Ontological Loop

The full ISPG ontology forms a self-consistent loop—not a one-way chain but a true cycle, because gravity feeds back into the vibration:

$$\nu_0 \xrightarrow{P=G_2(X)} \underset{\text{pressure}}{\mathcal{L}_\phi} \xrightarrow{e^{\pm\phi}} \underset{\text{geometry}}{ds^2} \xrightarrow{\nabla\phi} \underset{\text{force}}{F}$$

Feedback: the metric $e^{\pm\phi}$ modifies the wave equation $\square\phi$ (bi-conformal backreaction), which shifts the oscillon spectrum and the tail-emission power—closing the cycle (Quantum paper, §8.2).

Three geometries of rarefaction, one principle ($\nabla P \neq 0 \Rightarrow$ force):

1. **Gravity** : oscillon tail $|\nabla\varphi|^2 \neq 0 \rightarrow$ static $\rightarrow$ Bernoulli pressure
pressure drops deficit = g_N
: *(kinetic energy of the standing-wave tail reduces the medium's static pressure)*
2. **MOND** : vortex rotation pushes sub- $\rightarrow$ central rarefaction = g_h
stance outward centrifugally
: *(physical transport—vibrating medium expelled from centre)*
3. **Inertia** : acceleration $\rightarrow$ retarded self- $\rightarrow$ pressure asymmetry =
field $\rightarrow$ fore-aft $\delta\phi$ $-ma$
: *(temporal retardation—Bernoulli deficit not yet formed ahead, still present behind)*

Everything is one field, one vibration, one equation of state.

Vibration = pressure = geometry = gravity.

Matter = nonlinear self-trapping of vibration.

Inertia = retardation asymmetry of the same pressure deficit (Appendix 17; derived as $F = -m_{\text{eff}} a$ in ISPG_Quantum, Sec. 5).

MOND = macroscopic rarefaction of the same medium by rotation or convection (Appendix 9).

Dark energy = accumulated resonant-tail echoes (growing logarithmically, $w < -1$ in the conditional quantum-sector estimate; ISPG_Quantum, Sec. 6).

Dark-energy scale = perturbative-branch negative-feedback estimate, conditionally yielding $\rho_{\text{vac}} \sim 10^{-121} M_{\text{Pl}}^4$ while zero-point decoupling and non-perturbative normalization remain open (ISPG_Quantum, Sec. 6.3).

Spin-1/2 structural route = vortex winding number $n = \pm 1$, $J = n\hbar/2$ under the conditional spin bridge; Fermi–Dirac statistics and Pauli exclusion as a candidate Berry-phase/vortex-topology route (ISPG_Quantum, Sec. 10).

Gravitational phase/decoherence channel = leading coherent rate $\Gamma_\varphi = 2m_0|\Delta\Phi_N|/\hbar$; the ~ 0.5 Hz eV-photon coefficient is formal and short-flight limited, while irreversible decoherence is conditional on the tail/noise correlator (ISPG_Quantum, Sec. 9).

Neutrinos = polarisation-rotation waves (helical vortices, $n = \pm 1$) of the ν_0 -medium (Sec. 5).

Weak force = nonlinear torsion–transverse coupling (§5).

Gravitational waves = free transverse-traceless modes of the same medium; photon identification is deferred to a companion paper (Sec. 6.2).

Gluons = transverse waves confined by ν_N/ν_0 frequency mismatch; flux tube = forced-frequency bridge, $V \propto L$ (Sec. 6.3).

Three dimensions = minimum for $\ell = 0$ (scalar) + $\ell = 2$ (tensor) coexistence; 1D substrate $\rightarrow$ 3D by vibrational necessity (Sec. 7).

Conditional Higgs scaling: $v(\varphi) = v_0 e^{\varphi/2}$; the oscillon amplitude Φ_0 is the candidate ISPG order parameter, with the $\Phi_0 \rightarrow v_{\text{EW}}$ normalization left open (Sec. 7.6; ISPG_Quantum, Sec. 8).

5 Neutrinos as Polarisation-Rotation Waves

The Mathieu mass formula $m_N = m_1 b_N(q)$ organizes the massive charged-sector spectrum as a nearest- N empirical ladder (§3.12), but it assigns integer harmonic numbers $N = 1, 2, 3, \dots$ to its solutions. Neutrino masses ($m_\nu \lesssim 0.1$ eV) are a factor $\sim 10^6$ below the electron mass, corresponding to $N \ll 1$ —outside the integer-harmonic tower. This section proposes a resolution that preserves the single-field ontology and identifies neutrinos as the **fourth and final vibration mode** of the ν_0 -medium: *polarisation rotation*—the progressive turning of a transverse wave’s oscillation plane as it propagates.

The key insight is that “torsion” in the ISPG context does not require the substrate to possess physical thickness, nor does it require the scalar field to carry an intrinsic polarisation vector (which it does not: a real scalar $\phi(x)$ has no internal polarisation degrees of freedom). What is being rotated is not a polarisation plane in the vector-field sense but the *spatial argument* of a cylindrically-symmetric scalar mode about the propagation axis. In an expansion of ϕ in cylindrical coordinates $(r_\perp, \vartheta, z)$, the nonlinearity $G_2(X)$ admits a helical standing-wave solution

$$\delta\phi_\pm(r_\perp, \vartheta, z, t) = A(r_\perp) \cos[kz - \omega t \pm \vartheta], \quad (107)$$

i.e. the $\ell = \pm 1$ *azimuthal-winding* mode of the scalar substrate, whose constant-phase surfaces form a helix. Viewed in the field-configuration picture, this is precisely a vortex line with winding number $n = \pm 1$ (§5.4; cf. the complementary treatment of scalar vortices in the companion quantum paper [?]). The sense of the winding (clockwise vs. counterclockwise around the axis) furnishes a geometric chirality, and the vortex topology furnishes a half-integer phase under 2π rotation of the surrounding field configuration.

Status of the “polarisation-rotation” label. In the remainder of this section, “polarisation rotation” is retained only as a *visual metaphor* for the helical-argument mode of Eq. (107); it does not denote the rotation of an intrinsic polarisation vector in the sense of optics. The optical analogues of Faraday rotation and optical activity—where a genuine polarisation vector of A_μ rotates—apply to vector-field media; their role here is heuristic, not identificational. The formal object is the cylindrical $\ell = \pm 1$ mode, and all quantitative statements below are anchored to that structure.

5.1 Four modes of a wave-bearing medium

The scalar field ϕ in three emergent spatial dimensions (§7) supports perturbations characterised by their transformation under rotations about the propagation axis. In the ν_0 -medium there are exactly **four** independent perturbation classes:

1. **Longitudinal (compression):** the medium compresses and rarefies along the propagation direction. This is the $\ell = 0$ scalar mode $\rightarrow$ **gravity**.
2. **Transverse (displacement):** the field oscillates perpendicular to the propagation axis. This is the $\ell = 1$ dipole mode $\rightarrow$ **electromagnetism** (§7.5).
3. **Membrane (surface deformation):** higher multipole deformations of the field profile. These are the $\ell \geq 2$ tensor modes $\rightarrow$ **gluons** (confined; §6.3).
4. **Helical azimuthal winding ($\ell = \pm 1$ cylindrical mode):** a cylindrically-symmetric scalar mode whose phase winds by $\pm 2\pi$ around the propagation axis per wavelength, producing a helical constant-phase surface. This is not a polarisation vector rotating (the scalar has no such vector) but an azimuthal winding of the scalar argument itself, with topological charge $n = \pm 1$. We identify this mode with the **neutrino** and the **weak interaction**.

No fifth independent perturbation class exists in a scalar-field medium with two transverse dimensions. This is the ISPG analogue of the Standard Model’s four fundamental interactions: **each force is a perturbation mode of the same field.**

5.2 Why polarisation rotation is the neutrino

The polarisation-rotation mode supplies a structural candidate for several distinctive neutrino properties that the Standard Model treats as independent inputs:

1. **Intrinsic chirality.** Polarisation rotation has a definite handedness—the plane turns clockwise or counterclockwise as the wave advances. This is a *geometric* property of the helical wave itself, not an imposed quantum number. This gives a candidate geometric route to the observed chiral bias of weak neutrino interactions. For massive neutrinos the chirality/helicity distinction remains essential, and the precise map to the SM left-handed weak current is part of the open weak-sector derivation.
2. **Electrical neutrality.** Polarisation rotation is symmetric in the transverse plane: the oscillation visits both orthogonal directions equally. The time-averaged $\ell = 1$ dipole component is zero. The neutrino carries no electric charge.
3. **Extreme lightness.** Polarisation rotation requires coupling *between* the two transverse polarisations. In a perfectly isotropic linear medium this coupling is zero; it arises only through the X^2 nonlinearity of $G_2(X)$ acting on the oscillon background. The coupling strength is set by the background oscillation amplitude Φ_0 and the transverse–longitudinal aspect ratio $R_\perp/\lambda_0$, yielding an effective torsional rigidity (§5.3):

$$\kappa_T \sim c_s^2 \left(\frac{R_\perp}{\lambda_0} \right)^2 \ll c_s^2, \quad (108)$$

where c_s is the sound speed and $\lambda_0 = c/\nu_0$. The torsional mode frequency (and hence mass via $m = h\nu/c^2$) is suppressed by the same ratio. This supplies only a partial suppression; placing neutrino masses in the sub-eV range also requires the weak-trapping factor η_T in Eq. (123), whose first-principles value remains a future calculation.

4. **Extremely weak coupling.** Polarisation rotation neither compresses the medium (no direct gravitational coupling) nor creates a net transverse dipole (no electromagnetic coupling). It interacts with the other modes only through **nonlinear cross-terms** in the $G_2(X)$ action—the same X^2 term that couples the two transverse polarisations to each other and to the longitudinal mode. These couplings are suppressed by $(R_\perp/\lambda_0)^2$, the same small parameter that suppresses the mass. This is a candidate **weak-sector** coupling vocabulary: a nonlinear coupling between the polarisation-rotation sector and the scalar/vector sectors. The full V–A current, G_F , lepton-number rules, beta spectra, and parity-violation amplitudes remain part of the open weak-interaction derivation.
5. **Three flavours (consistency check, not prediction).** The polarisation-rotation mode lives on the oscillon background, which acts as a finite cavity of radius R_c . The cavity supports a discrete set of torsional standing waves with harmonic numbers $n = 1, 2, 3, \dots$ and mass eigenvalues $m_n \propto n^2 \omega_T$. The stability of the n -th harmonic is governed by the ratio of its decay width Γ_n to its mass m_n , with the dominant channel $\nu_n \rightarrow \nu_{n'} + \gamma$ (torsional $\rightarrow$ lower torsional + scalar/transverse radiation) scaling parametrically as

$$\Gamma_n \propto \frac{\alpha^2}{M^4} \omega_T^5 n^4 (n^2 - n'^2)^3, \quad (109)$$

where the n^4 comes from the azimuthal overlap integral (selection rule $\Delta n = \pm 1$) and $(n^2 - n'^2)^3$ from the phase space. The parameters α, M and the absolute normalisation

depend on the oscillon profile and are not fixed at the effective-field-theory level used here. The structural statement of this bullet is therefore: the bound-state spectrum admits a finite number of stable torsional harmonics, with the critical harmonic number $n_{\max}$ set by the ratio Γ_n/m_n computed on the full oscillon profile. Whether $n_{\max} = 3$, as observed phenomenologically, is a *consistency check* to be performed by the lattice programme; it is not derived here, and the present paper does not claim to predict the number three from first principles. Any quantitative lifetime estimates that would appear to single out $n = 4$ as the boundary are deferred to that dedicated calculation.

6. **Flavour oscillations.** The three torsional harmonics propagate in the same ν_0 -medium and share the same nonlinear background. Their eigenstates in the free-propagation basis (mass eigenstates) differ from their coupling basis to charged leptons (flavour eigenstates), because the coupling occurs at the oscillon core where the $G_2(X)$ nonlinearity is strongest. The mismatch is a candidate route to the PMNS mixing matrix and flavour oscillation. Deriving the oscillation Hamiltonian, mass-squared splittings, mixing angles, and CP phase remains future work.
7. **Parity violation of the weak interaction.** Circular polarisation is a *pseudoscalar* quantity: under spatial inversion ($\mathbf{x} \rightarrow -\mathbf{x}$), a left-handed helix becomes right-handed. Any interaction mediated by polarisation-rotation modes therefore violates parity *maximally*. The longitudinal, transverse (linear), and membrane modes are all parity-even, explaining why gravity, electromagnetism, and the strong force preserve parity while the weak force does not.

5.3 Formal derivation: the torsional mode from $G_2(X)$

We now show that the polarisation-rotation mode emerges directly from the $G_2(X)$ Lagrangian without introducing any new field or degree of freedom.

Setup.—Consider perturbations of the scalar field around the oscillon background $\phi_{\text{bg}}(t, r) = \Phi_0(r) \cos \omega_0 t$ (Eq. 82). A general transverse perturbation has two orthogonal polarisations. Writing the perturbation in cylindrical coordinates (r, ϑ, z) centred on the oscillon core:

$$\delta\phi = A(r) \cos(kz - \omega t + \vartheta) + B(r) \cos(kz - \omega t), \quad (110)$$

where A and B are the amplitudes of the two orthogonal transverse polarisations. The ratio A/B determines the polarisation state. For $A = B$ with a $\pi/2$ phase shift, the perturbation traces a **helix**:

$$\delta\phi_{\pm} = A(r) e^{i(kz - \omega t \pm \vartheta)}, \quad (111)$$

where $+$ is left-circular and $-$ is right-circular polarisation. The azimuthal factor $e^{\pm i\vartheta}$ is precisely the winding-number- (± 1) vortex phase of ISPG_Quantum, Sec. 10.

Mode coupling from $G_2(X)$.—The kinetic variable for the perturbed field is

$$X[\phi_{\text{bg}} + \delta\phi] = X_{\text{bg}} + \delta X^{(1)} + \delta X^{(2)} + \dots, \quad (112)$$

where $\delta X^{(1)}$ is linear in $\delta\phi$ and $\delta X^{(2)}$ is quadratic. The background contribution is (cf. Eq. 84):

$$X_{\text{bg}} = \frac{1}{2} \Phi_0^2 \omega_0^2 \sin^2(\omega_0 t). \quad (113)$$

In the action $G_2(X) = -\frac{M_{\text{Pl}}^2}{2} X + \frac{\alpha}{M^2} X^2$, the X^2 term evaluated on the perturbed field generates a quadratic Lagrangian for $\delta\phi$ whose “mass term” oscillates in time:

$$\mathcal{L}_{\delta\phi}^{(2)} = -\frac{M_{\text{Pl}}^2}{2} (\partial_\mu \delta\phi)^2 + \frac{2\alpha}{M^2} \left[2 X_{\text{bg}} (\partial_\mu \delta\phi)^2 + (\dot{\phi}_{\text{bg}} \delta\phi - \nabla \phi_{\text{bg}} \cdot \nabla \delta\phi)^2 \right]. \quad (114)$$

The first bracket provides a *global* time-dependent effective speed, identical to the longitudinal Mathieu driving of §3.13—this is $(R_\perp/\lambda_0)^0$ and gives no polarisation coupling.

Step 1: Decompose into polarisation channels.—Write the perturbation in the circular basis:

$$\delta\phi = \delta\phi_+ + \delta\phi_-, \quad \delta\phi_\pm \propto e^{\pm i\vartheta} f_\pm(r, z, t). \quad (115)$$

The second bracket in (114) contains the term $(\nabla\phi_{\text{bg}} \cdot \nabla\delta\phi)^2$; because $\phi_{\text{bg}} = \Phi_0(r) \cos\omega_0 t$ depends only on r , its gradient is radial: $\nabla\phi_{\text{bg}} = \Phi'_0(r) \hat{\mathbf{r}} \cos\omega_0 t$. Acting on $\delta\phi_\pm \propto e^{\pm i\vartheta}$:

$$\hat{\mathbf{r}} \cdot \nabla\delta\phi_\pm = \frac{\partial f_\pm}{\partial r} e^{\pm i\vartheta}, \quad \hat{\boldsymbol{\vartheta}} \cdot \nabla\delta\phi_\pm = \frac{\pm i f_\pm}{r} e^{\pm i\vartheta}. \quad (116)$$

The **radial** part $(\partial_r f_\pm)$ couples $\delta\phi_+$ and $\delta\phi_-$ only through terms proportional to $\Phi'_0(r)^2/r^2$ —the gradient of the background oscillon profile evaluated at the transverse scale $r \sim R_\perp$.

Step 2: Identify the suppression factor.—The background profile $\Phi_0(r) = A \text{sech}^2(r/R_c)$ has core radius $R_c \sim \hbar/(mc)$ (the Compton radius of the electron-mass oscillon, Sec. 3.13). For $r \gg R_c$ the profile decays exponentially. The polarisation-coupling terms involve the *azimuthal gradient* $f_\pm/r$, which at the oscillon boundary $r \sim R_c$ gives a coupling strength $\sim \Phi_0/R_c$. Relative to the longitudinal coupling $\sim \Phi_0\omega_0$, this introduces the ratio

$$\frac{R_\perp}{\lambda_0} \equiv \frac{1}{R_c \omega_0} = \frac{c}{R_c \omega_0 c} = \frac{\lambda_0}{2\pi R_c}, \quad (117)$$

where $\lambda_0 = c/\nu_0 = \pi\ell_P \approx 5 \times 10^{-35}$ m in the historical carrier-wavelength convention (superseded for the torsional suppression by λ_{osc} in Eq. (121) below) and $R_c = \hbar/(m_e c) = 3.86 \times 10^{-13}$ m is the electron Compton radius.

Step 3: Extract the torsional equation.—Define the slowly varying polarisation angle $\theta(z, t) \equiv \frac{1}{2} \arg(f_+/f_-)$, which measures the orientation of the polarisation plane. After projecting $\mathcal{L}_{\delta\phi}^{(2)}$ onto the θ -sector (integrating out the radial profile and the fast oscillation at ω_0), the effective equation is

$$\boxed{\frac{\partial^2 \theta}{\partial t^2} - \kappa_T \frac{\partial^2 \theta}{\partial z^2} + \Omega_T^2 [1 - 2q_T \cos(2\omega_0 t)] \theta = 0,} \quad (118)$$

where the parameters are determined by $G_2(X)$ and the oscillon profile:

- $\kappa_T = c_s^2 (R_\perp/\lambda_0)^2$ is the torsional rigidity—the speed at which a polarisation twist propagates. The factor $(R_\perp/\lambda_0)^2$ arises because the azimuthal gradient (116) introduces one power of $1/(R_c\omega_0)$ per derivative; the equation is second-order, giving the square.
- $\Omega_T = \omega_0 (R_\perp/\lambda_0)$ is the torsional gap frequency.
- $q_T = q (R_\perp/\lambda_0)^2$ is the torsional Mathieu parameter—the same q that governs the longitudinal mass spectrum, but suppressed by $(R_\perp/\lambda_0)^2$.

Eq. (118) is again a **Mathieu equation** for the torsional sector. Its stability bands determine the allowed torsional masses, just as the longitudinal Mathieu equation (§??) determines the quark and charged-lepton masses. The crucial difference: all torsional parameters carry the suppression factor $(R_\perp/\lambda_{\text{osc}})^2$, where λ_{osc} is the Compton wavelength of the oscillon on which the torsional mode propagates; torsional masses are suppressed by the same factor relative to longitudinal masses:

$$\boxed{\frac{m_\nu}{m_e} \sim \left(\frac{R_\perp}{\lambda_{\text{osc}}}\right)^2 \eta_T,} \quad (119)$$

with η_T the trapping fraction defined below. (Throughout the $(R_\perp/\lambda)^2$ expressions in Eqs. (??)–(118), the wavelength λ is to be read as λ_{osc} .)

Numerical evaluation.—The longitudinal wavelength that is relevant for the torsional suppression is *not* the carrier wavelength $\lambda_0 = c/\nu_0$ of the ν_0 -medium, but the Compton wavelength of the specific oscillon on which the torsional perturbation propagates:

$$\lambda_{\text{osc}} \equiv \hbar/(m_{\text{osc}}c). \quad (120)$$

The physical reason is that Eq. (118) describes a perturbation *on an oscillon background*, and the parametric driving in that equation is set by the oscillon’s own inverse Compton scale $1/\lambda_{\text{osc}}$, not by the medium carrier scale $1/\lambda_0$: the carrier oscillation enters only through the *driving amplitude* q , while the *length-scale ratio* that dresses the azimuthal gradient in Eq. (116) is $R_\perp$ measured against the oscillon’s own wavelength. Consequently the physically relevant aspect ratio is

$$\frac{R_\perp}{\lambda_{\text{osc}}} = \frac{\lambda_{\text{osc}}}{2\pi R_c} = \frac{\hbar/(m_{\text{osc}}c)}{2\pi \hbar/(m_e c)} = \frac{m_e}{2\pi m_{\text{osc}}}, \quad (121)$$

with $R_c = \hbar/(m_e c)$ the host-oscillon Compton radius. For $m_{\text{osc}} \sim m_e$ (the lightest charged fermion):

$$\frac{R_\perp}{\lambda_{\text{osc}}} \sim \frac{1}{2\pi} \approx 0.16. \quad (122)$$

The mass suppression from the aspect ratio alone is $(R_\perp/\lambda_{\text{osc}})^2 \sim 1/(2\pi)^2 \approx 0.025$. Acting on the electron scale this gives $m_e/(2\pi)^2 \simeq 13$ keV, i.e. about 1.6 orders of magnitude of suppression, not the full $m_\nu/m_e \sim 10^{-7}$ hierarchy. Reaching the sub-eV regime requires the trapping fraction to provide the dominant remaining suppression, typically $\eta_T \sim m_\nu/(13 \text{ keV}) \sim 10^{-6}$ for the observed neutrino scale. This remaining suppression arises because the torsional mode, unlike the longitudinal mode, is not self-trapped: it propagates *away* from the oscillon core, so the effective cavity size is not R_c but the much larger mean free path ℓ_{mfp} of the torsional wave in the ν_0 -medium. The trapping fraction

$$\eta_T = \frac{R_c}{\ell_{\text{mfp}}} \quad (123)$$

provides the missing suppression. The full neutrino mass formula is $m_\nu \sim m_e \times (R_\perp/\lambda_{\text{osc}})^2 \times \eta_T$, where η_T is determined by the nonlinear scattering cross-section of the torsional mode on the ν_0 -background.

Status.—The derivation of η_T from $G_2(X)$ requires computing the torsional mean free path via lattice simulation and is left to future work. The qualitative result—that neutrino masses are suppressed by both the aspect ratio *and* the weak trapping—is structural and does not depend on the numerical value of η_T .

5.4 Vortex topology and connection to ISPG_Quantum

The helical wave (111) is simultaneously a cylindrical $\ell = \pm 1$ azimuthal-winding mode (wave description) and a vortex line (field-configuration description). This duality connects the present analysis to the topological framework of ISPG_Quantum, Sec. 10, whose foundational assumptions (including the identification of a $U(1)$ -like phase on the oscillon) are inherited here rather than re-derived.

Identification.—A helical mode $\delta\phi_\pm \propto \cos(kz - \omega t \pm \vartheta)$ has a spatial argument that winds by $\pm 2\pi$ around the propagation axis per wavelength. This is exactly a vortex of winding number $n = \pm 1$ in the (ϑ, z) cylinder. By the results of ISPG_Quantum, Sec. 10:

- winding $n = \pm 1 \Rightarrow$ spin $J = \hbar/2$ (Eq. (10.4) of ISPG_Quantum);
- candidate Berry phase $\gamma = \pm\pi$ under exchange $\Rightarrow$ Fermi–Dirac statistics in the structural route (Eq. (10.12) of ISPG_Quantum; full Fock-space proof open);

- opposite-winding structural mirror $\Rightarrow$ antiparticle label (Sec. 10.2 of ISPG_Quantum).

The neutrino is therefore assigned the same structural fermion-label package used for electrons and quarks, subject to the conditional spin/statistics and antiparticle-status caveats in ISPG_Quantum. The distinction is *which sector* the vortex inhabits:

Particle	Vortex sector	Trapping strength
Electron, quarks	longitudinal / transverse	strong (Mathieu $q = 1.853$)
Neutrinos	polarisation rotation	weak ($q_T = q(R_\perp/\lambda_0)^2$)

All fermions are vortices; the neutrino’s uniqueness is that its vortex lives in the weakly coupled polarisation-rotation channel.

Chirality from topology.—Left-circular polarisation ($n = +1$) and right-circular ($n = -1$) are topologically distinct: one cannot be continuously deformed into the other without unwinding the vortex. This topological protection is a candidate route to an exact chirality label in the massless limit. For massive neutrinos, chirality and helicity must be distinguished: the small mass term Ω_T in Eq. (118) would allow mixing between $n = +1$ and $n = -1$ with the standard m_ν/E helicity-flip suppression. A full Weyl/Dirac spinor derivation remains part of the open weak-sector programme.

5.5 Beta decay in the torsional picture

The prototypical weak process—neutron beta decay—receives a transparent geometric interpretation:

$$n \rightarrow p + e^- + \bar{\nu}_e.$$

Inside the neutron, one d -quark oscillon carries a polarisation-rotation excitation (a helical component winding around its core). When this helical component radiates away, the released torsional energy propagates outward as an **antineutrino**—identified in this structural picture as an opposite-winding antineutrino label ($n = -1$). Simultaneously, the change in the quark’s transverse asymmetry ($-\frac{1}{3} \rightarrow +\frac{2}{3}$; i.e. $d \rightarrow u$) emits the difference as an **electron**: a transverse-mode oscillon carrying the released $\ell = 1$ asymmetry.

In this candidate picture, the W boson is the **massive mediator of the torsion–transverse coupling**: a nonlinear excitation that temporarily converts polarisation-rotation energy into transverse energy and vice versa. Its large mass ($m_W \approx 80$ GeV) reflects the suppressed coupling between these sectors. The Z boson is treated as the neutral analogue: a torsion–longitudinal coupling excitation.

5.6 The W and Z bosons and the $SU(2)$ doublet

In the Standard Model, $W^\pm$ and Z^0 are the massive gauge bosons of $SU(2)_L$. In the polarisation-rotation picture:

- The $W^\pm$ boson is identified heuristically with a localised excitation that *simultaneously* carries helical azimuthal-winding and transverse character—the bridge mode that converts a helical excitation into a transverse displacement. Its charge (± 1) reflects the transverse asymmetry it transfers. Its large mass arises parametrically because creating such a bridge requires overcoming the suppressed coupling $\sim (R_\perp/\lambda_{\text{osc}})^2$ between the two sectors.
- The Z^0 boson is the neutral bridge: it couples the helical-winding sector to the longitudinal (scalar) sector without transferring transverse asymmetry. Its mass is close to m_W because the same coupling suppression governs both channels.

Parametric status of the W, Z mass scale.—The present section fixes only the *parametric* structure of the W, Z mass scale: it is set by the product of the carrier scale and an inverse aspect-ratio factor built from $(\lambda_{\text{osc}}/R_{\perp})$, but neither the overall dimensional prefactor nor the absolute numerical value follows at this level of the derivation. In particular, the naive estimate $m_W \sim h\nu_0/c^2 \cdot (\lambda_{\text{osc}}/R_{\perp})$ does *not* reproduce the measured value $m_W \approx 80.4$ GeV—the carrier frequency ν_0 enters at the Planck scale, so the overall dimensional anchor must be fixed by a separate constitutive input from the main theory rather than read off from this estimate. The absolute determination of m_W, m_Z from the $G_2(X)$ background is a quantitative task deferred to a separate work built on the main theory’s action.

On the two-component space used here.—The pair (left-helical $n = +1$, right-helical $n = -1$) employed below is a *chirality* two-space of the helical-winding mode, not the weak-isospin doublet (ν_L, e_L) of the Standard Model: the two are related but formally distinct $(\text{SU}(2)_L$ pairs a left-handed neutrino with a left-handed charged lepton, not a particle with its antiparticle). What the construction below gives directly is therefore a chirality $\text{U}(2)$ acting on $(n = +1, n = -1)$; its identification with the Standard-Model $\text{SU}(2)_L$ —including the correct pairing of left-handed fermions into weak doublets and the assignment of hypercharges—requires an additional constitutive step that is not derived here and is deferred to a dedicated follow-up.

Chirality $\text{U}(2)$ from the 2-space.—Taking the chirality two-space $(n = +1, n = -1)$ as the working basis, the $G_2(X)$ nonlinearity provides a Hermitian coupling matrix between these two states. The most general 2×2 Hermitian matrix is parametrised by four real numbers,

$$H_{\text{coupling}} = a_0 \mathbb{I} + a_1 \sigma_1 + a_2 \sigma_2 + a_3 \sigma_3, \quad (124)$$

with σ_i the Pauli matrices on the chirality basis. Eq. (124) is therefore a chirality $\text{U}(2)$, whose $\text{SU}(2)$ subgroup mixes the two helicity-windings. Whether this chirality $\text{SU}(2)$ can be promoted, by the constitutive step referred to above, to the Standard-Model $\text{SU}(2)_L$ gauge group is what we mean below by “ $\text{SU}(2)$ sector”; it is not proven in the present section.

Parity-violation sketch.—Within the chirality two-space, flipping the winding number $n \rightarrow -n$ flips the relative coupling phase between helical and transverse sectors, so the two winding states enter the coupling asymmetrically. This provides a *candidate geometric source* of chiral asymmetry in the helical sector; the precise map from this asymmetry to the Standard-Model statement “only left-handed fermions couple to $W^{\pm}$ ” requires the constitutive $\text{SU}(2)$ -identification step above and is not completed here.

From global to local $\text{SU}(2)$.—The coupling matrix (124) depends on the oscillon profile $\Phi_0(r)$, which varies in space: the coefficients $a_i(\mathbf{x})$ inherit the spatial dependence of Φ_0 . The global chirality $\text{SU}(2)$ is thereby promoted to a *local* symmetry, with the oscillon-background gradients playing the role of the gauge connection—the same mechanism by which emergent gauge fields arise from spatially varying order parameters in condensed-matter systems (Berry connections in Bloch bands, spin-orbit coupling in topological insulators). The formal derivation of the Yang–Mills kinetic term from the $G_2(X)$ coupling structure is a future-work task.

Weinberg-angle analogue.—Within the chirality- $\text{SU}(2)$ framework of this section, the analogue of the weak mixing angle is the ratio of coupling strengths between the helical–transverse channel and the helical–longitudinal channel:

$$\tan \theta_W^{(\text{chir})} = \frac{g_1}{g_2} = \frac{\text{longitudinal coupling}}{\text{transverse coupling}}. \quad (125)$$

Both couplings arise from the same X^2 term but project onto different multipole sectors ($\ell = 0$ vs. $\ell = 1$); their ratio is therefore a calculable function of the oscillon profile $\Phi_0(r)$ and the Mathieu parameter q . Identifying $\theta_W^{(\text{chir})}$ with the measured Standard-Model Weinberg angle ($\sin^2 \theta_W = 0.231$) additionally requires the constitutive $\text{SU}(2)_L$ -identification step noted above; until that step is carried out, Eq. (125) is an *internal* ratio of this paper, not an experimental prediction.

5.7 Quantitative programme and future work

Established in this section:

- Polarisation-rotation mode emerges from $G_2(X)$ via the helical perturbation ansatz (Eq. 110); no new field required (§5.3).
- The torsional Mathieu equation (Eq. 118) is derived from $\mathcal{L}_{\delta\phi}^{(2)}$ by decomposing into circular-polarisation channels and projecting onto the polarisation-angle sector.
- The suppression factor $R_\perp/\lambda_{\text{osc}} = 1/(2\pi)$ is determined by the Compton radius of the oscillon, not by neutrino masses (Eq. 121).
- The additional suppression from weak torsional trapping (η_T , Eq. 123) accounts qualitatively for the observed $m_\nu/m_e \sim 10^{-7}$; the first-principles derivation of η_T is left to future work.
- Spin- $\frac{1}{2}$, Fermi–Dirac statistics, and chirality have a candidate structural route through vortex topology (§5.4; ISPG-Quantum, Sec. 10); the full Fock-space/spinor and massive-neutrino chirality derivations remain open.
- The two-component circular-polarisation space coupled by $G_2(X)$ supplies a candidate route toward the weak doublet structure (§5.6).
- The promotion of this candidate SU(2)-like structure to the SM local SU(2) $_L$ gauge theory via the spatial dependence of $\Phi_0(r)$ remains a future formal Yang–Mills derivation.

Quantitative predictions requiring lattice or numerical computation:

- **Neutrino masses.** The parametric structure $m_\nu \sim m_e (R_\perp/\lambda_{\text{osc}})^2 \eta_T$ is fixed by the present derivation: the aspect-ratio factor $R_\perp/\lambda_{\text{osc}} = 1/(2\pi)$ follows from the Compton radius of the oscillon (Eq. 121), and the trapping fraction η_T (Eq. 123) is defined through the nonlinear scattering cross-section of torsional modes on the ν_0 -background. The evaluation of η_T from $G_2(X)$, together with the relevant boundary/profile data, requires a dedicated lattice computation and is the topic of a separate work. Before that calculation the three neutrino masses are not independently predicted; after it, the parametric form can be tested without adding an extra phenomenological mass factor.
- **PMNS matrix.** The overlap integrals between the three torsional eigenstates and the charged-lepton oscillon coupling modes determine the mixing angles θ_{12} , θ_{23} , θ_{13} and the CP phase δ .
- **Mass ordering.** The torsional Mathieu spectrum predicts whether the hierarchy is normal or inverted, depending on the sign of q_T corrections from the oscillon boundary conditions.
- **Majorana vs. Dirac.** A circularly polarised vortex is symmetric under combined charge conjugation and parity (CP) if and only if the boundary conditions identify $n = +1$ with $n = -1$ at the oscillon core. This identification produces a Majorana mass; its absence preserves a Dirac mass. The distinction is determined by the topology of the oscillon core—accessible via lattice simulation.
- **Weinberg angle.** θ_W follows from the ratio of torsion–longitudinal to torsion–transverse couplings (Eq. 125), calculable from the oscillon profile. The prediction $\sin^2 \theta_W \stackrel{?}{=} 0.231$ is a decisive quantitative test.

- **Yang–Mills from $G_2(X)$.** Deriving the $F_{\mu\nu}^a F^{a\mu\nu}$ kinetic term from the position-dependent coupling matrix $H_{\text{coupling}}(\mathbf{x})$.

This mechanism, once numerically confirmed, completes the ISPG vibration programme: four perturbation modes of a single scalar field account for all four fundamental forces—gravity, electromagnetism, the strong force, and the weak force—from a single field, a single vibration frequency ν_0 , and a single nonlinear interaction $G_2(X)$.

6 Massless Sector: Transverse-Traceless Modes and Gluon Confinement

The preceding sections addressed all *massive* particles. The bi-conformal medium also supports *massless* transverse modes. This section outlines the structural classification that emerges naturally from the oscillon picture. The massless exterior sector corresponds to the gravitational-wave (spin-2) part of the theory; the identification of the electromagnetic photon inside this framework is a separate geometric question whose full development is the topic of a companion paper. The confined counterpart of the free transverse mode is identified with the gluon, and the structural confinement mechanism is developed below.

6.1 Four modes of a single medium

The bi-conformal metric $ds^2 = -e^\phi c^2 dt^2 + e^{-\phi} d\sigma^2$ admits two types of perturbation: scalar ($\delta\phi$) and transverse-traceless tensor (δg_{ij}^{TT} , two polarizations). An oscillon, as a localized nonlinear structure, divides the medium into two geometric regions: the **interior** (the core where $G_2(X)$ nonlinearity is strong) and the **exterior** (the asymptotic region where the field is approximately linear).

Relation to the cylindrical ℓ -decomposition of §5. The “four modes” enumerated below—(scalar, tensor) $\times$ (interior, exterior)—are a *different* decomposition of the full perturbation content from the cylindrical-multipole classification used for the scalar sector in §6.1 (longitudinal $\ell = 0$, transverse $\ell = 1$, membrane $\ell \geq 2$, helical $\ell = \pm 1$ winding). They are complementary, not redundant: the ℓ -decomposition there is internal to the scalar sector and classifies the cylindrical angular structure of $\delta\phi$, whereas the scalar-vs.-tensor split used here sits one level higher, at the level of the metric perturbations of the bi-conformal background. In particular, the tensor sector (rows 3–4 of the table below) is not part of the scalar- ℓ tower of §6.1; conversely, the scalar helical $\ell = \pm 1$ mode of §6.1 sits inside row 1 (scalar, interior) of the table below. No unified single “four-mode” enumeration is asserted to cover both decompositions simultaneously.

Combining two perturbation types with two geometric regions yields exactly four vibration modes:

Mode : Type / Region	→ Particle	→ Properties
1 : scalar, localized (core)	→ oscillon (matter)	→ $m > 0$, integer harmonics
2 : torsional (twist)	→ neutrino	→ $m \approx 0$, chiral, three flavours
3 : tensor, free (exterior)	→ gravitational wave (spin-2) ¹	→ $m = 0$, propagating
4 : tensor, confined ($\nu_N \neq \nu_0$)	→ gluon	→ $m = 0$, confined by frequency gap

Two statuses are kept distinct in this section: (i) the imported standard photon field A_μ used for present electromagnetic propagation calculations, and (ii) the future programme of deriving an emergent helicoidal/transverse-vector medium mode that would reproduce that photon. Only the first status is used below.

No additional fields or symmetry postulates are introduced. The four categories exhaust the perturbation content of the bi-conformal metric around an oscillon background: scalar vs. tensor ($\times 2$), interior vs. exterior ($\times 2$).

6.2 Transverse-traceless modes and the gravitational-wave sector

A transverse-traceless perturbation of the metric propagating in the exterior region satisfies the linearized wave equation with zero effective mass. It travels at c and carries two independent polarization states. Within the present geometric framework these modes correspond to gravitational waves—the massless spin-2 sector already established in the main theory [49]; they do not give rise to a photon, whose identification is a separate geometric question treated below.

Propagation.—The wave travels at c , the transverse wave speed of the ν_0 -string. Because the string is uniform (one ν_0 , one tension), the speed is the same everywhere and for all frequencies—no dispersion. The two polarizations correspond to the two independent directions in the transverse plane.

Speed of light as a fundamental limit.—The speed c is the propagation speed of perturbations on the medium that *is* space. Nothing can travel through a medium faster than the medium’s own wave speed; c is therefore a geometric maximum, not an empirical accident.

Photon identification.—The identification of the electromagnetic photon within the ISPG medium is not part of the present paper. The electromagnetic sector is carried by the standard $U(1)$ gauge field A_μ in the usual minimally coupled form (see [49], Appendix 5); its embedding into the oscillon frequency-to-mass framework is a separate question whose full development is the topic of a companion paper. In the remainder of this paper, “massless transverse mode” refers to the gravitational-wave sector; any statement about photon propagation in a gravitational field (lensing, redshift, Shapiro delay) rests on the standard A_μ treatment of the main theory, not on a direct identification of the photon with the TT metric mode.

6.3 Gluon: confined transverse wave

Inside the oscillon core the medium vibrates at the oscillon’s own frequency $\nu_N = m_N c^2 / h \neq \nu_0$. Transverse-traceless perturbations born in this region inherit ν_N as their characteristic frequency. When such a perturbation reaches the core boundary and attempts to propagate into the exterior, it encounters a medium whose background vibration is ν_0 —a fundamentally different rhythm.

Confinement by frequency incompatibility.—The ν_0 -background suggests a dispersive medium for transverse modes tuned to $\nu_N \neq \nu_0$. Because the $G_2(X)$ nonlinearity couples all oscillation modes to the background, a tensor wave whose frequency is not commensurate with ν_0 is a natural candidate for a non-propagating or evanescent exterior mode. The inharmonicity criterion therefore supplies a structural route to gluon confinement by *frequency mismatch*. A complete proof of the band gap and decay length requires the exterior $G_2(X)$ dispersion relation and boundary matching problem to be solved.

Why scalar modes are ν_0 -harmonic but tensor modes are not.—This is the crucial asymmetry. Approximate the oscillon core as a spherical cavity of radius R . Both scalar and tensor perturbations satisfy radial wave equations inside the cavity, but with different angular momentum constraints:

¹The electromagnetic photon is a $U(1)$ gauge boson (A_μ) treated in the main theory [49] in the standard minimally coupled form; its embedding into the present oscillon/frequency-to-mass picture is deferred to a companion paper.

- **Scalar perturbations** ($\ell = 0$ allowed). The radial eigenvalues are the zeros of the spherical Bessel function $j_0(kR) = 0$, located at

$$k_n R = n\pi, \quad n = 1, 2, 3, \dots \quad (126)$$

The resulting frequencies $\omega_n = n\pi c/R$ form an **integer-harmonic series**. This is why the Mathieu mass spectrum produces ν_0 -harmonics: the scalar cavity is one-dimensional in its radial structure, analogous to a vibrating string.

- **Tensor perturbations** ($\ell \geq 2$ required, since transverse-traceless modes carry spin-2). The radial eigenvalues are the zeros of $j_\ell(kR) = 0$ for $\ell \geq 2$. For $\ell = 2$, the first three zeros are:

$$k_n R \approx 5.763, 9.095, 12.323, \dots \quad (127)$$

These are **not integer multiples of π** : the ratio of the lowest tensor eigenvalue to the scalar fundamental is

$$\frac{k_1^{(\ell=2)}}{k_1^{(\ell=0)}} = \frac{5.763}{\pi} \approx 1.835, \quad (128)$$

a non-integer ratio. The tensor modes form an **inharmonic series**—analogous to the overtones of a drum, which are not integer multiples of the fundamental (unlike a string).

This is the mathematical origin of the confinement criterion: scalar modes (matter) and free transverse-traceless modes (gravitational waves) live in an integer-harmonic tower built on ν_0 and propagate freely through the ν_0 -background. Tensor cavity modes (gluons) have frequencies that are *incommensurate* with the integer harmonic tower:

$$\boxed{\frac{\omega_{\text{gluon}}}{\nu_0} = \frac{j_{\ell,n}}{\pi} \notin \mathbb{Z} \quad (\ell \geq 2).} \quad (129)$$

A wave whose frequency is not a ν_0 -harmonic is expected to couple poorly to coherent propagation in the ν_0 -background; whether it decays evanescently, and with what decay length, is the exterior band-gap calculation left for future work. A useful acoustic analogy: the ν_0 -background is a *guitar string*—a one-dimensional resonator whose overtones are exact integer multiples of the fundamental. The oscillon interior is a *handpan* (a curved metal membrane)—a two-dimensional resonator whose overtones are governed by Bessel zeros and are *not* integer multiples of the fundamental. A vibration born on the handpan cannot sustain itself on the guitar string because the two harmonic series are incommensurate. This “string-membrane asymmetry” ($j_{0,n}/\pi \in \mathbb{Z}$ vs. $j_{\ell \geq 2,n}/\pi \notin \mathbb{Z}$) is a theorem of Bessel function theory. The identification of this asymmetry with physical confinement is a structural mechanism: it becomes a derivation only after the exterior propagation problem for the full medium is closed.

This immediately explains two key features of the strong interaction:

1. **Confinement.** In this candidate mechanism, a single gluon cannot exist in free space because the ν_0 -background does not support coherent propagation of the inharmonic tensor-cavity mode. Photons *can* propagate freely because, in the present treatment, they are described as the imported $U(1)_{\text{EM}}$ gauge mode A_μ rather than as localized resonant excitations; a direct medium-mode identification (helicoidal/Kelvin/vector-mode candidate) remains future work.
2. **Flux tube and linear potential.** When two quarks (sub-structures within the oscillon) are separated, the region between them must sustain the interior frequency ν_N against the exterior ν_0 pressure. Maintaining ν_N in a strip of width $\sim R_{\text{core}}$ and length L costs energy proportional to L :

$$V(L) \approx \sigma L, \quad \sigma \sim \frac{G_2''(X_0) \nu_N^2}{c^2} R_{\text{core}}^2, \quad (130)$$

giving a route to the linear confining potential used in lattice QCD. If this flux-tube picture is dynamically realized, then when $V(L)$ exceeds $2m_q c^2$, the tube breaks and a new quark–antiquark pair materializes—the analogue of QCD string breaking.

Physical picture.—The oscillon interior is a “bubble” of frequency ν_N immersed in a ν_0 -ocean. Gluons are transverse vibrations that can live inside the bubble but drown in the ocean. Pulling two quarks apart stretches the bubble into a tube; the ocean presses on the tube walls from all sides, and the energy cost grows linearly with length.

Numerical verification.—The inharmonicity criterion was verified numerically (see `python/gluon_modes.py`). The glueball spectrum predicted by the Bessel-zero ratios $j_{\ell,n}/j_{2,1}$ can be compared directly with lattice-QCD glueball benchmarks (Morningstar & Peardon 1999; Chen et al. 2006):

State	M_{lat} (MeV)	ratio _{lat}	ratio _{ISPG}	Δ
0^{++}	1710	1.000	1.000 ($\ell=2, n=1$)	0%
2^{++}	2390	1.398	1.420 ($\ell=4, n=1$)	1.6%
2^{-+}	3100	1.813	1.807 ($\ell=3, n=2$)	0.3%
3^{++}	3670	2.146	2.138 ($\ell=2, n=3$)	0.4%

(131)

Four of the six lowest glueball states match to $\leq 2\%$; the remaining two (0^{-+} and 1^{+-}) deviate by $\sim 5\%$, likely because the free-cavity Bessel zeros neglect the $G_2(X)$ self-interaction corrections. The fundamental ratio

$$\frac{M(0^{++})}{M(p)} = \frac{1710}{938} = 1.823 \approx \frac{j_{2,1}}{\pi} = 1.835 \quad (\Delta = 0.66\%), \quad (132)$$

provides a suggestive numerical consistency check.

Scope of the ratio comparison.—The internal ratios of the glueball spectrum are fixed entirely by the Bessel-zero ratios $j_{\ell,n}/j_{2,1}$ once the $(\ell, n) \leftrightarrow J^{PC}$ assignments of Eq. (131) are accepted; in that limited sense the four-state comparison uses no additional tunable coefficient. It is not a standalone experimental confirmation of QCD glueballs. Two caveats must be stated honestly. (i) The $(\ell, n) \leftrightarrow J^{PC}$ assignment is itself an educated identification of spherical-cavity modes with lattice-QCD glueball states; alternative assignments would produce different ratios and have not been ruled out here. (ii) The ratio-to-proton in Eq. (132) uses $M(p) = 938$ MeV as an external dimensionful input—the proton mass is not derived in this section; it is only being used to set the absolute scale. The internal glueball-ratio pattern is therefore a coefficient-free lattice-benchmark comparison under the chosen assignments, while the proton-anchor agreement in Eq. (132) is a *consistency check* against that anchor, not an independent prediction of $M(p)$ itself.

The eightfold structure.—The Standard Model has exactly eight gluons, corresponding to the eight generators of SU(3). In the spherical-cavity approximation, the lowest TT tensor energy level ($\ell = 2, n = 1$) has two polarisation families (even and odd parity), each with $2\ell + 1 = 5$ orientations, giving $2 \times 5 = 10$ modes. A baryon oscillon, however, contains three sub-oscillons (quarks) arranged in a triangle, reducing the symmetry from SO(3) to the discrete group D_3 . Under this restriction the $\ell = 2$ representation decomposes as

$$D^2|_{D_3} = A_1 \oplus E \oplus E = 1 + 2 + 2. \quad (133)$$

The C_3 selection rule ($e^{im_p \cdot 2\pi/3}$ sums to zero unless $m_p = 0, \pm 3$) restricts the perturbation to the $Y_{4,0}$ (oblate) and $Y_{4,\pm 3}$ (triangular) spherical harmonics. Computing the Gaunt integrals exactly yields the D_3 perturbation matrix (see `python/gluon_proof.py`):

$$H_{D_3} = \varepsilon \begin{pmatrix} g_0^{(-2)} & 0 & 0 & g_3 & 0 \\ 0 & g_0^{(-1)} & 0 & 0 & g_3 \\ 0 & 0 & g_0^{(0)} & 0 & 0 \\ g_3 & 0 & 0 & g_0^{(+1)} & 0 \\ 0 & g_3 & 0 & 0 & g_0^{(+2)} \end{pmatrix}, \quad (134)$$

where $g_0^{(m)}$ are the Gaunt coefficients for $Y_{4,0}$ and $g_3 = -0.238$ is the $Y_{4,\pm 3}$ coupling. The matrix has block-diagonal structure: $m = 0$ is decoupled (A_1 irrep) while $\{m = -2, m = +1\}$ and $\{m = -1, m = +2\}$ form two 2×2 blocks whose eigenvalues are pairwise degenerate (E irreps). The resulting eigenvalues confirm the D_3 decomposition:

Subgroup	deg.	λ	status
$E_\downarrow$	2	-0.145	confined
A_1	1	+0.242	confined
$E_\uparrow$	2	+0.347	resonant

(135)

When the nonperturbative parameter ε is large enough that $E_T + \varepsilon \lambda(E_\uparrow) \approx E_S^{(n=2)} = (2\pi)^2$, the upper E pair enters resonance with the nearest scalar eigenvalue. The $\ell = 2$ tensor fundamental at $(kR)^2 = j_{2,1}^2 = 33.22$ lies below the scalar $n = 2$ level at $(2\pi)^2 = 39.48$; the gap is $\Delta = 6.26$. The Gaunt eigenvalue spread is 0.492, so resonance occurs at $\varepsilon \approx \Delta/0.492 \approx 13$. A perturbative estimate gives $\varepsilon_{\text{pert}} \approx 0.5$, so the required value lies in a genuinely nonperturbative regime. An exploratory script `python/gluon_epsilon.py` was intended to estimate this scale by comparing circular and D_3 -deformed two-dimensional cavities, but its current circular-cavity benchmark is not calibrated: it does not reproduce the expected low-lying Bessel spectrum before the triangular deformation is added. Consequently its previously quoted ε_{2D} and dressed ε_{3D} values are not used as evidence here. The reliable quantitative target is the analytic requirement $\varepsilon \approx \Delta/0.492 \approx 13$; deriving whether the real 3D oscillon reaches this target requires a corrected finite-element or spectral eigenproblem.

Non-independence note.—If a future 2D-to-3D dressing calculation uses $j_{2,1}/\pi = 1.835$, it will share the same Bessel-zero input that appears in Eq. (132). Such a result should therefore be counted as a single structural thread, not as an independent second numerical prediction, until the full 3D eigenproblem is solved directly.

At resonance the upper E pair acquires $\sim 50\%$ scalar admixture and radiates through the ν_0 -exterior in ~ 2 oscillation periods. The A_1 mode ($\sim 10\%$ leak) and $E_\downarrow$ pair ($< 1\%$ leak) remain confined. The total count is

$$\boxed{\underbrace{5}_{\text{odd parity}} + \underbrace{2}_{E_\downarrow} + \underbrace{1}_{A_1} = 8 = \dim[\text{SU}(3)_{\text{adj}}] = 3^2 - 1.} \quad (136)$$

The count of eight confined modes in the baryon-cavity problem matches the eight generators of $\text{SU}(3)_{\text{adj}}$.

Scope of the “8” count.—What the calculation above rigorously establishes is the mode count “8” in a *specific* cavity problem: the TT perturbations of a D_3 -symmetric (triangular) baryon-like oscillon, after the Gaunt-coefficient restriction of the $\ell = 2$ multiplet. That count *coincides* with $\dim \text{SU}(3)_{\text{adj}} = 3^2 - 1 = 8$. It does *not*, however, derive the Standard-Model gauge group $\text{SU}(3)_c$, which must be *universal* (the same eight gauge bosons act on $q\bar{q}$ mesons as on qqq baryons), whereas the D_3 -triangle argument is specific to three-quark hadronic geometry: a $q\bar{q}$ meson would sit in the D_2 dihedral symmetry class and would produce a different mode count from this construction. A genuine derivation of $\text{SU}(3)_c$ as a gauge group must therefore come from a hadron-independent structure of the $G_2(X)$ sector, not from a specific baryonic cavity geometry; the numerical coincidence “ $3^2 - 1 = 8$ ” recorded here is a *consistency check* of the baryon-sector mode count against the known SM adjoint dimension, and the gauge-group derivation proper is a task deferred to subsequent work.

The next computational step is to solve the full coupled perturbation equation

$$\square h_{ij}^{TT} + \mathcal{M}^2[\nu_N, \nu_0, G_2] h_{ij}^{TT} = 0 \quad (137)$$

inside a three-dimensional three-body oscillon with evanescent exterior boundary conditions. This 3D finite-element calculation will determine the required nonperturbative splitting directly

and test whether the eight surviving modes' coupling algebra reproduces $SU(3)$. If so, the colour gauge group would emerge geometrically from the oscillon's D_3 cavity structure.

6.4 Summary and future programme

The single-field, single-medium ontology of ISPG accommodates all particle categories through the vibration-mode classification:

- **Massive particles** (§3.12): scalar, localized $\rightarrow$ oscillon harmonics, $m_N = m_1 b_N(q)$.
- **Neutrinos** (§5): polarisation-rotation wave (helical vortex with $n = \pm 1$) $\rightarrow$ candidate chirality route and $m_\nu \ll m_e$ after the open η_T suppression; the weak-sector coupling is modeled as nonlinear coupling between torsional and other modes.
- **Gravitational waves** (§6.2): tensor, free $\rightarrow$ propagating transverse-traceless mode (spin-2), $m = 0$. The electromagnetic photon (A_μ , $U(1)$ gauge boson) is treated in the main theory [49]; its embedding into the present oscillon framework is the topic of a companion paper.
- **Gluons** (§6.3): tensor, confined by ν_N/ν_0 frequency mismatch $\rightarrow$ interior transverse mode, $m = 0$, confined; flux tube = frequency bridge, $V(L) \propto L$.

This classification is **exhaustive**: the ν_0 -string with transverse extent supports exactly four vibration types (longitudinal, transverse, torsional, membrane), and the oscillon divides space into two regions (interior, exterior). No fifth vibration mode exists without introducing new fields or extra dimensions—four forces from four modes.

The quantitative development of Sections 6.2–6.3 (derivation of coupling constants, $SU(3)$ emergence, electroweak unification) constitutes a major programme for future work.

7 Why Three Dimensions? Emergence of Spatial Dimensionality from a One-Dimensional Substrate

The preceding sections established that the ν_0 -background has integer harmonics $\omega_n = n\pi c/R$ (the zeros of $j_0(kR)$), while tensor cavity modes have non-integer (Bessel) harmonics. This spectral fact has a deeper consequence that we now develop: *the integer-harmonic structure of the scalar sector is the signature of a fundamentally one-dimensional substrate*, and the three spatial dimensions we observe emerge as a geometric necessity of vibration.

7.1 The scalar sector is one-dimensional

The spherical Bessel function of order zero is

$$j_0(x) = \frac{\sin x}{x}. \quad (138)$$

Its zeros lie at $x = n\pi$, $n = 1, 2, 3, \dots$ —*identical* to the eigenvalues of a one-dimensional vibrating string of length L clamped at both ends:

$$k_n L = n\pi, \quad \omega_n = n\omega_1. \quad (139)$$

This is not an analogy but a mathematical identity. The radial equation for $\ell = 0$ modes in a spherical cavity, after the standard substitution $u(r) = r\psi(r)$, is the one-dimensional wave equation:

$$\frac{d^2 u}{dr^2} + k^2 u = 0, \quad u(0) = u(R) = 0. \quad (140)$$

The isotropic ($\ell = 0$) character of the scalar field ϕ eliminates all angular dependence; three spatial coordinates collapse into a single radial variable, and the mode structure becomes *exactly* that of a string clamped at two endpoints.

Scope of the claim.—The statement is a theorem about the $\ell = 0$ mode spectrum, not an assertion that the physical universe “is” a literal one-dimensional string. The $u = r\psi$ substitution and the resulting one-dimensional wave equation hold for *any* three-dimensional spherically symmetric Helmholtz problem; this is a standard mathematical identity, not an ISPG-specific result. What the theorem therefore establishes, for the *scalar sector* of the medium—the sector responsible for gravity and particle masses—is only that its $\ell = 0$ spectrum is controlled by a one-dimensional Sturm–Liouville structure. The higher- ℓ sectors (transverse, torsional, membrane) supply the additional degrees of freedom for electromagnetism, weak and strong interactions; the full field $\phi(\mathbf{x}, t)$ lives on a three-dimensional spatial domain throughout. The “substrate is one-dimensional” reading that appears in the remainder of this section is a proposed *ontological interpretation* built on top of the spectral theorem, not an additional mathematical consequence of it; we flag it explicitly as an interpretive posture, not a derived result, and its experimental implications are separated out in §7.4.

Because there is only one field ϕ and its isotropic mode is one-dimensional, the universal frequency ν_0 is automatically the same everywhere: one string has one pitch. No synchronisation mechanism is needed. This resolves the question of why ν_0 is a universal constant—it is the vibration frequency of a single quantum entity, not of many independent oscillators.

7.2 Sequential emergence of dimensions

A clamped string (1D) can vibrate only if it is free to displace in at least one transverse direction. This displacement opens a second dimension:

1. **First dimension:** the axis of tension—the direction along which the “string” is stretched (radial coordinate of the $\ell = 0$ mode). This is the longitudinal (scalar) direction.
2. **Second dimension:** the string vibrates transversely—it moves “up” and “down” perpendicular to the tension axis. A plane (2D) is created. In the spherical-harmonic language, this introduces the polar angle θ and with it the $\ell = 1$ (dipole) modes.
3. **Third dimension:** on the 2D membrane so created, a second transverse vibration mode can exist—one that is perpendicular to both the tension axis and the first transverse direction. In spherical harmonics this is the azimuthal angle φ and corresponds to $\ell = 2$ (quadrupole, tensor) modes. These are the **gluon-candidate tensor modes** discussed in Section 6.3.

The chain is



The present construction uses the angular sphere S^2 of ordinary three-dimensional space, with coordinates (θ, φ) . A hypothetical d -dimensional construction would instead use S^{d-1} , and the angular argument alone does not exclude $d > 3$. Restricting the construction to exactly $d = 3$ therefore requires the minimality/empirical input stated below, or a future stability/graph-reconstruction argument.

Why exactly 3 and not 2 or 4?—A purely (2+1)-dimensional universe has no local propagating spin-2/tensor degrees of freedom in the standard gravitational degree-count; the transverse-traceless polarizations vanish at two spatial dimensions. This is a gauge/dynamical obstruction, not a harmonic-existence obstruction: harmonics on S^1 exist for arbitrary integer winding.

Three spatial dimensions is therefore the **minimum** in which both the scalar sector and propagating tensor degrees of freedom coexist:

$$d = 3 = \underbrace{1}_{\text{longitudinal } (\ell=0, \text{ scalar})} + \underbrace{2}_{\text{transverse } (\ell=2, \text{ tensor required})}. \quad (142)$$

The number of spatial dimensions is constrained from below by the requirement that the substrate support tensor vibrations. In the present interpretive picture, the $\ell = 2$ tensor sector—and, in the strong-sector application, the gluon-candidate sector of Section 6.3—is the structure that motivates the third spatial dimension. The full gluon/confinement dynamics, however, remain open until the exterior band-gap problem and the dynamic $SU(3)_c$ vertices are derived. The confinement criterion then receives a second interpretation: it is the mismatch between a 2D membrane-like vibration and a 1D string background, precisely the “string-membrane asymmetry” of Section 6.3.

Lower bound vs. unique selection.—Eq. (142) is strictly a *lower bound*: $d = 3$ is the smallest integer for which the propagating transverse- traceless spin-2 sector is available in the usual gravitational degree count. Larger spatial dimensions ($d = 4, 5, \dots$) also support propagating tensor modes and therefore pass the same structural test. The argument above does not by itself exclude $d > 3$; the selection of exactly $d = 3$ over $d > 3$ requires an additional principle—an Occam-style minimality postulate, stability arguments from the main theory, or empirical input from the observed angular structure of matter and gravity. We adopt Occam minimality as a working postulate in the remainder of this paper and label it as such, rather than as a theorem derived from the $G_2(X)$ action alone.

7.3 The “processor” picture: a single quantum entity

The one-dimensional reduction raises an ontological question: is space a collection of many vibrating elements, or is it a **single quantum entity** that manifests at every point?

A computer processor is physically one device, yet it “serves” the entire machine: every memory address, every running process. It has a clock frequency—a single fundamental tick—and in each tick it updates the state at a particular address. The ISPG medium may be analogous: ϕ is a single quantum field—not a collection of point particles—and ν_0 is its clock frequency. Non-locality (quantum entanglement) is then trivial: there is no distance to bridge between two “parts” of a single entity. The speed of light c is the maximum rate at which the entity can update different “addresses” in its quantum state.

Mathematically, the field $\phi(x, t)$ already encodes this picture: a single function evaluated at all spacetime points. The ontological step is to take the formalism literally— ϕ is not “a value at each point of a pre-existing space” but rather *the single quantum object whose superposition state we parametrize by x* . The apparent spatial extension of the universe is the quantum state-space of one entity.

This interpretation changes no equation but resolves several puzzles:

- **Universality of ν_0 :** one entity, one frequency.
- **Quantum non-locality:** two entangled particles share a state of one field; no signal needs to “travel.”
- **Continuity of space:** there are no gaps between particles—there is one object whose excitation pattern we call “geometry.”

Whether the “one entity” interpretation is merely a restatement of quantum field theory or carries testable consequences (e.g. via Bell-inequality violations in gravitational degrees of freedom) is addressed in the next subsection.

7.4 From one entity to three-dimensional space: the entanglement route

The claim that ϕ is a single quantum entity whose entanglement structure generates the appearance of spatial extension can be made precise using three results, the first two of which are already established in the literature.

Step 1: vacuum entanglement obeys an area law.—For any quantum field in its vacuum state, the entanglement entropy between a region A and its complement scales as the *area* of the boundary, not the volume:

$$S_{\text{ent}}(A) = \frac{\mathcal{A}(\partial A)}{4\ell_{\text{eff}}^2} + \text{sub-leading}, \quad (143)$$

where ℓ_{eff} is a UV cutoff length. This is the Bombelli–Koul–Lee–Sorkin area law [91], confirmed for free and interacting fields alike. In ISPG the role of ℓ_{eff} is played by the oscillon core radius $R \sim c/\nu_0$.

Step 2: entanglement area as a route to geometric area (Ryu–Takayanagi [89, 90]).—In holographic theories [93] the entanglement entropy of a boundary region A equals the area of the minimal surface in the bulk that is homologous to A :

$$S_{\text{ent}}(A) = \frac{\text{Area}(\gamma_A)}{4G_N}, \quad (144)$$

where γ_A is the Ryu–Takayanagi surface and G_N is Newton’s constant. This is a guide rather than a theorem already proved for an arbitrary $G_2(X)$ scalar vacuum: the ISPG-specific tensor-network/MERA reconstruction of distances, angles and curvature from $|\Omega\rangle_\phi$ remains a future calculation. The ER = EPR conjecture [92] is therefore used here as geometric intuition, not as a completed derivation.

Step 3 (ISPG-specific).—The single field ϕ , vibrating at ν_0 with the $G_2(X)$ nonlinearity, produces a vacuum state $|\Omega\rangle$ whose entanglement structure satisfies three consistency conditions. We check each as scaffolding for the future reconstruction:

1. **Bekenstein–Hawking matching.** The UV cutoff of the ISPG medium is the Planck-scale lattice spacing $a = \pi \ell_P$. (This is the shortest wavelength the medium supports; it coincides numerically with the classical estimate $\lambda_0^{(\text{cl})}$ of Eq. 54, but the physical reason it enters here is that ℓ_P is the gravitational UV cutoff, not that the classical $\nu_0^{(\text{cl})}$ sets the dynamics—the quantum value ν_0 governs all oscillonic physics as stated in §??.) The Srednicki area law for a free scalar gives $S = \alpha_0 \mathcal{A}/a^2$ with $\alpha_0 \approx 0.30$. With $a = \pi \ell_P$:

$$S_{\text{free}} = \frac{0.30 \mathcal{A}}{\pi^2 \ell_P^2} = \frac{0.030 \mathcal{A}}{\ell_P^2}. \quad (145)$$

The Bekenstein–Hawking entropy requires $S_{\text{BH}} = \mathcal{A}/(4\ell_P^2)$, i.e. $\alpha = \pi^2/4 = 2.467$. The $G_2(X)$ nonlinearity provides the enhancement through two mechanisms:

- *Sound-speed reduction.* For a k-essence Lagrangian $G_2(X)$, perturbations propagate at the effective sound speed (established in §3.13):

$$c_s^2 = \frac{G_{2,X}}{G_{2,X} + 2X G_{2,XX}} = \frac{1}{1 + 4q}. \quad (146)$$

The entanglement entropy of a scalar with sound speed $c_s < c$ scales as $1/c_s$ (because the effective UV cutoff in momentum space is $k_{\text{max}} = \omega_{\text{max}}/c_s$, increasing the number of modes per unit area). With $q = 1.853$:

$$F_{c_s} = \frac{1}{c_s} = \sqrt{1 + 4q} = \sqrt{8.412} \approx 2.900. \quad (147)$$

- *Floquet sideband enhancement.* The oscillating background $\phi_0 \cos(\omega_0 t)$ means that vacuum fluctuations $\delta\phi$ satisfy a Mathieu equation (§??). By the Floquet–Bloch theorem, each stable mode is not a pure plane wave but a superposition of frequency sidebands:

$$\delta\phi_k(t) = e^{i\nu_k t} \sum_{n=-\infty}^{\infty} c_{2n}^{(k)} e^{2in\omega_0 t}, \quad (148)$$

where ν_k is the Floquet exponent and the Fourier coefficients c_{2n} satisfy the Mathieu three-term recurrence.

For a free field ($q = 0$): only $c_0 \neq 0$, $|c_0|^2 = 1$ —one channel per mode. For $q \neq 0$: the parametric driving distributes spectral weight among sidebands. Each sideband at frequency $\nu_k + 2n\omega_0$ is an independent entanglement channel across any spatial bipartition, because entanglement entropy is additive over orthogonal frequency components.

The enhancement factor is the **inverse participation ratio** (IPR) of the ground-state Mathieu function $\text{ce}_0(z, q)$:

$$F_{\text{bands}} = \frac{1}{|A_0^{(0)}(q)|^2}, \quad (149)$$

where $A_0^{(0)}$ is the zeroth Fourier coefficient of ce_0 , normalised so that $|A_0|^2 + 2 \sum_{r=1} |A_{2r}|^2 = 1$. Physically: $|A_0|^2$ is the fraction of spectral weight remaining in the unshifted ($n = 0$) channel; $1/|A_0|^2$ counts how many effective channels share the weight.

The coefficient $A_0^{(0)}(q)$ is determined by solving the Mathieu recurrence

$$a_0 A_0 = q A_2, \quad (a_0 - 4n^2) A_{2n} = q(A_{2n-2} + A_{2n+2}), \quad (150)$$

with the eigenvalue $a_0(q)$ fixed by the convergence condition $A_{2n} \rightarrow 0$ as $n \rightarrow \infty$. At $q = 1.853$ the numerical solution gives

$$|A_0^{(0)}(1.853)|^2 \approx 0.353, \quad F_{\text{bands}} \approx 2.84. \quad (151)$$

This value can be verified independently: the recurrence (150) is the same system already solved in §3.12—no new input is required.

The total enhancement is

$$F = F_{c_s} \times F_{\text{bands}} = \sqrt{1 + 4q} \times \frac{1}{|A_0^{(0)}(q)|^2} \approx 2.90 \times 2.84 = 8.22, \quad (152)$$

giving the effective Srednicki coefficient $\alpha_{\text{eff}} = 0.30 \times 8.22 = 2.47 \approx \pi^2/4 = 2.467$ and therefore

$$S_{\text{ent}} = \frac{\alpha_{\text{eff}}}{\pi^2 \ell_P^2} \mathcal{A} = \frac{\pi^2/4}{\pi^2 \ell_P^2} \mathcal{A} = \frac{\mathcal{A}}{4 \ell_P^2} = S_{\text{BH}}. \quad (153)$$

Both factors— F_{c_s} and F_{bands} —are determined by the single parameter $q = 1.853$, which is itself fixed by the muon mass (§3.13). No additional free parameter enters the entropy calculation.

Status of the match.—Equation (153) should be read as a *consistency check*, not as a first-principles derivation of S_{BH} . Two ingredients in the enhancement factor require further substantiation:

- $F_{c_s} = 1/c_s$ is assumed on the ground that the effective momentum cutoff scales as $k_{\text{max}} = \omega_{\text{max}}/c_s$; the rigorous extension of Srednicki’s flat-space, $c_s = c$ area-law derivation to an oscillating $G_2(X)$ background is not carried out here.

- $F_{\text{bands}} = 1/|A_0^{(0)}(q)|^2$ assumes that the Floquet sidebands contribute as *independent* entanglement channels. Whether the sidebands behave as orthogonal Fock-space factors or as phase-coherent Fourier components of a single mode depends on the full Bogoliubov structure of the time-periodic vacuum; that calculation is a self-contained technical task and the topic of a separate work.

The 0.1% agreement $\alpha_{\text{eff}} \approx \pi^2/4$ is therefore a numerical coincidence of the current parametric estimates: it is *compatible* with the Bekenstein–Hawking coefficient but does not by itself constitute a derivation of it. This matches the general epistemic posture of the paper: the BH match is part of the area-law scaffolding, not a core gravitational prediction of the main theory [49].

2. **Dimensional selection.** The entanglement pattern is that of a (3+1)-dimensional manifold because the $\ell = 0$ (scalar) and $\ell = 2$ (tensor) sectors coexist *only* for $d \geq 3$ spatial dimensions (§7.2), and the Occam minimum selects $d = 3$.
3. **Scalar–tensor dichotomy.** The vacuum two-point function (verified numerically in `python/entanglement_proof.py`, Section 3) shows:

$$\langle \delta\phi(\mathbf{x}) \delta\phi(\mathbf{0}) \rangle \propto \frac{1}{|\mathbf{x}|^2} \quad (\text{power law, long-range}), \quad (154)$$

$$\langle h_{ij}^{TT}(\mathbf{x}) h_{kl}^{TT}(\mathbf{0}) \rangle \propto \frac{e^{-\kappa|\mathbf{x}|}}{|\mathbf{x}|} \quad (\text{exponential, confined}), \quad (155)$$

with $\kappa = 2\pi\sqrt{\nu_N^2 - \nu_0^2}/c$. Using $\nu_N/\nu_0 = j_{2,1}/\pi = 1.835$, $\kappa R_{\text{osc}} \approx 9.66$: at $r = 5 R_{\text{osc}}$ the scalar correlator is 10^{-2} while the tensor is 10^{-20} —a factor 10^{18} suppression. Confinement is an entanglement property: tensor modes decohere exponentially outside the oscillon, while scalar modes maintain long-range quantum correlations that weave the spatial fabric.

The reconstruction chain (schematic).—Steps 1–3, taken together with the RT reconstruction step that *remains to be carried out* (below), would establish that the three spatial dimensions are an **output**, not an input, of the formalism. Schematically:

$$\boxed{\text{one } \phi \xrightarrow{\nu_0 \text{ vibration}} |\Omega\rangle \xrightarrow{\text{entanglement}} S = \frac{\mathcal{A}}{4\ell_P^2} \xrightarrow{\text{RT}} g_{\mu\nu} \text{ (3+1)}} \quad (156)$$

The final arrow—the explicit tensor-network reconstruction of $g_{\mu\nu}$ from the entanglement pattern of $|\Omega\rangle_\phi$ —is the *missing* step; steps 1–3 assemble the scaffolding (area-law matching, dimensional lower bound, scalar–tensor correlator dichotomy) under which that reconstruction could be attempted, but the reconstruction itself is not executed in this paper. The slogan “space is the entanglement shadow of ϕ ” should therefore be read as the *target* of the programme, not as a theorem established by Eqs. (153)–(155); its status is the same as that of ER = EPR in the general holography literature.

Observational context.—Three independent observations are *compatible* with the single-entity reading developed above, though each is equally compatible with a conventional quantum-field-theory reading—they constrain, rather than select between, the two viewpoints:

- **Universality of ν_0 :** if ν_0 is taken as genuinely universal then no synchronisation mechanism is required. This is descriptive of any single-field theory (including ordinary QFT) and does not by itself pick out the “one entity” interpretation.
- **Spatial continuity:** Fermi-LAT constrains Lorentz-violating dispersion of gamma rays from cosmological distances down to the Planck scale. A single continuous entity is consistent with this constraint, but so is any standard continuum field theory with exact Lorentz

symmetry; the observation excludes strongly granular models, not the two readings being compared here.

- **Quantum non-locality:** Bell-inequality violations (Aspect, Clauser, Zeilinger, Nobel 2022) are reproduced by the Fock-space structure of ordinary entangled multi-particle states in standard QFT; they are equally consistent with the single-entity reading, but do not discriminate between the two.

These observations therefore *allow* the single-entity interpretation without uniquely picking it out over the standard-QFT reading. A genuinely discriminating signature of the single-entity picture (if any) would live in gravitational degrees of freedom and is an open question, not a consequence of the observations listed above.

The next step is the full Ryu–Takayanagi reconstruction: given $|\Omega\rangle_\phi$ and its entanglement structure, build the (3+1)-dimensional metric $g_{\mu\nu}$ explicitly using tensor-network (MERA) techniques. We emphasise that this reconstruction is **not** carried out in the present paper: steps 1–3 above assemble necessary ingredients (area law, dimensional lower bound, confinement dichotomy), but the explicit construction of $g_{\mu\nu}$ from $|\Omega\rangle$ is a self-contained tensor-network computation and is the topic of a separate work. Until that computation is carried out, the claim “space is the entanglement shadow of ϕ ” has the status of a structural conjecture supported by the scaffolding of Eqs. (153)–(155), not of a derived theorem.

A speculative corollary: other notes, other universes.—If our cosmos is a string vibrating at the note ν_0 , nothing in the formalism forbids the existence of other “strings” vibrating at different fundamental frequencies ν'_0 . Each such string would generate its own harmonic tower, its own oscillon spectrum, and therefore its own particle physics and chemistry. Two strings at incommensurate frequencies would be mutually transparent—unable to exchange coherent energy—and would coexist in the same “space” without interacting, much as two radio stations on different frequencies share the same air. In this picture, a multiverse is not a collection of spatially separated bubble universes but a *superposition of co-located vibrational modes at different fundamental pitches*—each pitch defining a self-contained physics, invisible to the others.

7.5 Electric charge as transverse symmetry breaking

The dimensional-emergence picture suggests a natural origin for electric charge that requires no new fields or parameters.

The transverse plane and its symmetry.—The ν_0 -string vibrates transversely, creating a two-dimensional plane (§7.2). This vibration is **symmetric**: the string displaces equally in all transverse directions. The oscillon—a localized, nonlinear resonance of the same string—may or may not preserve this symmetry:

- **Symmetric oscillon** ($\ell = 0$ only): the localized vibration pattern has no preferred transverse direction. In the ideal fundamental-mode limit it has no transverse dipole and is a candidate electrically neutral mode. → **candidate electrically neutral**.
- **Asymmetric oscillon** ($\ell = 1$ dipole component): the $G_2(X)$ nonlinearity, or the internal structure of the resonance, breaks the transverse reflection symmetry. The oscillon vibrates “more” in one transverse direction than the opposite. This dipolar pattern is the candidate geometric seed for coupling to transverse waves. → **candidate electrically charged**.

This is a charge-like structural assignment, not yet a derivation of the electromagnetic charge operator. A Noether current, an emergent gauge field A_μ , the Maxwell kinetic term, universal coupling to matter, and the normalisation of the elementary charge e remain part of the gauge-completion programme.

The **sign** of the charge is the direction of the transverse asymmetry. Because the transverse plane is two-dimensional, the asymmetry direction can be parametrised by an angle $\theta \in [0, 2\pi)$. Rotating all asymmetry directions by the same angle leaves the physics unchanged—this is a **global U(1)-like transverse-angle symmetry**. Promoting this global symmetry to a local electromagnetic $U(1)_{\text{EM}}$ gauge symmetry—that is, deriving the photon A_μ and Maxwell dynamics from the medium—is the gauge-completion programme and is not carried out here.

Consequences.— The consequences below are qualitative pictures that follow from the candidate transverse-asymmetry route. They are not derivations of the Coulomb $1/r^2$ law, the Lorentz force, linear superposition, or the exact charge-sign convention.

1. **Two signs:** the asymmetry has two stable orientations (related by π rotation) $\rightarrow$ positive and negative charge.
2. **Charge conservation:** total transverse asymmetry cannot change; creating a $+$ asymmetry requires simultaneously creating $- \rightarrow$ pair production.
3. **Antimatter:** both orientations are equally probable $\rightarrow$ every particle has an antiparticle with opposite charge.
4. **Opposite charges attract:** a $+$ and a $-$ oscillon together restore transverse symmetry between them (destructive interference of asymmetries) $\rightarrow$ lower energy $\rightarrow$ attraction.
5. **Like charges repel:** two $+$ oscillons reinforce the asymmetry (constructive interference) $\rightarrow$ higher energy $\rightarrow$ repulsion.
6. **Ideal zero-dipole modes are photon-inactive at this level:** a fundamental symmetric ($\ell = 0$) mode with no transverse dipole would not couple through this candidate channel. Neutral composites, atoms, magnetic moments, polarizabilities, and electromagnetic form factors are outside this simplified statement.
7. **Fractional charges (quarks):** inside a baryon the D_3 symmetry (§6.3) allows the asymmetry direction to lock at 120 intervals ($2\pi/3$ phase steps), yielding charges quantised in thirds. The detailed mapping to $+2/3$ and $-1/3$ requires solving the coupled scalar–vector mode equation inside the D_3 cavity—a task for future work (see the charge programme below).

Electromagnetic force as transverse Bernoulli effect.—Gravity in ISPG arises from a pressure deficit in the *scalar* (longitudinal) sector: oscillon tails deplete the ν_0 background, lowering the local pressure and producing attraction (§4). The electromagnetic force is conjectured to follow from the analogous Bernoulli mechanism operating in the *transverse* sector. At this stage this is a qualitative candidate picture, not a derivation of Maxwell electrodynamics or the Coulomb/Lorentz force law:

- **Opposite charges:** between a $+$ and a $-$ oscillon the transverse asymmetries are anti-parallel and partially cancel (destructive interference). The transverse energy density between them is *reduced* $\rightarrow$ lower transverse pressure $\rightarrow$ the oscillons are pushed together. **Attraction.**
- **Like charges:** between two $+$ oscillons the asymmetries are parallel and reinforce (constructive interference). The transverse energy density between them is *increased* $\rightarrow$ higher transverse pressure $\rightarrow$ the oscillons are pushed apart. **Repulsion.**

The unifying picture—four forces as four vibration modes of the ν_0 -string—is:

Gravity	= longitudinal (compression)	$\ell = 0$ scalar Bernoulli,	(157)
Electromagnetism	= transverse (displacement)	$\ell = 1$ transverse Bernoulli,	
Weak force	= torsional (twist)	chiral, pseudoscalar,	
Strong force	= membrane (2D vibration)	$\ell \geq 2$ confined tensor.	

Magnetism.—A magnetic field is not a separate phenomenon; it is the relativistic manifestation of the electric (transverse) asymmetry pattern when the charged oscillon *moves*. Because ISPG already derives Lorentz time dilation from frequency invariance (§1), the full electromagnetic tensor $F_{\mu\nu}$ emerges automatically: a static transverse asymmetry is an electric field; the same asymmetry seen from a boosted frame acquires a magnetic component. No additional mechanism is required.

Future work: from qualitative to quantitative electromagnetism.—The structural elements above (symmetry breaking $\rightarrow$ charge, 2D transverse plane $\rightarrow$ U(1), Bernoulli attraction/repulsion) are fixed by the geometry and require no free parameters. What remains is to close two quantitative gaps:

1. **Charge assignment.** The $G_2(X)$ nonlinearity or the Mathieu mode structure must determine which oscillons develop an $\ell = 1$ transverse component (charged) and which remain purely $\ell = 0$ (neutral). Concretely, the Mathieu stability chart for the $\ell = 1$ sector has its own critical q threshold; oscillons whose N falls in the $\ell = 1$ stable band are charged. This is a well-posed eigenvalue problem once the $\ell = 1$ effective potential is extracted from $G_2(X)$.
2. **Fine-structure constant.** α_{EM} should emerge as the ratio of the $\ell = 1$ dipole amplitude to the $\ell = 0$ monopole amplitude:

$$\alpha_{\text{EM}} = \frac{|\Phi_{\ell=1}|^2}{|\Phi_{\ell=0}|^2} \stackrel{?}{=} \frac{1}{137.036}. \quad (158)$$

Current exploratory scans in `SM/python/scan_alpha*.py` do not yet obtain the observed value; deriving the dipole-to-monopole amplitude ratio that yields $\alpha_{\text{EM}} \simeq 1/137$ remains future work. This ratio is computable from the same lattice simulation that determines q and η_T . It constitutes the decisive quantitative test of the electromagnetic sector.

Both computations are lattice tasks within the existing $G_2(X)$ framework; no new theoretical ingredient is required.

7.6 The Higgs boson as amplitude fluctuation

The companion paper gives the bi-conformal scaling of an effective electroweak VEV [100–104] as conformal symmetry breaking—showing that $v(\varphi) = v_0 e^{\varphi/2}$, recovering the broken-phase electroweak gauge-boson mass matrix once the SM electroweak sector is supplied, and recording the gravitational correction $\delta m_H^2 \sim 10^{-34} m_H^2$ —is given in the companion paper [51] (ISPG_Quantum, Sec. 8). Here we present the oscillon-level picture that motivates the candidate ISPG replacement.

Throughout this paper a single theme recurs: **mass is the magnitude of vibration**. An oscillon’s mass is its harmonic number N ; $m_N = m_1 b_N(q)$ is, at its core, a statement about how vigorously the ν_0 -string vibrates in the trapped region. This theme identifies the target that a Higgs-replacement mechanism must reproduce.

The oscillon amplitude as candidate order parameter.—The ν_0 -string vibrates everywhere with amplitude Φ_0 :

$$\phi(x, t) \sim \Phi_0 \cos(2\pi\nu_0 t). \quad (159)$$

If $\Phi_0 = 0$ (the string is at rest), the $G_2(X)$ nonlinearity has nothing to act on: no self-focusing occurs, no oscillons form, and every excitation propagates as a massless wave. The moment $\Phi_0 \neq 0$, the nonlinear term $\alpha X^2/M^2$ activates, oscillons condense, and particles acquire mass. This is the ISPG analogue of the role played by the Standard Model’s Higgs vacuum expectation value v : with $v = 0$ the SM fermions and gauge bosons are massless; with $v \neq 0$ they acquire mass proportional to their coupling.

The normalization target is:

$$\boxed{\Phi_0 \longrightarrow v_{\text{EW}} = 246 \text{ GeV} \quad (\text{open coarse-graining map}).} \quad (160)$$

Here Φ_0 is dimensionless oscillon-profile data, whereas v_{EW} is an imported electroweak energy scale until the normalization map is derived.

Effective amplitude symmetry breaking.—The $G_2(X)$ kinetic structure suggests an equilibrium amplitude Φ_0 (Eq. 99). The linear term $-M_{\text{Pl}}^2 X/2$ penalises large amplitudes; the nonlinear term $\alpha X^2/M^2$ rewards them. The competition produces an effective potential $V_{\text{eff}}(\Phi_0)$ with a minimum at $\Phi_0 \neq 0$:

$$V_{\text{eff}}(\Phi_0) \sim -\mu^2 \Phi_0^2 + \lambda \Phi_0^4, \quad \mu^2, \lambda > 0,$$

where μ^2 and λ are determined by α , M , M_{Pl} , and ω_0 in the effective amplitude description. This is the **Mexican-hat analogy**: the $G_2(X)$ kinetic structure suggests an effective amplitude potential in the broken-phase regime, but a genuine non-derivative $\text{SU}(2)$ -doublet Higgs potential is not derived here. The string “chooses” to vibrate at $\Phi_0 \neq 0$ rather than remaining at rest: this is the candidate ISPG analogue of spontaneous symmetry breaking.

The Higgs boson.—A localised perturbation in which the vibration amplitude deviates from its equilibrium value, $\Phi_0 \rightarrow \Phi_0 + h(x, t)$, oscillates around the potential minimum. The frequency of this oscillation is the **candidate amplitude-mode Higgs mass**:

$$m_H^{(\text{amp})} = \sqrt{V_{\text{eff}}''(\Phi_0)}. \quad (161)$$

The observed value $m_H = 125 \text{ GeV}$ is currently a target/ladder hit, not a first-principles prediction from this effective amplitude potential. In signal-processing language: the ν_0 -vibration is the **carrier wave**; the Higgs boson is its **amplitude modulation** (AM).

Coupling proportional to mass.—Heavier particles are higher harmonics of the ν_0 -string. Higher harmonics depend more sensitively on the nonlinear trapping strength, which is set by Φ_0 . A fluctuation $h(x, t)$ therefore affects heavy oscillons more than light ones:

$$g_{Hff} \propto \frac{\partial m_N}{\partial \Phi_0}, \quad (162)$$

which must be tested by explicitly differentiating the Mathieu law $m_N = m_1 b_N(q(\Phi_0))$. SM-like proportionality to m_N is not automatic; it requires verifying that $(\partial b_N / \partial q)(\partial q / \partial \Phi_0)$ reproduces the Yukawa coupling pattern. This derivative test is open future work.

Goldstone bosons and W/Z masses.—The field ϕ is a single *real* scalar, but its transverse perturbations carry two circular-polarisation degrees of freedom $\delta\phi_{\pm} \propto e^{\pm i\vartheta}$ (§5.3). The full low-energy field content is therefore:

$$\underbrace{\Phi_0}_{\text{amplitude (real)}} + \underbrace{\theta_1, \theta_2, \theta_3}_{\text{polarisation angles (SU(2) phase space)}}, \quad (163)$$

where the three θ_i parametrise the orientation of the polarisation ellipse in the two-dimensional transverse plane and its coupling to the longitudinal sector (§5.6). No new field is introduced: the θ_i are collective coordinates of the same ϕ .

Fluctuations of Φ_0 cost energy (the effective amplitude curvature) $\rightarrow$ **candidate amplitude-mode Higgs boson**. Fluctuations of θ_i are the proposed collective route to the three would-be Goldstone directions. A completed E7 construction must still show that these collective coordinates form the correct $SU(2)_L \times U(1)_Y/U(1)_{\text{EM}}$ coset, couple gauge-invariantly to the vector sector, and reproduce the standard broken-phase mass matrix. Only after that step can they be identified with Goldstone modes that are “eaten” by $W^\pm$ and Z^0 . At present the observed W/Z masses remain conditional on the imported SM electroweak data and on the open $\Phi_0 \rightarrow v_{\text{EW}}$ normalization.

The electroweak coincidence.—The geometric mean of the Planck mass and the electron mass is $\sqrt{m_P m_e} \approx 79$ GeV, remarkably close to $m_W = 80$ GeV (§3.12). In the ISPG picture this points to a common structural origin: the electroweak scale v_{EW} is an open normalization target for Φ_0 , which is fixed by $\omega_0 = 2\pi\nu_0$ and the $G_2(X)$ parameters (Eq. 99)—the same quantities that determine m_e (the first stable harmonic) and m_P (the Planck scale). All three scales share a common origin.

Future work.—The Higgs mass would be obtained from $\sqrt{V''_{\text{eff}}(\Phi_0)}$ once the effective amplitude potential, the $\Phi_0 \rightarrow v_{\text{EW}}$ normalization, and the Φ_0 -dependence of the Mathieu coefficient $b_N(q(\Phi_0))$ are derived. The observed $m_H = 125$ GeV is currently a target/ladder hit, not a first-principles prediction. The precise mapping between Φ_0 and $v_{\text{EW}} = 246$ GeV through the bi-conformal normalisation conventions is a natural part of that calculation.

7.7 Summary

Dimensional emergence in ISPG.

1. The ν_0 -background has an integer-harmonic spectrum (j_0 zeros = $n\pi$)—the signature of a 1D string.
2. Vibration of the string requires one transverse direction $\rightarrow$ 2D.
3. Tensor ($\ell \geq 2$) modes require a second transverse direction $\rightarrow$ 3D.
4. Three dimensions is the **minimum** for both scalar and tensor modes; no further dimension is generated.
5. Confinement is the incompatibility between the 2D vibration (gluon) and the 1D background (ν_0 -string).
6. ν_0 is universal because the substrate is a single quantum entity (one string, one pitch).
7. Space = entanglement shadow of ϕ : $|\Omega\rangle \rightarrow S \propto A \rightarrow g_{\mu\nu}$ (Ryu–Takayanagi route, §7.4).
8. Electric charge = transverse symmetry breaking of the oscillon; $U(1)$ = rotational freedom of the 2D transverse plane (§7.5).
9. Neutrino = polarisation-rotation wave (helical vortex, $n = \pm 1$); weak-sector candidate = nonlinear torsion–transverse coupling; parity violation has a candidate geometric source in torsional chirality (§5).
10. **Four forces = four vibration modes** of one string: longitudinal (gravity), transverse (EM), torsional (weak), membrane (strong).
11. Conditional scaling: $v(\varphi) = v_0 e^{\varphi/2}$; the oscillon amplitude Φ_0 is the candidate ISPG analogue of the Higgs order parameter, with $\Phi_0 \rightarrow v_{EW}$ an open normalization target. $G_2(X)$ suggests an effective amplitude potential; a non-derivative $SU(2)$ -doublet potential is not derived here (§7.6).

Companion-paper results and candidate routes (ISPG-Quantum):

12. Spin-1/2 structural route = vortex winding number, $J = n\hbar/2$ under the conditional spin bridge; Fermi–Dirac/Pauli structure as a candidate route from Berry phase, with the rigorous QFT/Fock-space proof open (Sec. 10).
13. Effective Higgs-sector scaling: $v(\varphi) = v_0 e^{\varphi/2}$; the broken-phase mass matrix is recovered once the SM electroweak sector and imported v_0 are supplied (Sec. 8).
14. Koide formula $Q = 2/3$: numerical support from the E1D/rank-1 cavity analysis on the MASS/Mathieu lepton ladder, 0.009% accuracy; the 3D-to-1D bridge from raw cavity eigenfrequencies to lepton masses remains open (Sec. 7.2).
15. Inertia: $F = -m_{\text{eff}} a$ from retarded oscillon potential; $m_{\text{inertial}} \equiv m_{\text{gravitational}}$ (Sec. 5).
16. Dark-energy scale estimate (conditional): $\rho_{\text{vac}} \sim 10^{-121} M_{\text{Pl}}^4$ on the perturbative branch (Sec. 6).
17. Gravitational phase/decoherence: $\Gamma_\varphi = 2m_0|\Delta\Phi_N|/\hbar$ as the leading coherent rate; the eV-photon ~ 0.5 Hz coefficient is short-flight limited, and irreversible decoherence is conditional on the tail/noise correlator (Sec. 9).

This section is primarily conceptual; the mathematical foundations—Bessel-function identities, gravitational-wave counting in d dimensions, the spherical-cavity decomposition—are all standard results. The new content is the *chain of inference*: from $j_0(x) = \sin(x)/x$ to a lower-bound/minimality route to the observed three spatial dimensions and to a structural confinement criterion for tensor modes. The quantitative programme—proving that the one-dimensional radial/Mathieu mass ladder and the full 3D cavity dynamics reduce to one another,

with an explicit $N \leftrightarrow (n, \ell)$ map and the observed 3+1-dimensional spacetime—is a goal for future work.

8 Future Directions

Completed (in this paper)

- ✓ Oscillon wave equation solution: $m_N = m_1 b_N(q)$, Koide 0.02%, muon $< 0.001\%$ (§3.10, §3.12, §3.13).
- ✓ q is universal—sector-specific q is excluded by the electron stability condition.
- ✓ Glueball mass-ratio pattern from Bessel-function zeros: $\leq 2\%$ for four lattice-benchmark assignments, with no additional coefficient once the assignments are chosen; proton anchoring remains an external scale check (§6.3).
- ✓ Bekenstein–Hawking entropy matching: $S_{\text{ent}} = \mathcal{A}/(4\ell_P^2)$ from $G_2(X)$ sound-speed reduction and Mathieu-band summation (§7.4).

Companion-paper results and conditional estimates (ISPG_Quantum)

- ▷ **Spin-1/2 and Fermi–Dirac statistics (candidate topological route):** $J = n\hbar/2$ from vortex winding under the conditional spin bridge; Berry phase $= \pm\pi$ for $n = \pm 1$ exchange as a structural Pauli/Fermi–Dirac route; the full QFT/Fock-space proof remains imported/open (Sec. 10).
- ▷ **Effective Higgs-sector scaling:** $v(\varphi) = v_0 e^{\varphi/2}$ as conditional conformal scaling of an imported electroweak scale; the $\Phi_0 \rightarrow v_{\text{EW}}$ normalization and first-principles Higgs-mass derivation remain open (Sec. 8).
- ✓ **Koide formula:** $Q = 2/3$ numerical support via rank-1 perturbation and Hartree analysis on the MASS/Mathieu lepton ladder, 0.009% accuracy; the 3D-to-1D bridge from raw cavity eigenfrequencies (which give $Q \approx 0.336$, not $2/3$) remains open (Sec. 7.2).
- ✓ **Inertia and equivalence principle:** $F = -m_{\text{eff}} a$ derived from retarded potential of accelerated oscillon; $m_{\text{inertial}} \equiv m_{\text{gravitational}}$ (Sec. 5).
 - **Dark-energy scale estimate (conditional):** echo-background energy self-regulated by a negative-feedback loop, yielding $\rho_{\text{vac}} \sim 10^{-121} M_{\text{Pl}}^4$ on the perturbative branch, while zero-point decoupling and non-perturbative normalization remain open (Sec. 6).
 - **Gravitational phase/decoherence:** $\Gamma_\varphi = 2m_0|\Delta\Phi_N|/\hbar$ is the leading coherent rate; the eV-photon ~ 0.5 Hz coefficient is short-flight limited, and irreversible decoherence is conditional on the tail/noise correlator (Sec. 9).
- ✓ **Gravitational birefringence:** one-loop vacuum polarisation prediction, distinguishable from QED and Lorentz-violation birefringence (Sec. 11).

Highest priority: would elevate fits to predictions

- **Lattice simulation of k-oscillons** with $G_2(X)$ on the bi-conformal background—to compute β , Φ_0 , and verify $\beta\Phi_0^2 = 0.853$ independently of the muon mass (Eq. 99). *This is the single most important outstanding calculation.*
- $N = 4$ **resonance search** at ~ 329 keV (§3.12). Experimental detection or exclusion directly tests the Mathieu stability structure.

Future work directions

- **Neutrino programme:** derive the torsional rigidity κ_T from $G_2(X)$; solve Eq. (??) for the three torsional eigenvalues and PMNS mixing angles; determine mass ordering and Majorana/Dirac character; derive the Weinberg angle from the torsion–transverse coupling ratio (§5).
- **Massless sector:** derive photon coupling constants; solve Eq. (137) for the oscillon cavity eigenmodes to verify whether exactly eight confined tensor modes emerge with $SU(3)$ algebra; compute σ from $G_2(X)$ and verify $V(L) \propto L$ (§6).
- **Photon identification:** determine how the scalar-metric medium produces spin-1 transverse excitations (photons) in addition to the established spin-2 TT modes (classical gravitational waves of the medium itself, not a separate quantum excitation) (§6).
- **Electroweak programme:** derive $SU(2)_L \times U(1)_Y$ from the torsion–transverse coupling structure; compute the Weinberg angle θ_W ; verify W/Z masses from $(R_\perp/\lambda_0)$ suppression (§5, §7.5).
- **Higgs-mass programme:** build the first-principles amplitude-potential calculation of $V''_{\text{eff}}(\Phi_0)$ using α and M from the lattice simulation, and test whether it reproduces the $m_H = 125$ GeV target (§7.6).
- **Electric charge and $\alpha_{\text{EM}} \approx 1/137$:** attempt to derive from the dipole-to-monopole amplitude ratio; test whether D_3 phase locking can reproduce quark charges $+2/3, -1/3$ (§7.5).
- **Pre-geometric derivation of dimensionality:** reformulate §7 in a tensor-network or graph-theoretic framework; complete the Ryu–Takayanagi reconstruction (§7.4).
- **Discrete substrate:** lattice or graph reproduction of the bi-conformal metric in the continuum limit.

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